

Course-End Project

Securing Linux Servers Using Honeypots and IP Blocking

Objective

To deploy a honeypot and configure defensive mechanisms like SSH hardening and automated IP blocking to detect, analyze, and mitigate SSH brute-force attacks on public-facing Linux servers

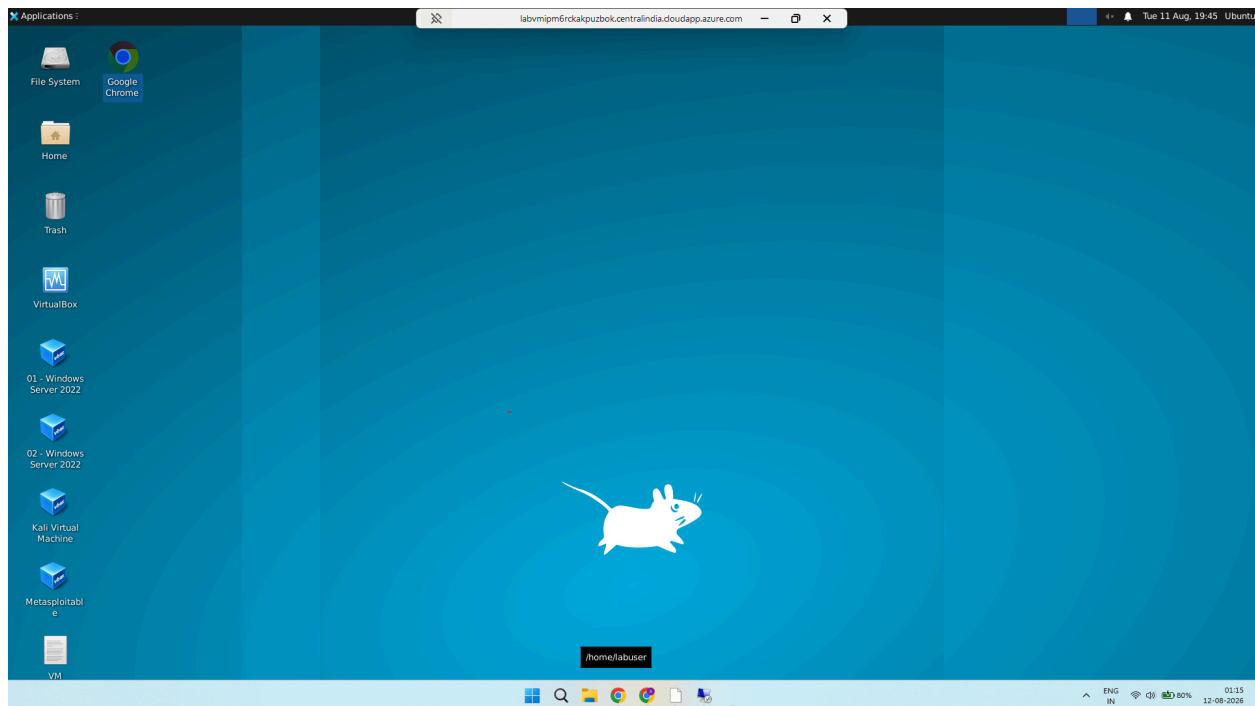
Tasks

The following tasks outline the process of deploying a honeypot and configuring SSH hardening and IP blocking to detect and mitigate brute-force attacks on Linux servers:

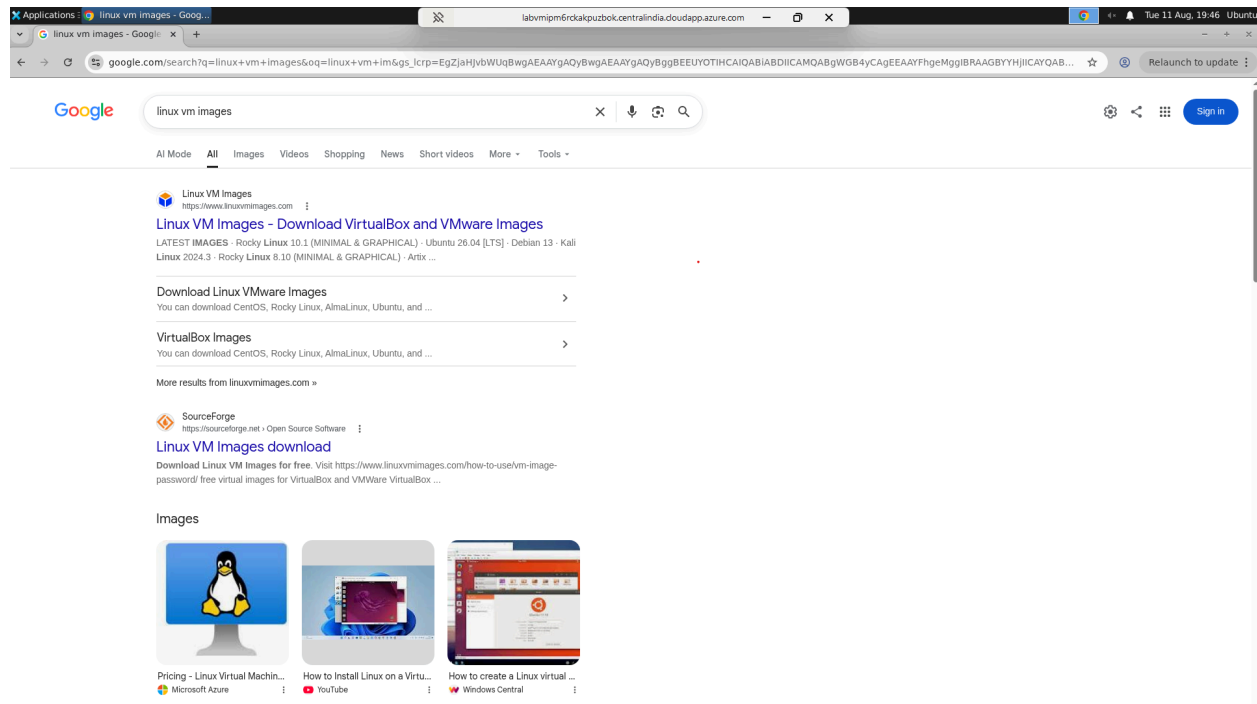
1. Deploy honeypot to simulate SSH attacks and collect data
2. Harden SSH configuration to secure the server
3. Implement IP blocking to mitigate repeated attacks
4. Analyze attack data and generate security test cases

Task 1: Deploy Honeypot to Simulate SSH Attacks and Collect Data

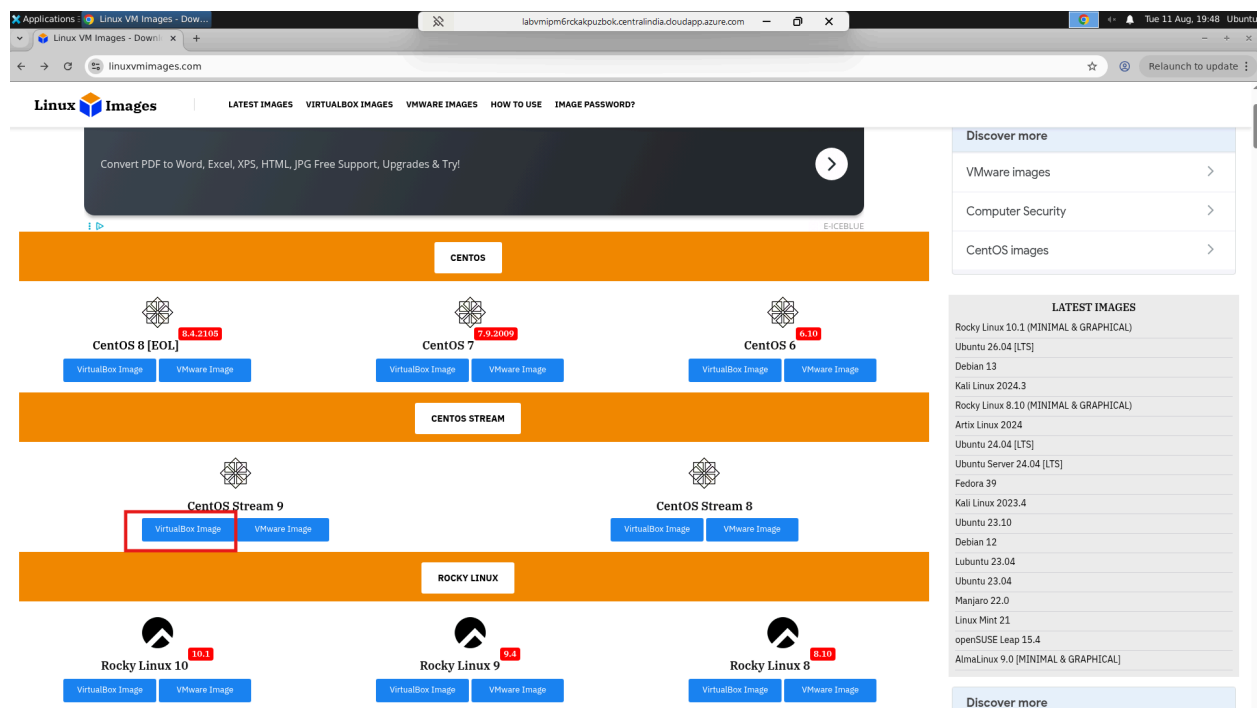
1.1 Click on the Google Chrome icon on the desktop to open the browser



1.2 Search for Linux VM Images in Google and click on the link Linux VM Images download VirtualBox and VMware Images from the search results.



1.3 Click on the VirtualBox Image under the CentOS Stream 9 section to download the VirtualBox image for CentOS



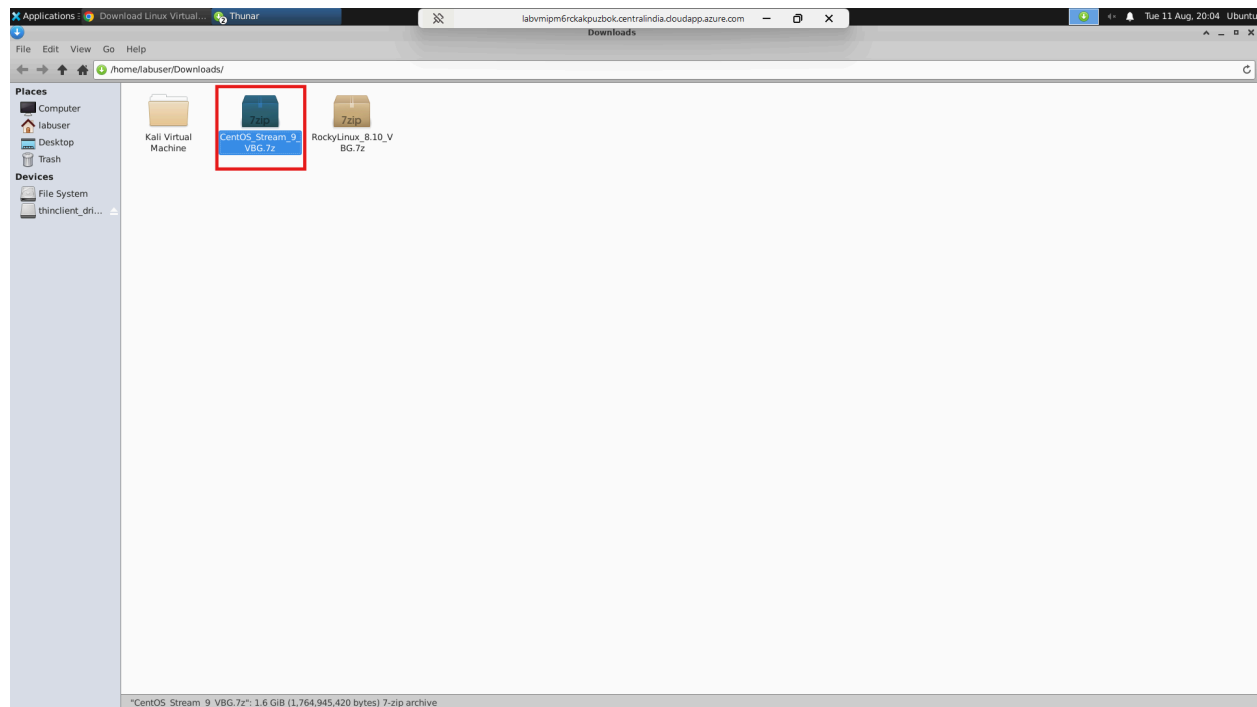
1.4 Click on the Download Links under the Contents section to access the download links for the CentOS Stream 9 VirtualBox image

The screenshot shows the Linux Images website with the CentOS Stream 9 VirtualBox image page. The page features a sidebar with navigation links, a main content area with a large image of a person in a lab coat, and a 'CONTENTS' section. The 'CONTENTS' section includes links for 'Introduction', 'Download Links', 'System Details & Login Information', 'Import VM Image', and 'Conclusion'. The 'Download Links' link is highlighted with a red box. Below the 'CONTENTS' section, there is an 'Introduction' section with text about CentOS Stream 9. A 'New Features' section lists five features: 1. Kernel v5.14, Python 3.9; 2. The default desktop environment is GNOME 4; 3. Integrated OpenSSL 3; 4. Smart Card; 5. SSH root password. A 'Boost .NET Word to PDF, JPG' advertisement is visible at the bottom of the page. The URL in the browser is linuxvmimages.com/images/centosstream-9/.

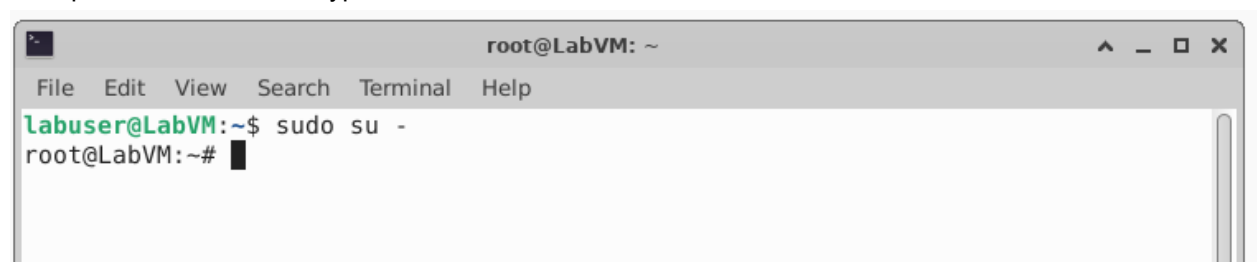
1.5 Click on the VirtualBox Image (1.7 GB) under the Graphical Installation section to download the CentOS Stream 9 image

The screenshot shows the Linux Images website with the Rocky Linux 8.10 VirtualBox image page. The page features a sidebar with navigation links, a main content area with a large image of a mountain, and a 'Graphical Installation' section. The 'Graphical Installation' section includes links for 'VirtualBox Image (1.6 GB)' and 'VMware Image (1.6 GB)'. The 'VirtualBox Image (1.6 GB)' link is highlighted with a red box. Below the 'Graphical Installation' section, there is a 'Rocky Linux 8.6' section. A 'URZĄDZENIA DO ŁODÓW I SLUSHY' advertisement is visible at the bottom of the page. The URL in the browser is linuxvmimages.com/images/rockylinux-8/#download-links.

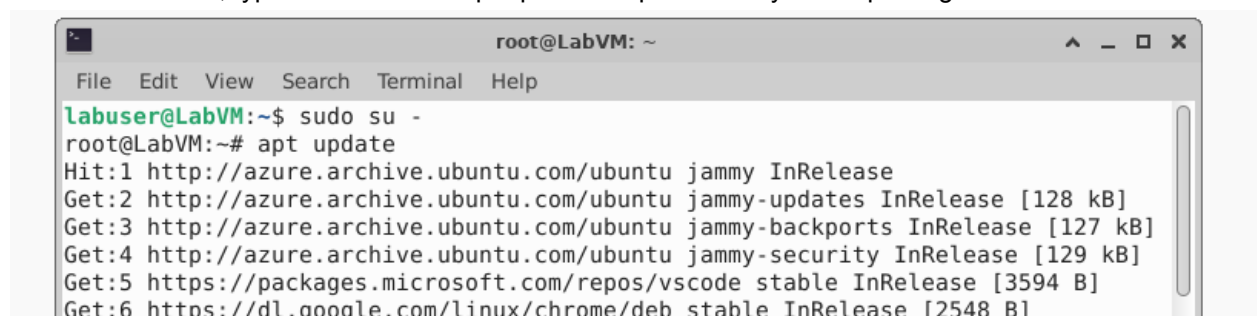
1.6 Open the file explorer and locate the downloaded CentOS_Stream_9_VBM.7z file



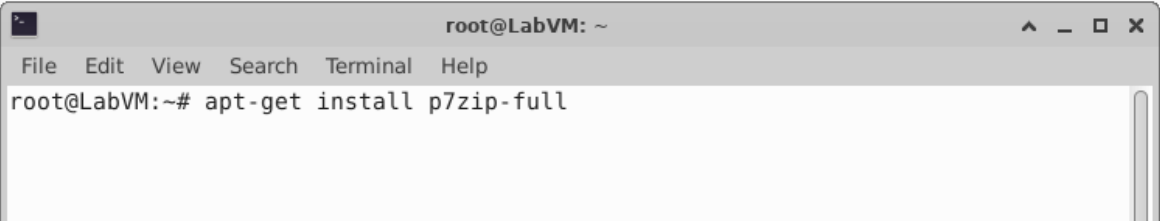
1.7 Open the terminal and type the command `sudo su -` to switch to the root user



1.8 In the terminal, type the command `apt update` to update the system's package information

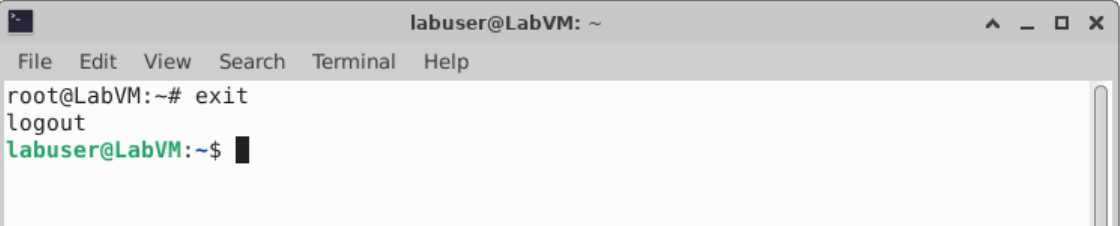


1.9 Type the command `apt-get install p7zip-full` to install the necessary software for extracting the downloaded file



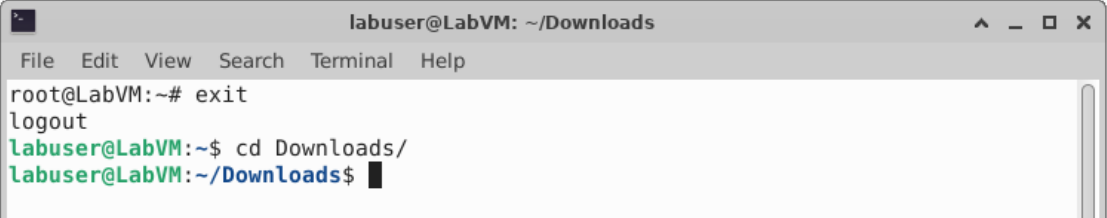
```
root@LabVM: ~  
File Edit View Search Terminal Help  
root@LabVM:~# apt-get install p7zip-full
```

1.10 Log out from the root user by typing the command `exit` in the terminal to return to the regular user



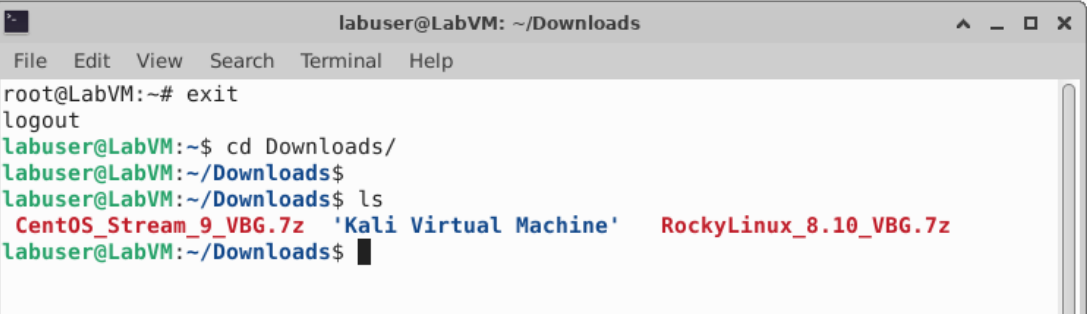
```
labuser@LabVM: ~  
File Edit View Search Terminal Help  
root@LabVM:~# exit  
logout  
labuser@LabVM:~$
```

1.11 Navigate to the Downloads directory by typing the command `cd ~/Downloads` in the terminal



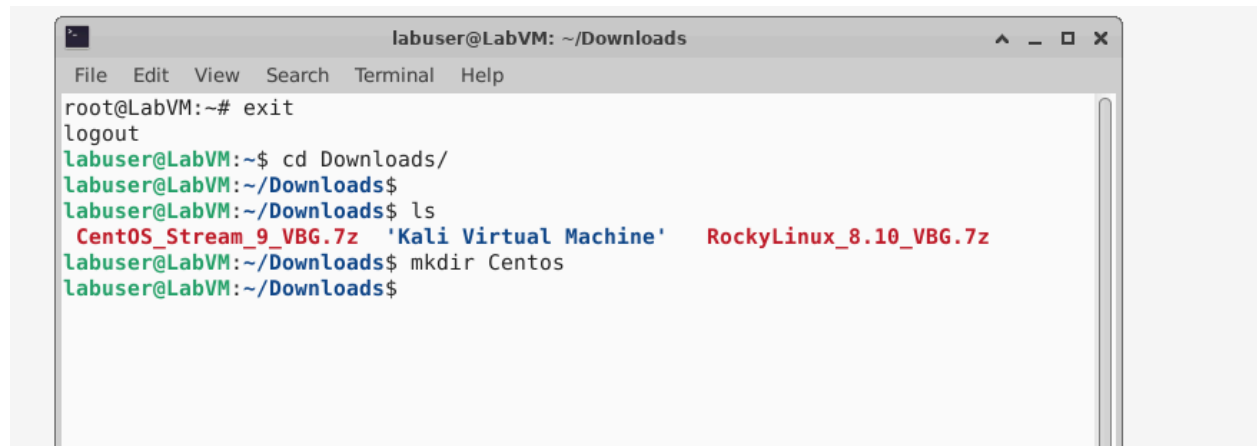
```
labuser@LabVM: ~/Downloads  
File Edit View Search Terminal Help  
root@LabVM:~# exit  
logout  
labuser@LabVM:~$ cd Downloads/  
labuser@LabVM:~/Downloads$
```

1.12 List the contents of the Downloads directory by typing the command `ls` in the terminal



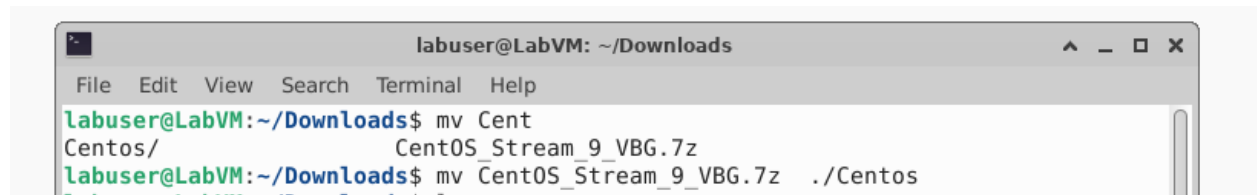
```
labuser@LabVM: ~/Downloads  
File Edit View Search Terminal Help  
root@LabVM:~# exit  
logout  
labuser@LabVM:~$ cd Downloads/  
labuser@LabVM:~/Downloads$  
labuser@LabVM:~/Downloads$ ls  
CentOS_Stream_9_VBG.7z 'Kali Virtual Machine' RockyLinux_8.10_VBG.7z  
labuser@LabVM:~/Downloads$
```

1.13 Create a new directory named Centos by typing the command mkdir Centos in the terminal



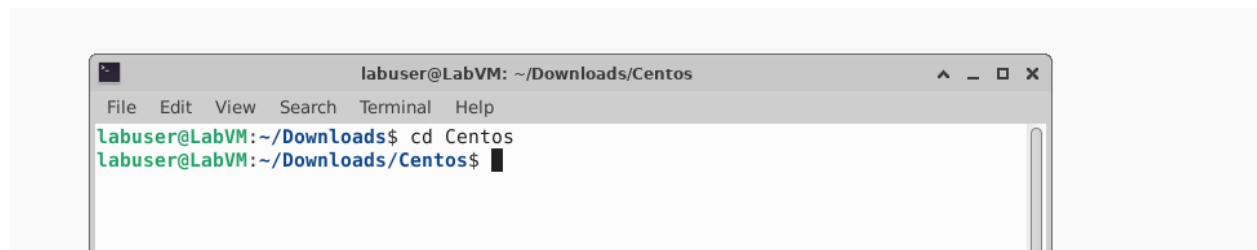
```
labuser@LabVM: ~/Downloads
File Edit View Search Terminal Help
root@LabVM:~# exit
logout
labuser@LabVM:~$ cd Downloads/
labuser@LabVM:~/Downloads$
labuser@LabVM:~/Downloads$ ls
CentOS_Stream_9_VBG.7z 'Kali Virtual Machine' RockyLinux_8.10_VBG.7z
labuser@LabVM:~/Downloads$ mkdir Centos
labuser@LabVM:~/Downloads$
```

1.14 Use the following command to move the CentOS_Stream_9_VBM.7z file into the Centos directory:
mv CentOS_Stream_9_VBM.7z ./Centos



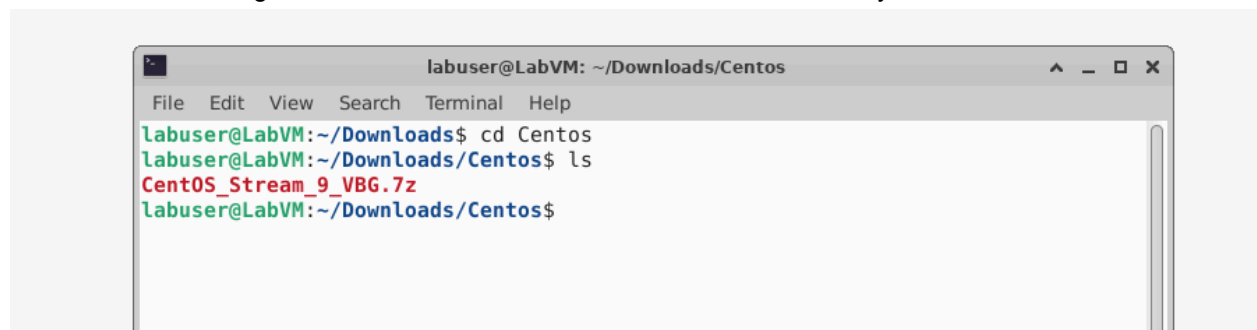
```
labuser@LabVM: ~/Downloads
File Edit View Search Terminal Help
labuser@LabVM:~/Downloads$ mv Cent
Centos/
labuser@LabVM:~/Downloads$ mv CentOS_Stream_9_VBG.7z ./Centos
```

1.15 Use the following command to navigate into the Centos directory: cd Centos



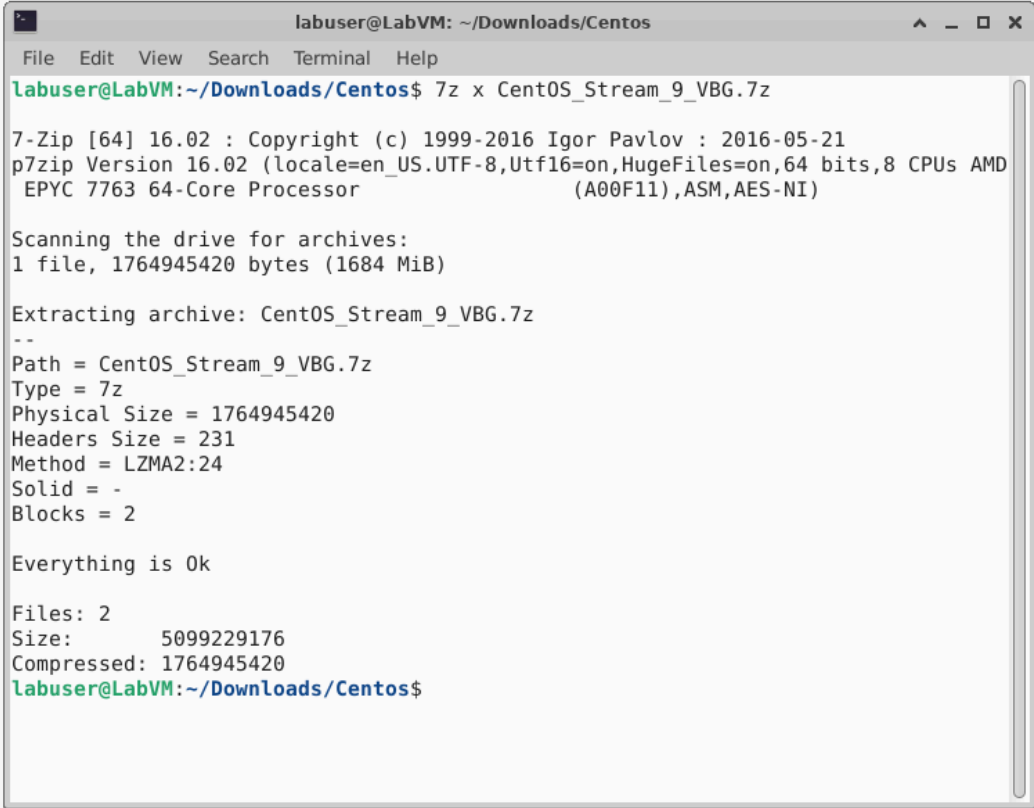
```
labuser@LabVM: ~/Downloads/Centos
File Edit View Search Terminal Help
labuser@LabVM:~/Downloads$ cd Centos
labuser@LabVM:~/Downloads/Centos$
```

1.16 Use the following command to list the contents of the Centos directory: ls



```
labuser@LabVM: ~/Downloads/Centos
File Edit View Search Terminal Help
labuser@LabVM:~/Downloads$ cd Centos
labuser@LabVM:~/Downloads/Centos$ ls
CentOS_Stream_9_VBG.7z
labuser@LabVM:~/Downloads/Centos$
```

1.17 Use the following command to extract the CentOS_Stream_9_VBM.7z file: 7z x CentOS_Stream_9_VBM.7z



```
labuser@LabVM: ~/Downloads/Centos
File Edit View Search Terminal Help
labuser@LabVM:~/Downloads/Centos$ 7z x CentOS_Stream_9_VBG.7z

7-Zip [64] 16.02 : Copyright (c) 1999-2016 Igor Pavlov : 2016-05-21
p7zip Version 16.02 (locale=en_US.UTF-8,Utf16=on,HugeFiles=on,64 bits,8 CPUs AMD
EPYC 7763 64-Core Processor               (A00F11),ASM,AES-NI)

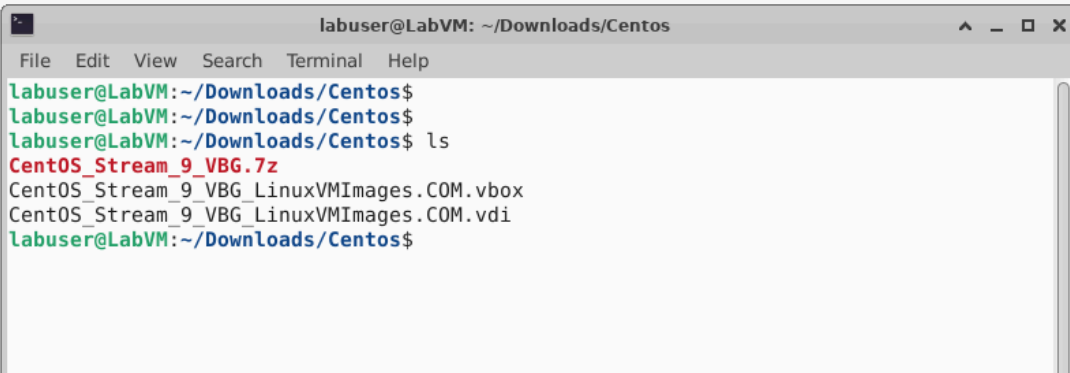
Scanning the drive for archives:
1 file, 1764945420 bytes (1684 MiB)

Extracting archive: CentOS_Stream_9_VBG.7z
--
Path = CentOS_Stream_9_VBG.7z
Type = 7z
Physical Size = 1764945420
Headers Size = 231
Method = LZMA2:24
Solid = -
Blocks = 2

Everything is Ok

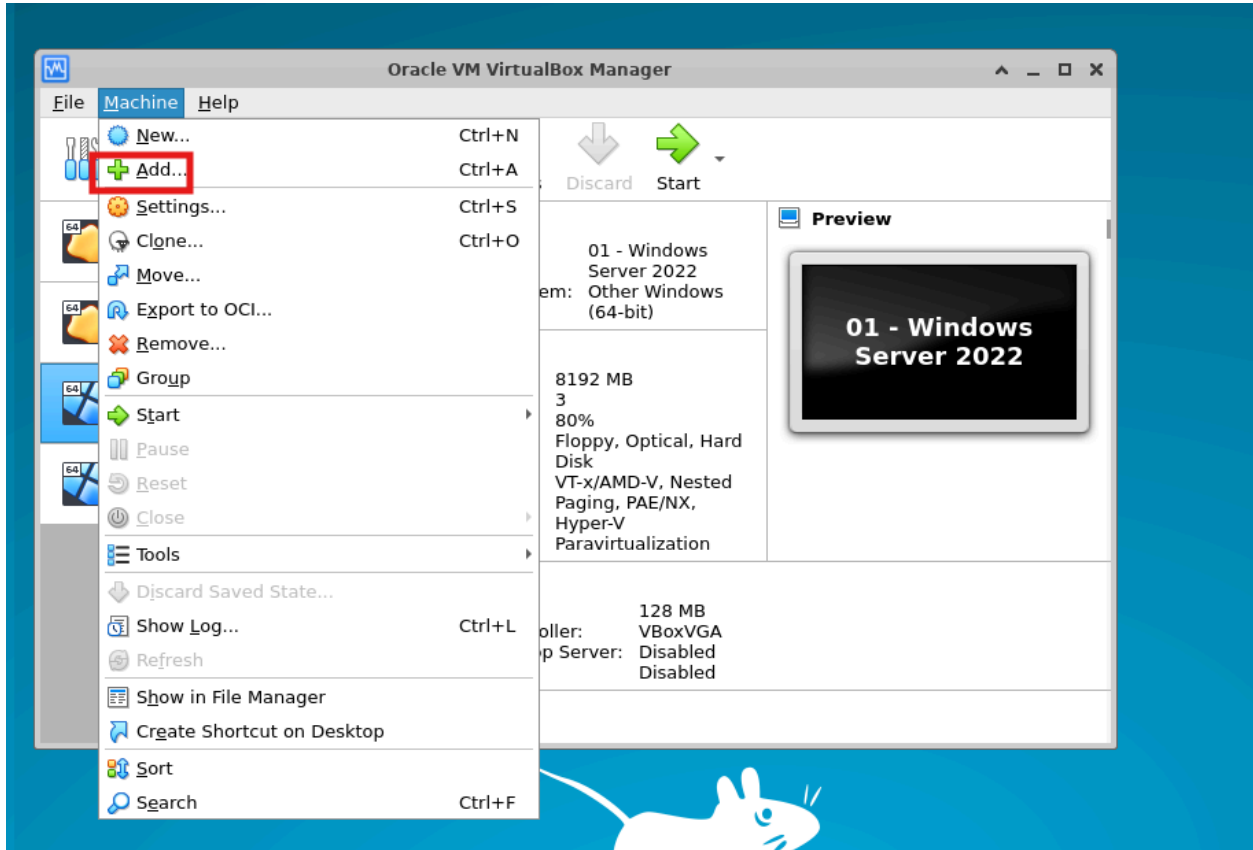
Files: 2
Size:      5099229176
Compressed: 1764945420
labuser@LabVM:~/Downloads/Centos$
```

1.18 Use the following command to list the contents of the "Centos" directory and confirm the extracted files: ls

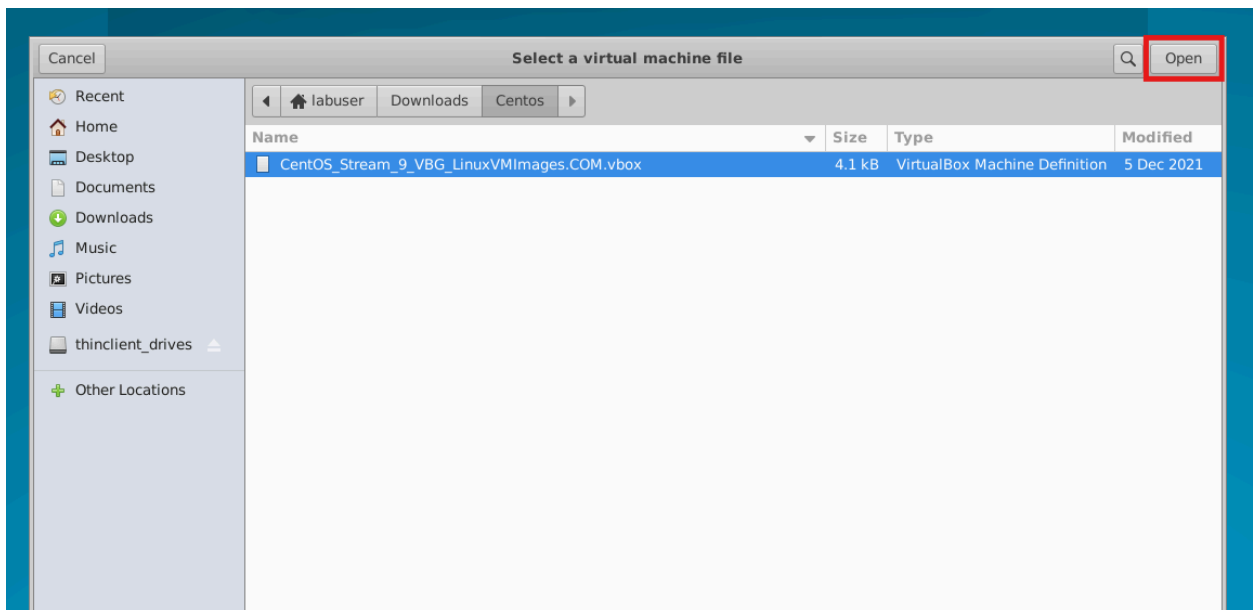


```
labuser@LabVM: ~/Downloads/Centos
File Edit View Search Terminal Help
labuser@LabVM:~/Downloads/Centos$
labuser@LabVM:~/Downloads/Centos$
labuser@LabVM:~/Downloads/Centos$ ls
CentOS_Stream_9_VBG.7z
CentOS_Stream_9_VBG_LinuxVMImages.COM.vbox
CentOS_Stream_9_VBG_LinuxVMImages.COM.vdi
labuser@LabVM:~/Downloads/Centos$
```

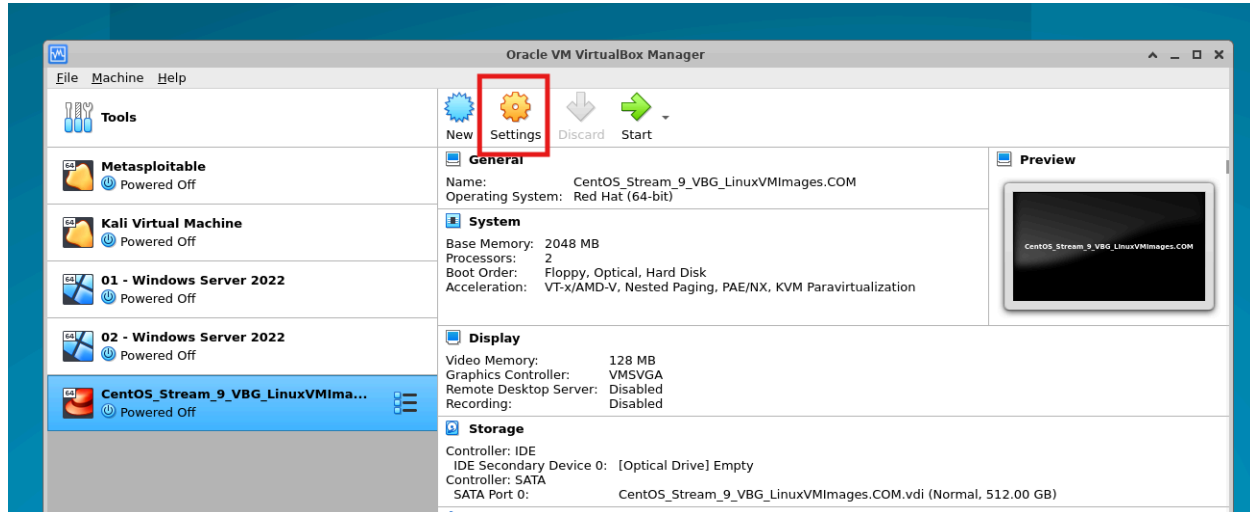
1.19 Open Oracle VM VirtualBox and start the Windows Server 2022 virtual machine by selecting it and click on Add



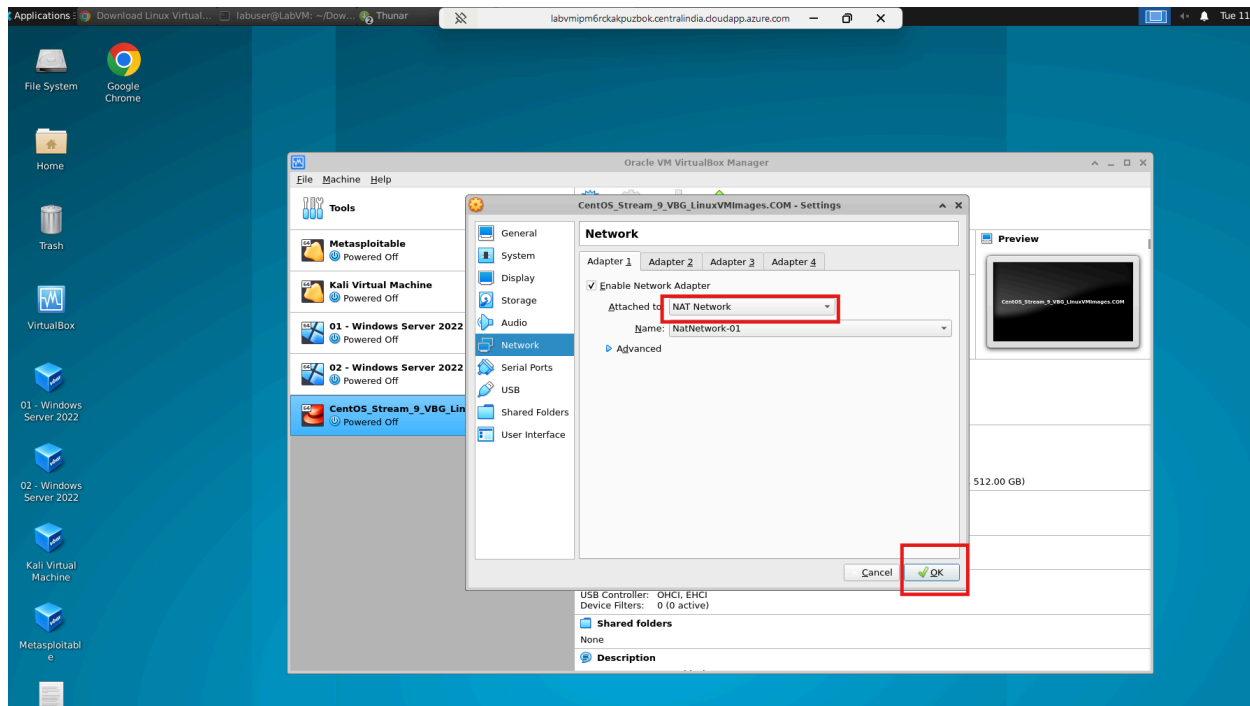
1.20 Select the CentOS_Stream_9_VBM_LinuxVMImages.COM.vbox file from the Centos directory in Oracle VM VirtualBox and click Open to add the CentOS virtual machine



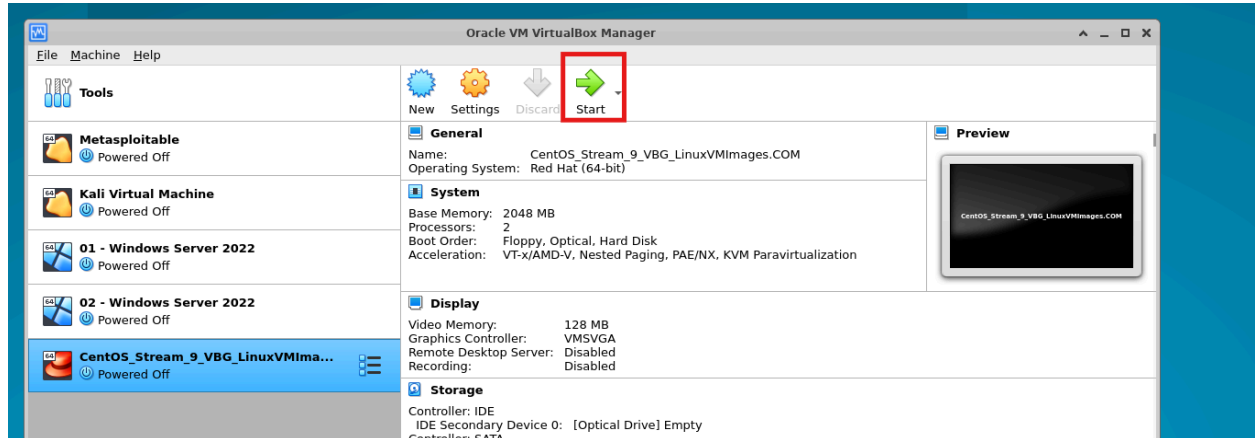
1.21 Select the CentOS_Stream_9_VBG_LinuxVMImages.COM virtual machine in Oracle VM VirtualBox and click Settings



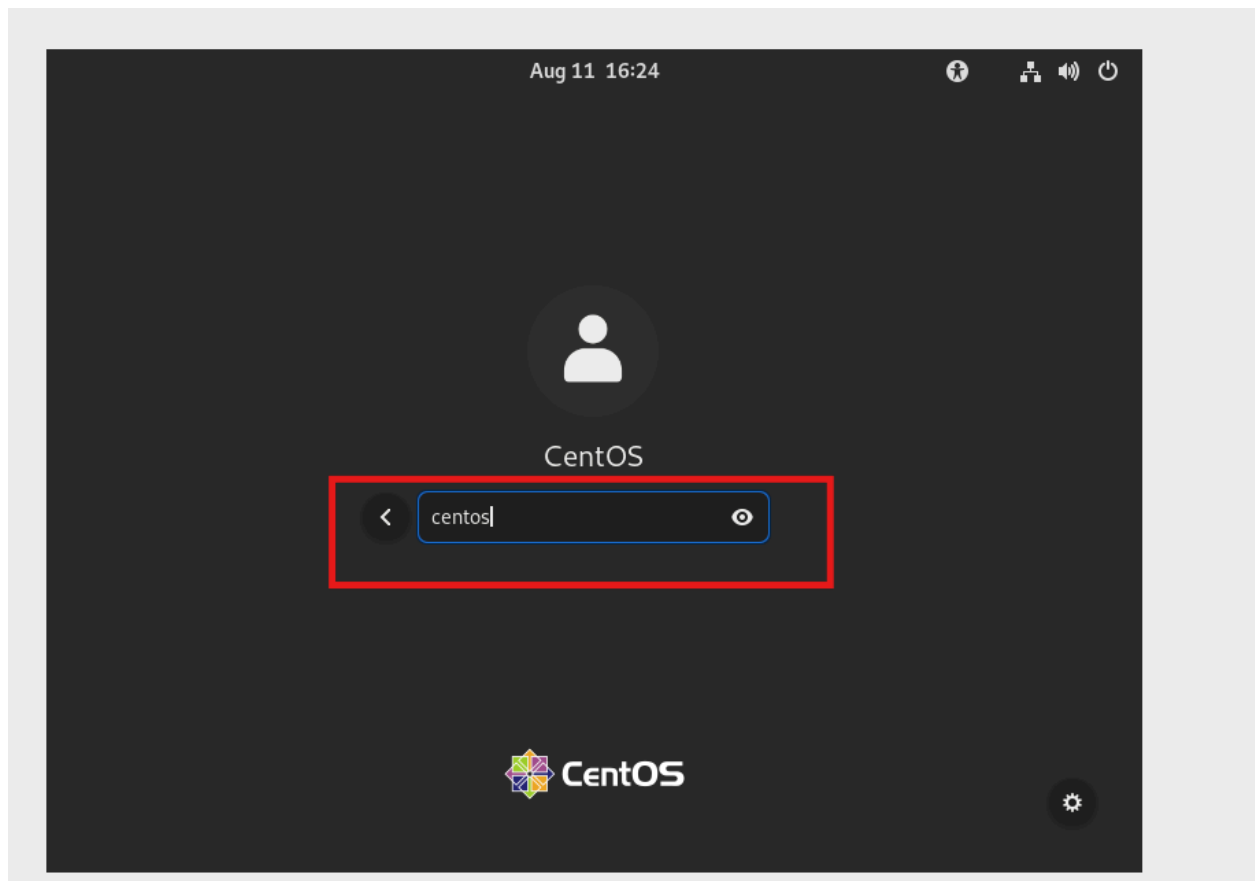
1.22 Go to the Network settings of the virtual machine, enable the network adapter, and set it to NAT Network. then, click OK to save the changes



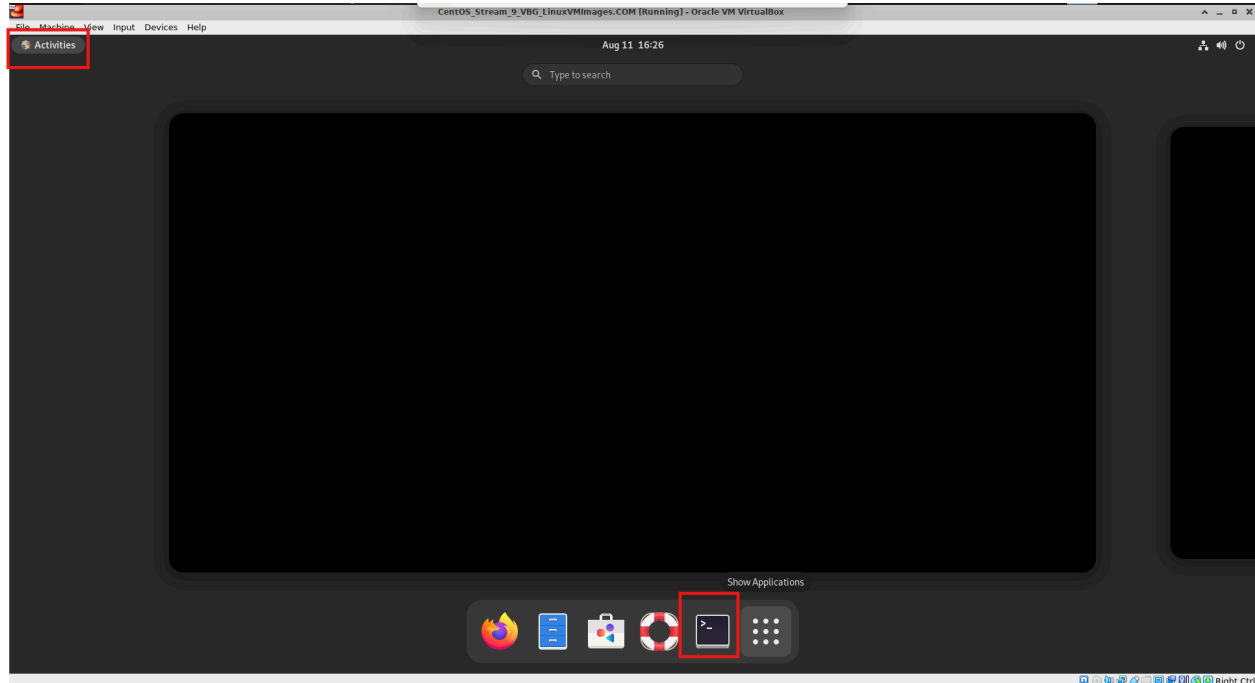
1.23 Click on the Start button in Oracle VM VirtualBox to power on the CentOS_Stream_9_VBM_LinuxVMImages.COM virtual machine



1.24 Enter the username centos in the login screen and press Enter to log into the CentOS system



1.25 Click on the Activities button at the top left corner, then open the Terminal by selecting the terminal icon from the dock at the bottom.



1.26 Use the following command to update the system packages: `yum update` in the terminal

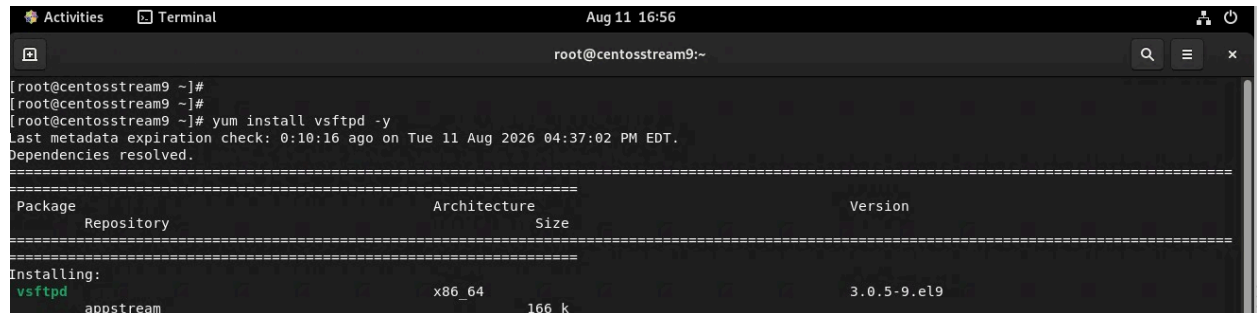
```
root@centosstream9:~  
[root@centosstream9 ~]# yum update  
CentOS Stream 9 - BaseOS                1.8 MB/s | 9.0 MB    00:05  
CentOS Stream 9 - AppStream              8.3 MB/s | 28 MB    00:03  
Last metadata expiration check: 0:00:01 ago on Tue 11 Aug 2026 04:27:55 PM EDT.  
Dependencies resolved.  
=====
```

Package	Arch	Version	Repo	Size
Installing:				
kernel	x86_64	5.14.0-734.el9	baseos	699 k
kernel-devel	x86_64	5.14.0-734.el9	appstream	22 M
mesa-compat-libxatracker	x86_64	25.0.7-3.el9	appstream	1.3 M
replacing mesa-libxatracker	x86_64	21.1.5-2.el9		
samba-ndr-libs	x86_64	4.24.5-1.el9	baseos	1.2 M

1.27 Use the following command to install the OpenSSH server: `yum install openssh-server`

```
root@centosstream9:~  
[root@centosstream9 ~]#  
[root@centosstream9 ~]#  
[root@centosstream9 ~]# yum install openssh-server  
Last metadata expiration check: 0:08:41 ago on Tue 11 Aug 2026 04:37:02 PM EDT.  
Package openssh-server-9.9p1-11.el9.x86_64 is already installed.  
Dependencies resolved.  
Nothing to do.  
Complete!  
[root@centosstream9 ~]#
```

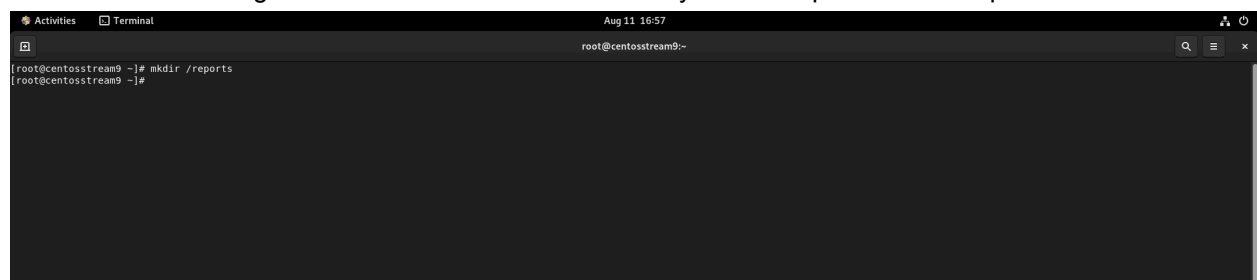
1.28 Use the following command to install the VSFTP server: `yum install vsftpd -y`



```
root@centosstream9:~  
[root@centosstream9 ~]#  
[root@centosstream9 ~]#  
[root@centosstream9 ~]# yum install vsftpd -y  
Last metadata expiration check: 0:10:16 ago on Tue 11 Aug 2026 04:37:02 PM EDT.  
Dependencies resolved.  
=====
```

Package	Repository	Architecture	Size	Version
Installing: vsftpd	appstream	x86_64	166 k	3.0.5-9.el9

1.29 Use the following command to create a new directory named reports: `mkdir /reports`



```
root@centosstream9:~  
[root@centosstream9 ~]# mkdir /reports  
[root@centosstream9 ~]#
```

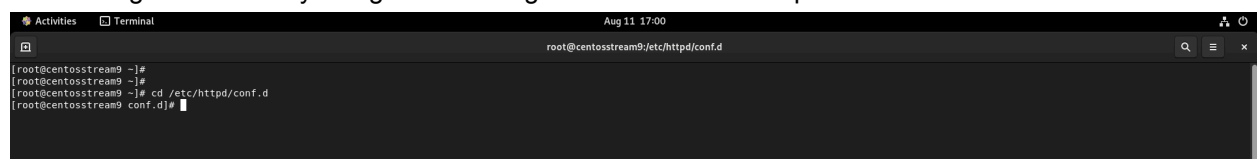
1.30 Use the following command to install the Apache HTTP server: `yum install httpd`



```
root@centosstream9:~  
[root@centosstream9 ~]#  
[root@centosstream9 ~]#  
[root@centosstream9 ~]# yum install httpd  
Last metadata expiration check: 0:21:25 ago on Tue 11 Aug 2026 04:37:02 PM EDT.  
Dependencies resolved.  
=====
```

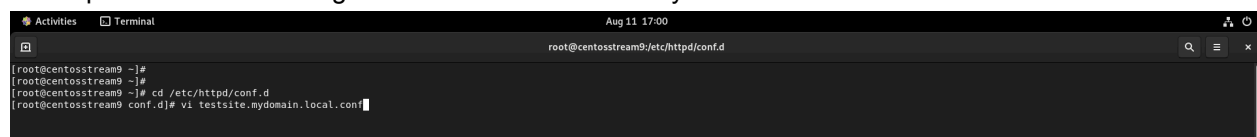
Package	Architecture	Version	Repository	Size
Installing: httpd	x86_64	2.4.62-14.el9	appstream	47 k
Installing dependencies:				
apr	x86_64	1.7.0-12.el9	appstream	123 k
apr-util	x86_64	1.6.1-23.el9	appstream	95 k
apr-util-bdb	x86_64	1.6.1-23.el9	appstream	13 k
centos-logos-httpd	noarch	90.9-1.el9	appstream	1.5 M
httpd-core	x86_64	2.4.62-14.el9	appstream	1.5 M
mod_ssl	x86_64	2.4.62-14.el9	appstream	13 k

1.31 Change the directory using the following command: `cd /etc/httpd/conf.d`



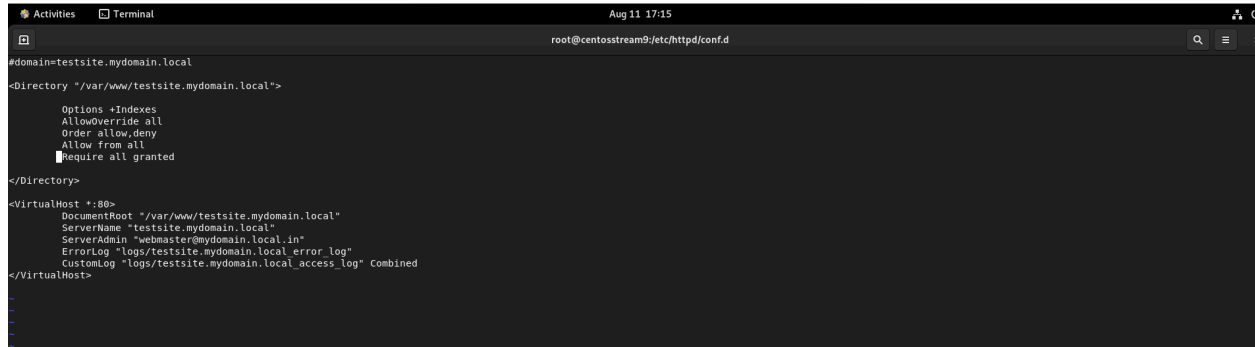
```
root@centosstream9:/etc/httpd/conf.d  
[root@centosstream9 ~]#  
[root@centosstream9 ~]#  
[root@centosstream9 ~]# cd /etc/httpd/conf.d  
[root@centosstream9 conf.d]#
```

1.32 Open the vi editor using the command: `vi testsite.mydomain.local.conf`



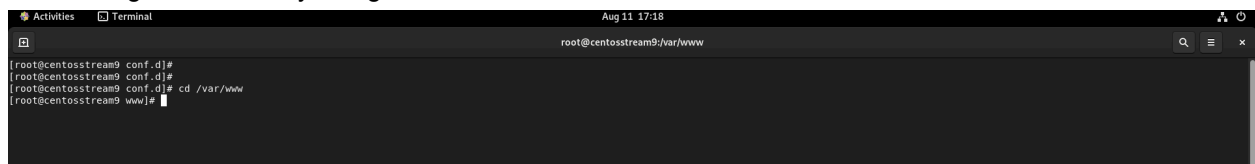
```
root@centosstream9:/etc/httpd/conf.d  
[root@centosstream9 ~]#  
[root@centosstream9 ~]#  
[root@centosstream9 ~]# cd /etc/httpd/conf.d  
[root@centosstream9 conf.d]# vi testsite.mydomain.local.conf
```


1.33 Press i to enter insert mode and paste the following script for configuring the virtual host as shown in the file. then, save and exit the editor



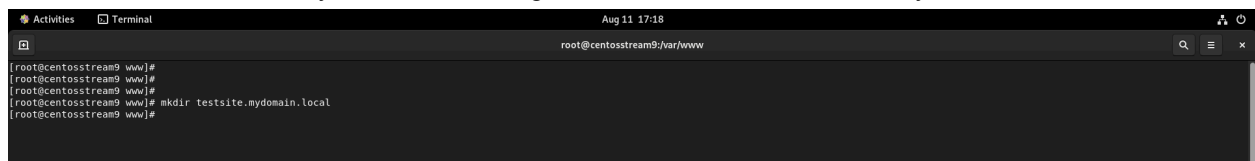
```
root@centosstream9:/etc/httpd/conf.d
#domain=testsite.mydomain.local
<Directory "/var/www/testsite.mydomain.local">
    Options +Indexes
    AllowOverride all
    Order allow,deny
    Allow from all
    Require all granted
</Directory>
<VirtualHost *:80>
    DocumentRoot "/var/www/testsite.mydomain.local"
    ServerName "testsite.mydomain.local"
    ServerAdmin "webmaster@mydomain.local.in"
    ErrorLog "logs/testsite.mydomain.local_error_log"
    CustomLog "logs/testsite.mydomain.local_access_log" Combined
</VirtualHost>
```

1.34 Change the directory using the command: cd /var/www



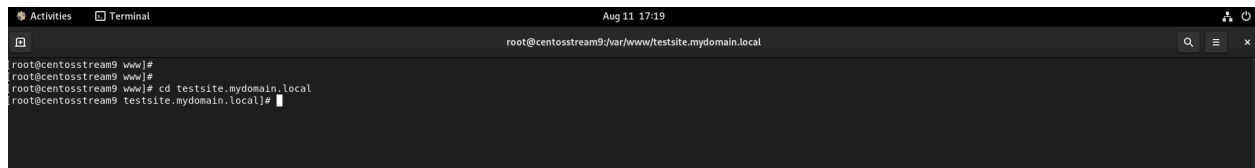
```
root@centosstream9:/var/www
[root@centosstream9 conf.d]#
[root@centosstream9 conf.d]#
[root@centosstream9 conf.d]# cd /var/www
[root@centosstream9 www]#
```

1.35 Create a new directory inside www using the command: mkdir testsite.mydomain.local



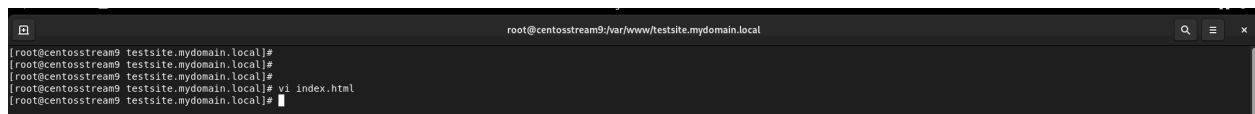
```
root@centosstream9:/var/www
[root@centosstream9 www]#
[root@centosstream9 www]#
[root@centosstream9 www]# mkdir testsite.mydomain.local
[root@centosstream9 www]#
```

1.36 Change the directory using the command: cd testsite.mydomain.local



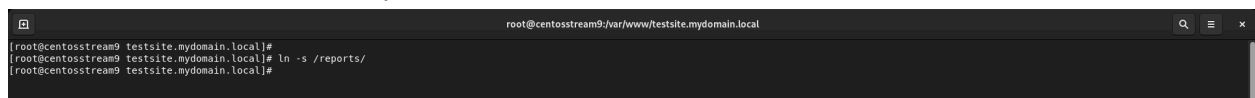
```
root@centosstream9:/var/www/testsite.mydomain.local
[root@centosstream9 www]#
[root@centosstream9 www]#
[root@centosstream9 www]# cd testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
```

1.37 Open the index.html file using the command: vi index.html



```
root@centosstream9:/var/www/testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# vi index.html
[root@centosstream9 testsite.mydomain.local]#
```

1.38 Use the command: ln -s /reports/



```
root@centosstream9:/var/www/testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# ln -s /reports/
[root@centosstream9 testsite.mydomain.local]#
```

1.39 Start the Apache HTTP Server (httpd) using the command: systemctl start httpd

```
root@centosstream9:/var/www/testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start httpd
[root@centosstream9 testsite.mydomain.local]#
```

1.40 Enable the Apache HTTP Server (httpd) service using the command: s

```
root@centosstream9:/var/www/testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# systemctl start httpd
[root@centosstream9 testsite.mydomain.local]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@centosstream9 testsite.mydomain.local]#
```

1.41 Navigate to this <http://testsite.mydomain.local> website using the command: firefox <http://testsite.mydomain.local>

```
root@centosstream9:/var/www/testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local
Parent 119973, Main Thread) WARNING: Failed to create DBus proxy for org.ally.Bus: Failed to execute child process "dbus-launch" (No such file or directory)
: 'glib warning', file /build/ldir/build/8/BUILD/firefox-140.13.0/toolkit/xre/nsSigHandlers.cpp:201
** (firefox:119973): WARNING **: 17:32:01.307: Failed to create DBus proxy for org.ally.Bus: Failed to execute child process "dbus-launch" (No such file or directory)
```

1.42 Now navigate to this website <http://testsite.mydomain.local/reports>



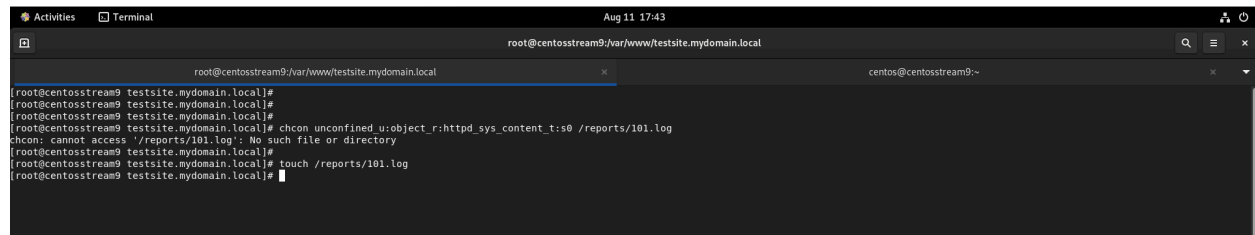
1.43 Use the command: chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/

```
root@centosstream9:/var/www/testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/
[root@centosstream9 testsite.mydomain.local]#
```

1.44 Use the command: chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log

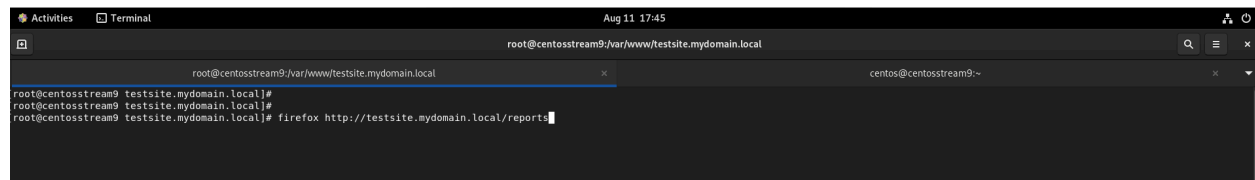
```
root@centosstream9:/var/www/testsite.mydomain.local
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:httpd_sys_content_t:s0 /reports/101.log
chcon: cannot access '/reports/101.log': No such file or directory
[root@centosstream9 testsite.mydomain.local]#
[root@centosstream9 testsite.mydomain.local]#
```

1.45 Now create a file using the command: touch /reports/101.log

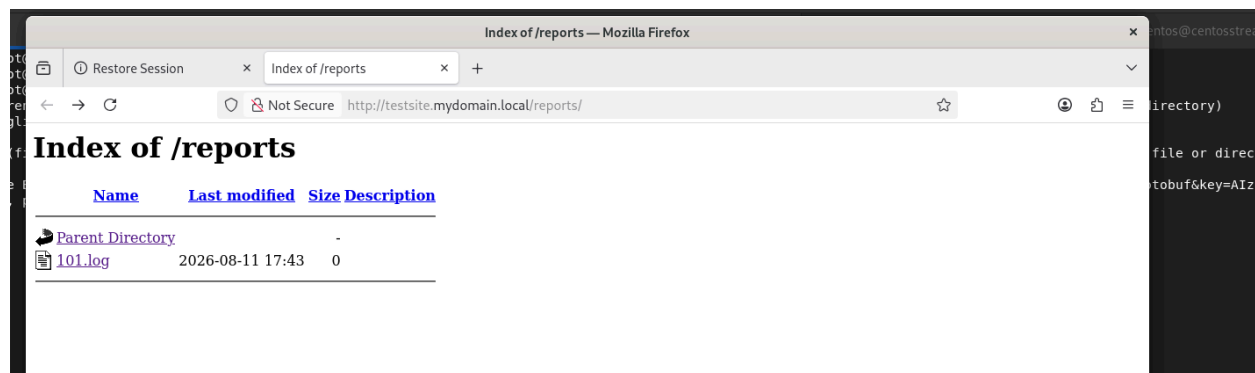


```
root@centosstream9:/var/www/testsite.mydomain.local
root@centosstream9 testsite.mydomain.local]#
root@centosstream9 testsite.mydomain.local]#
root@centosstream9 testsite.mydomain.local]# chcon unconfined_u:object_r:http_sys_content_t:s0 /reports/101.log
chcon: cannot access '/reports/101.log': No such file or directory
root@centosstream9 testsite.mydomain.local]#
root@centosstream9 testsite.mydomain.local]# touch /reports/101.log
root@centosstream9 testsite.mydomain.local]#
```

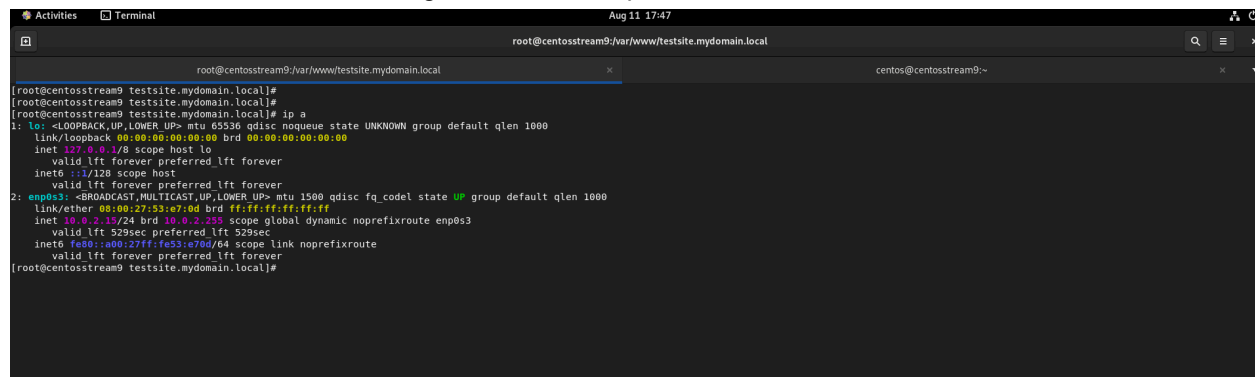
1.46 Open this <http://testsite.mydomain.local/reports> website using the command: firefox <http://testsite.mydomain.local/reports>



```
root@centosstream9 testsite.mydomain.local]#
root@centosstream9 testsite.mydomain.local]#
root@centosstream9 testsite.mydomain.local]# firefox http://testsite.mydomain.local/reports
```



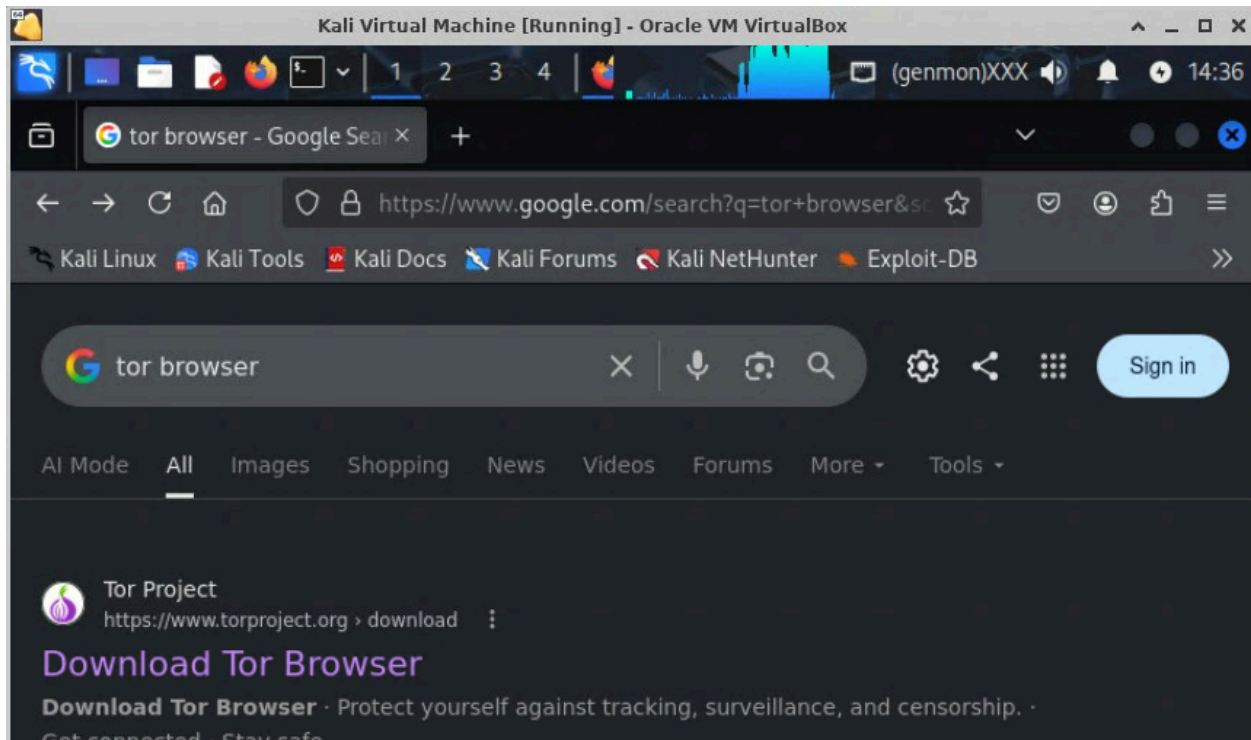
1.47 Check the IP of kali Linux using the command: ip a



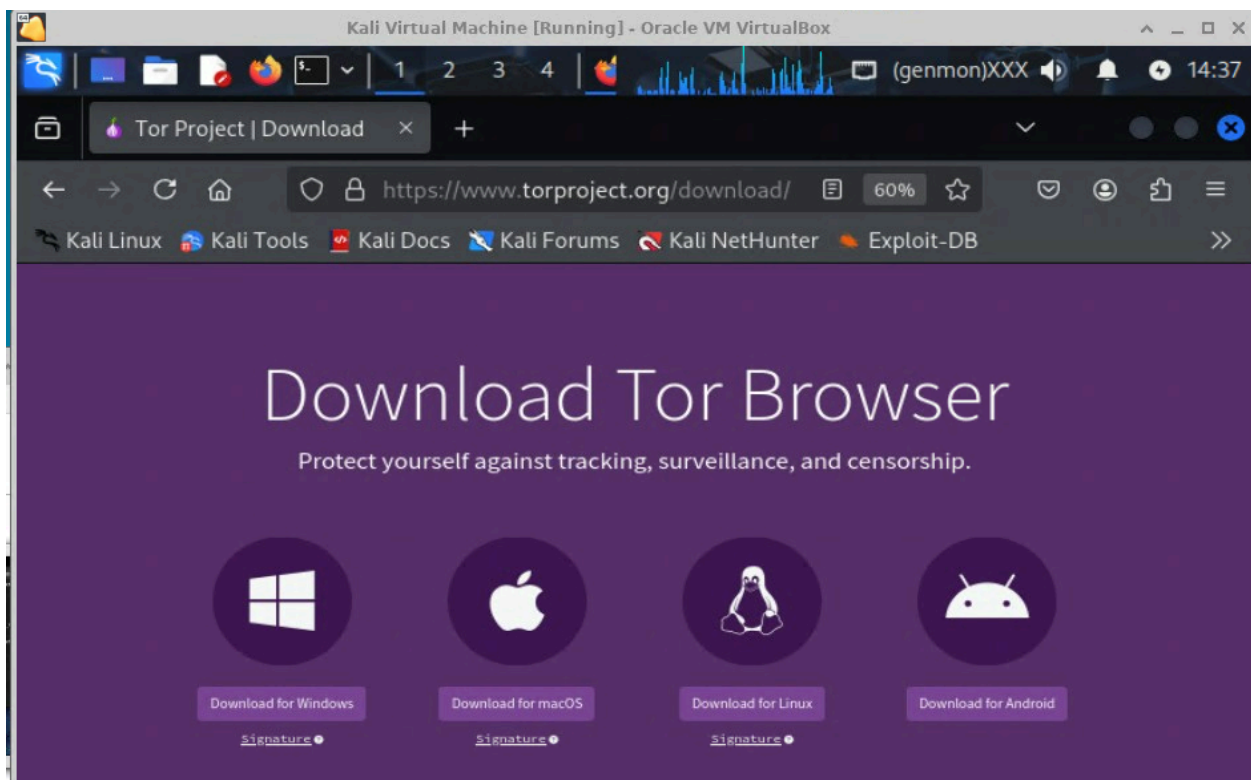
```
root@centosstream9:/var/www/testsite.mydomain.local
root@centosstream9 testsite.mydomain.local]#
root@centosstream9 testsite.mydomain.local]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:e7:0d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 529sec preferred_lft 529sec
    inet6 fe80::a00:27ff:fe53:e70d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@centosstream9 testsite.mydomain.local]#
```

Task 2: Harden SSH Configuration to Secure the Server

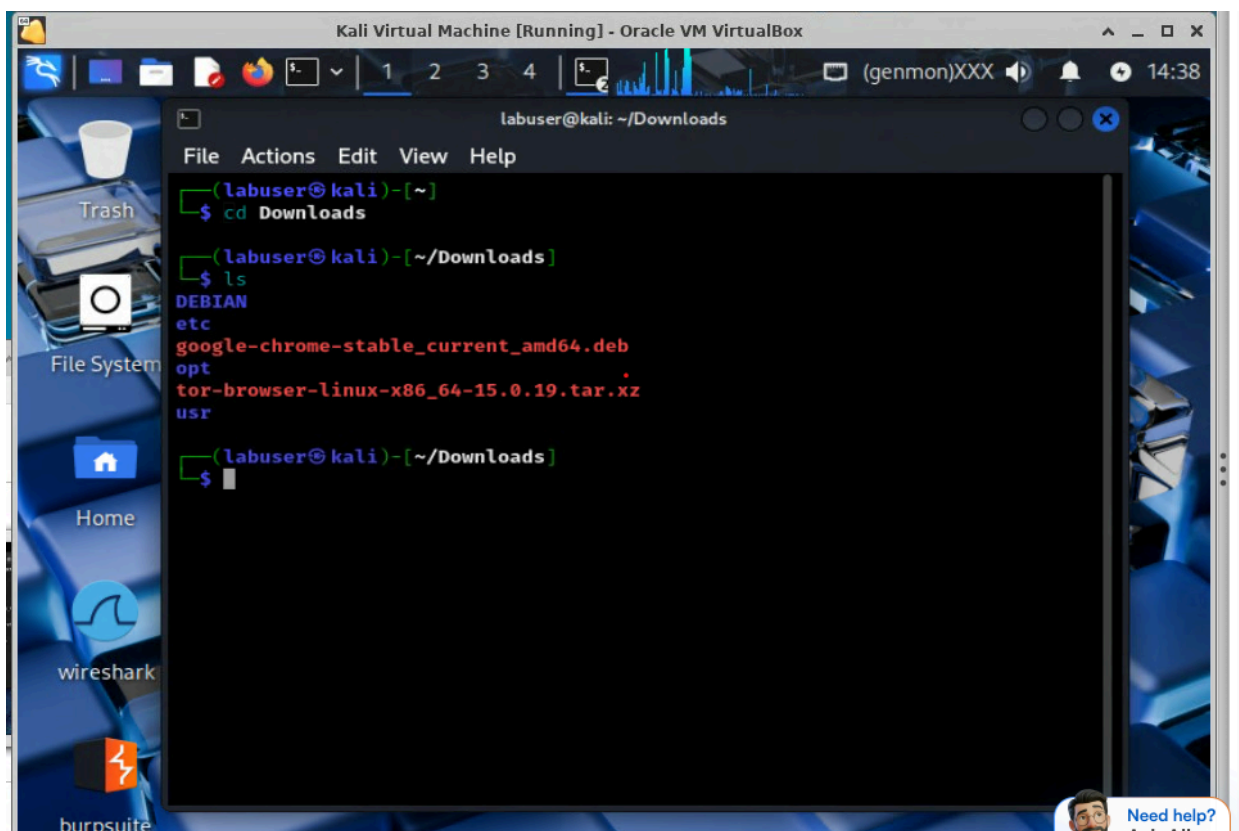
2.1 Open the firefox and search tor browser then click on the Download link



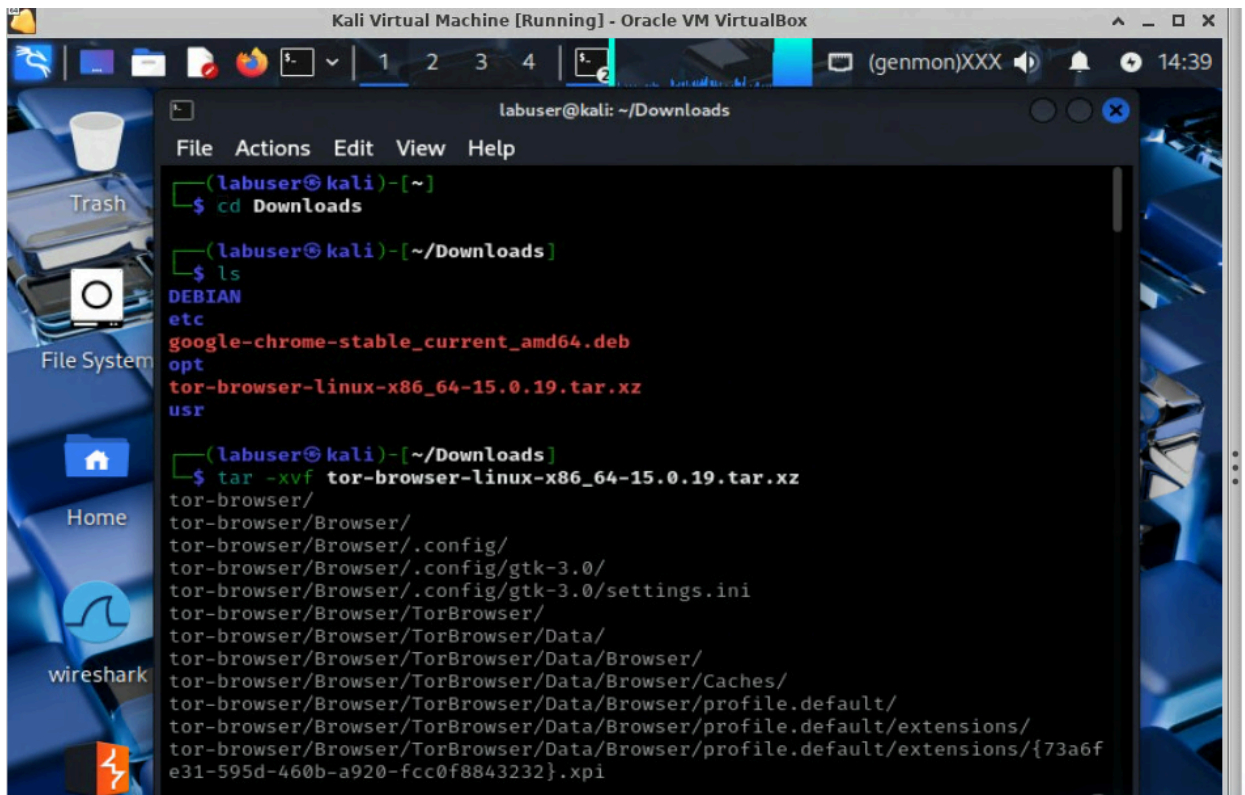
2.2 Click on Download for Linux



2.3 Open the terminal and navigate to Downloads directory using the command: `cd Downloads` And use the command: `ls` to see the file.



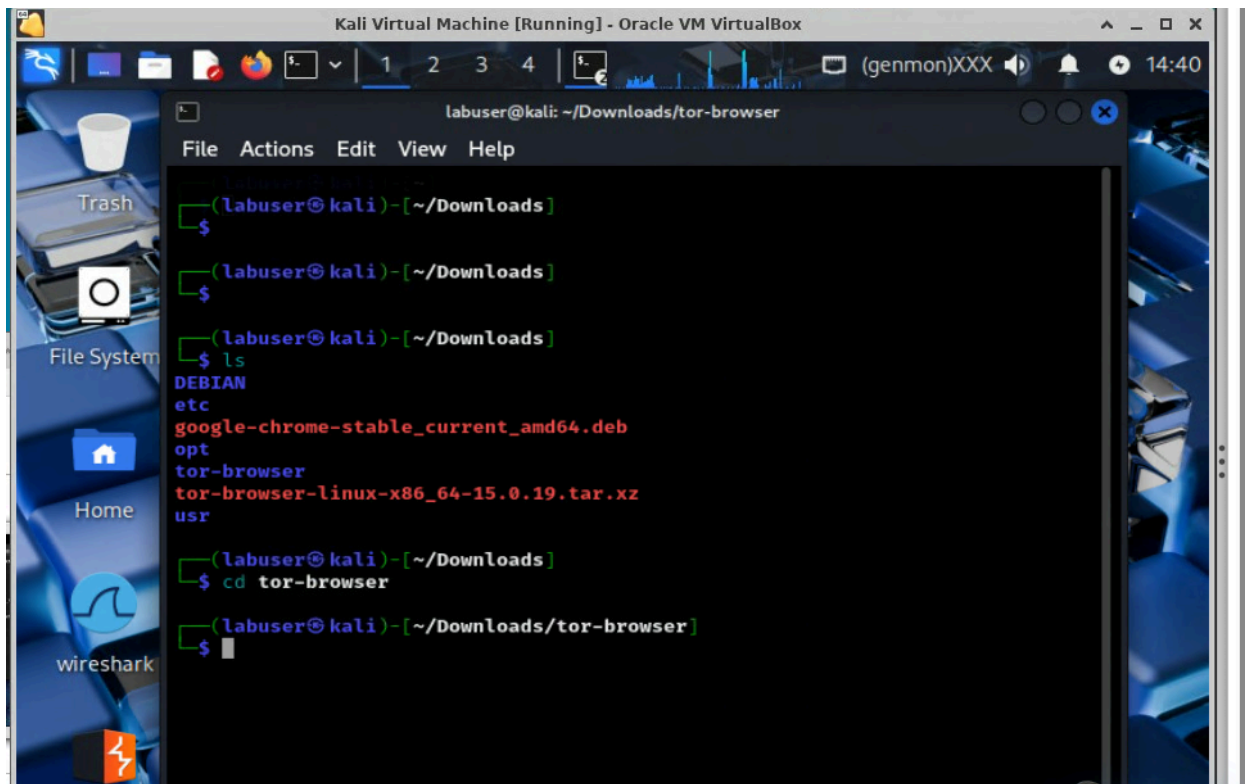
2.4 Now Extract the file using the command: `tar -xvf tor-browser-linux-x86_64-14.0.3.tar.xz`



The screenshot shows a Kali Linux terminal window titled "labuser@kali: ~/Downloads". The terminal displays the following commands and output:

```
(labuser@kali)~  
$ cd Downloads  
(labuser@kali)~/Downloads  
$ ls  
DEBIAN  
etc  
google-chrome-stable_current_amd64.deb  
opt  
tor-browser-linux-x86_64-15.0.19.tar.xz  
usr  
(labuser@kali)~/Downloads  
$ tar -xvf tor-browser-linux-x86_64-15.0.19.tar.xz  
tor-browser/  
tor-browser/Browser/  
tor-browser/Browser/.config/  
tor-browser/Browser/.config/gtk-3.0/  
tor-browser/Browser/.config/gtk-3.0/settings.ini  
tor-browser/Browser/TorBrowser/  
tor-browser/Browser/TorBrowser/Data/  
tor-browser/Browser/TorBrowser/Data/Browser/  
tor-browser/Browser/TorBrowser/Data/Browser/Caches/  
tor-browser/Browser/TorBrowser/Data/Browser/profile.default/  
tor-browser/Browser/TorBrowser/Data/Browser/profile.default/extensions/  
tor-browser/Browser/TorBrowser/Data/Browser/profile.default/extensions/{73a6fe31-595d-460b-a920-fcc0f8843232}.xpi
```

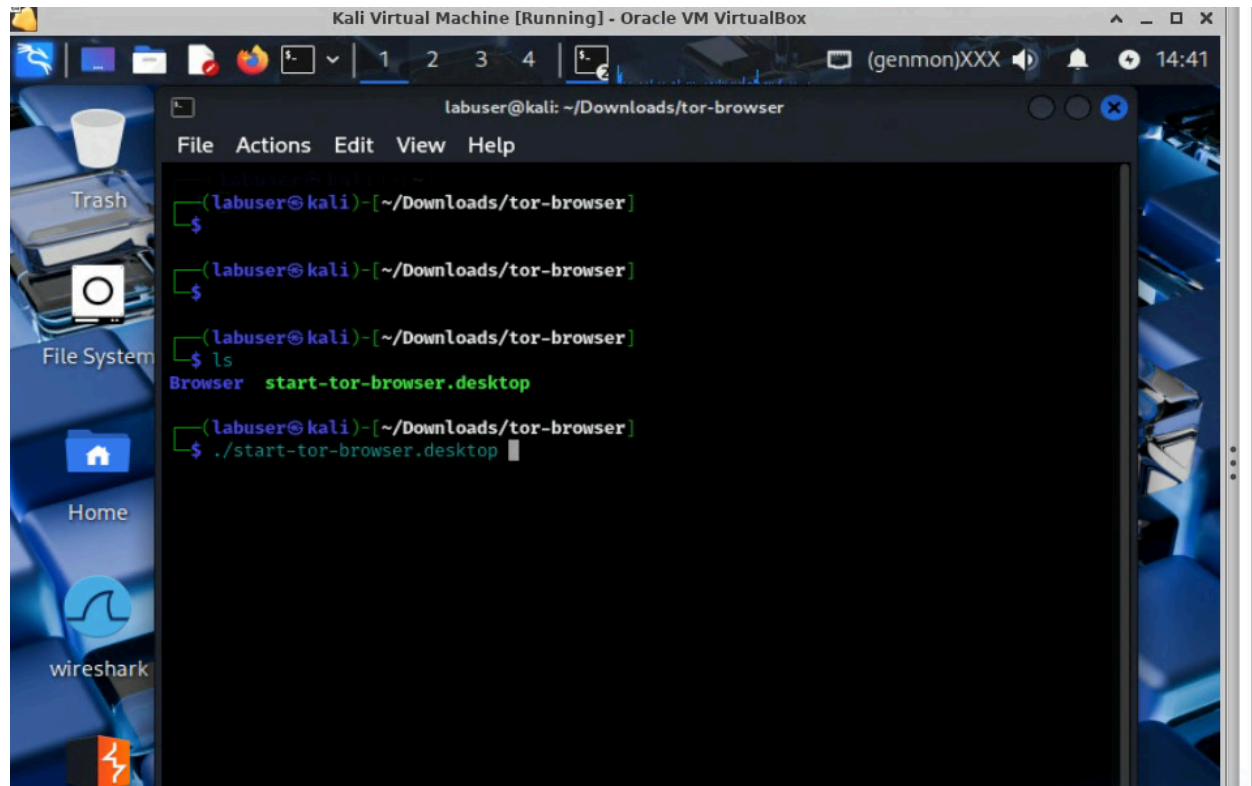
2.5 Use the command: `ls` to see the file



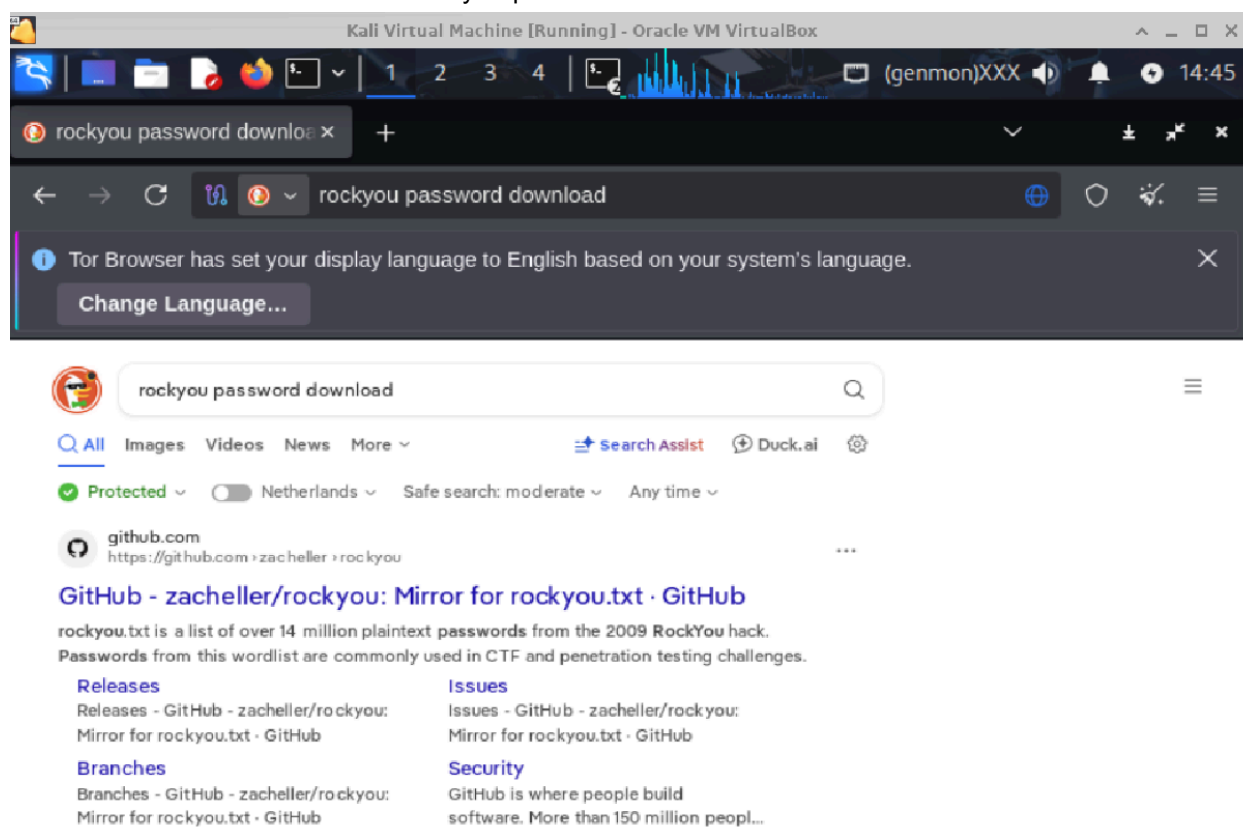
The screenshot shows a Kali Linux terminal window titled "labuser@kali: ~/Downloads/tor-browser". The terminal displays the following commands and output:

```
(labuser@kali)~/Downloads  
$  
(labuser@kali)~/Downloads  
$  
(labuser@kali)~/Downloads  
$ ls  
DEBIAN  
etc  
google-chrome-stable_current_amd64.deb  
opt  
tor-browser  
tor-browser-linux-x86_64-15.0.19.tar.xz  
usr  
(labuser@kali)~/Downloads  
$ cd tor-browser  
(labuser@kali)~/Downloads/tor-browser  
$
```

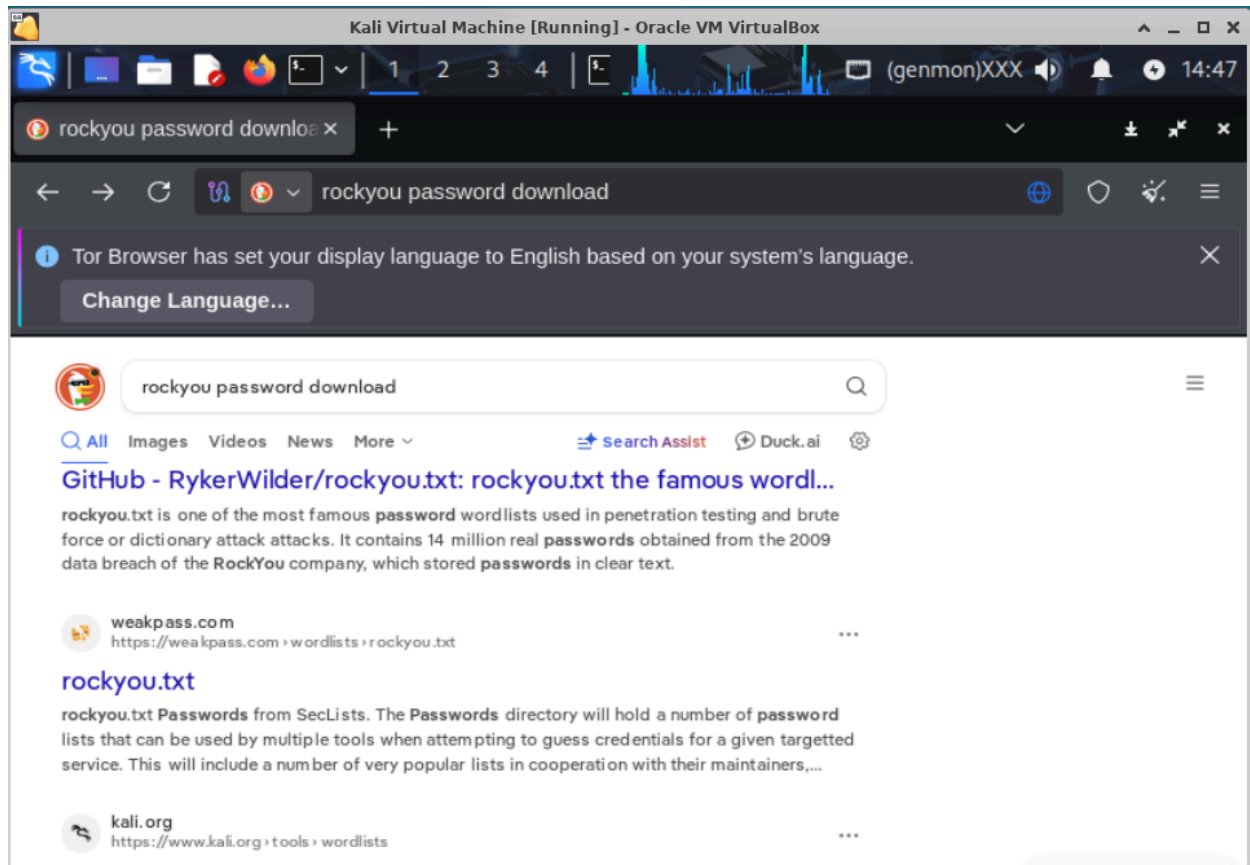
2.6 Change the directory using the command: `./start-tor-browser.desktop`



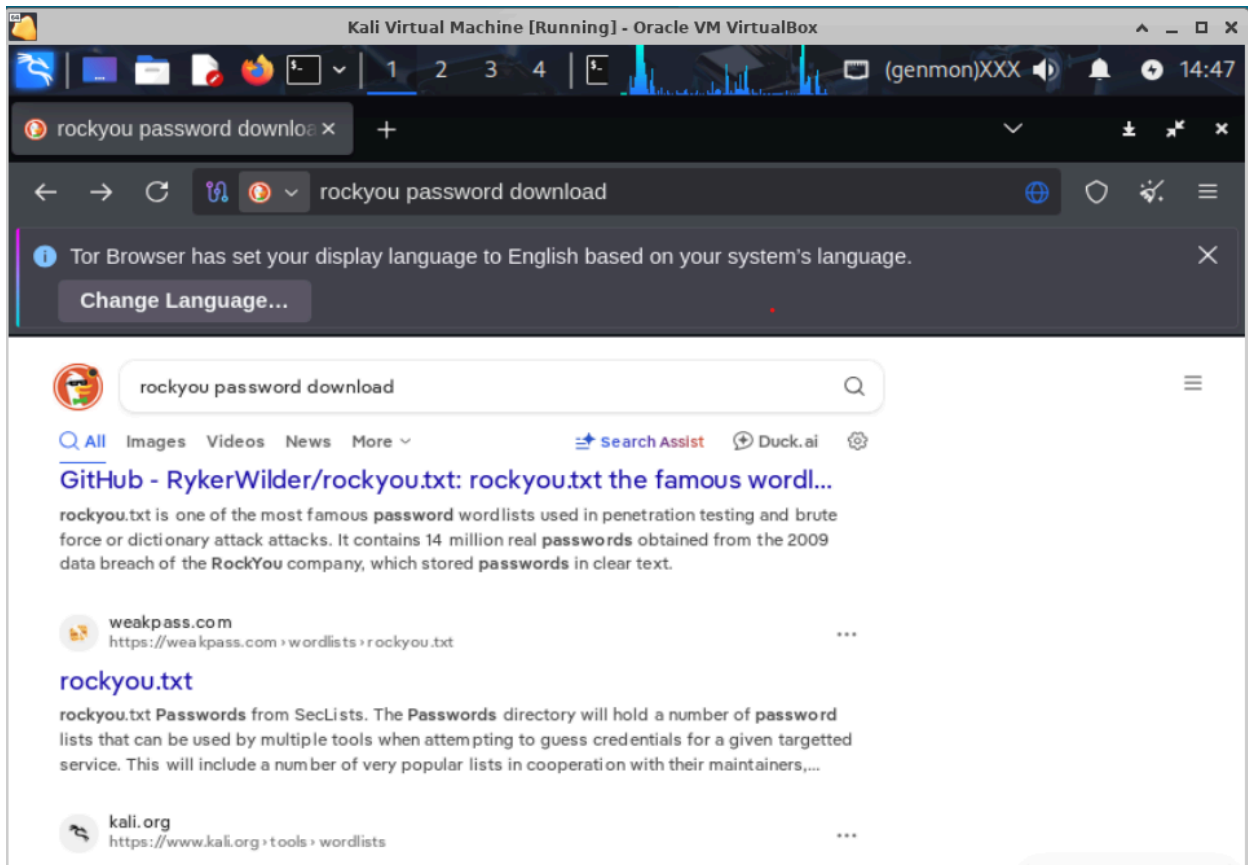
2.7 Now search in the tor browser rockyou password download



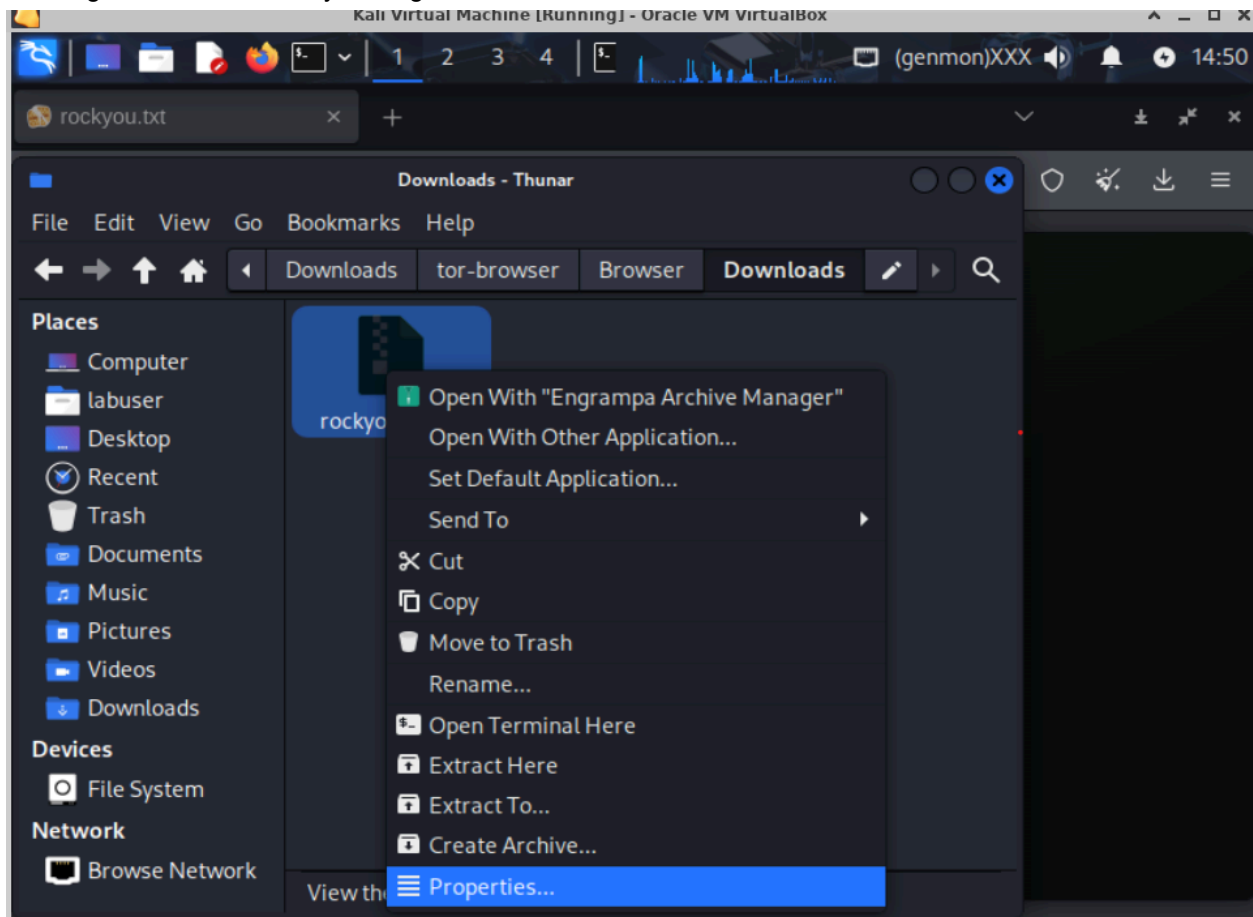
2.8 Click on rockyou.txt – Weakpass



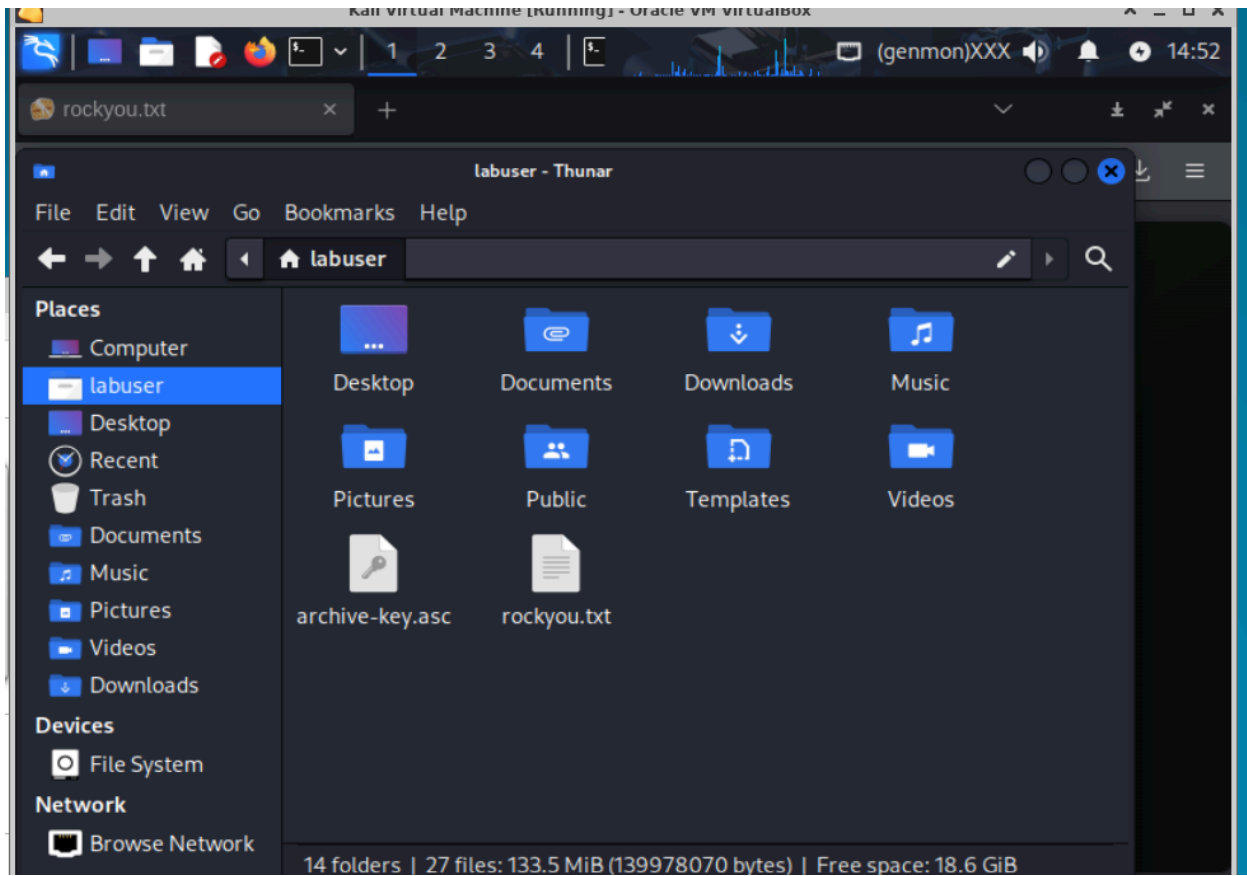
2.9 Click on rockyou.txt to download the file



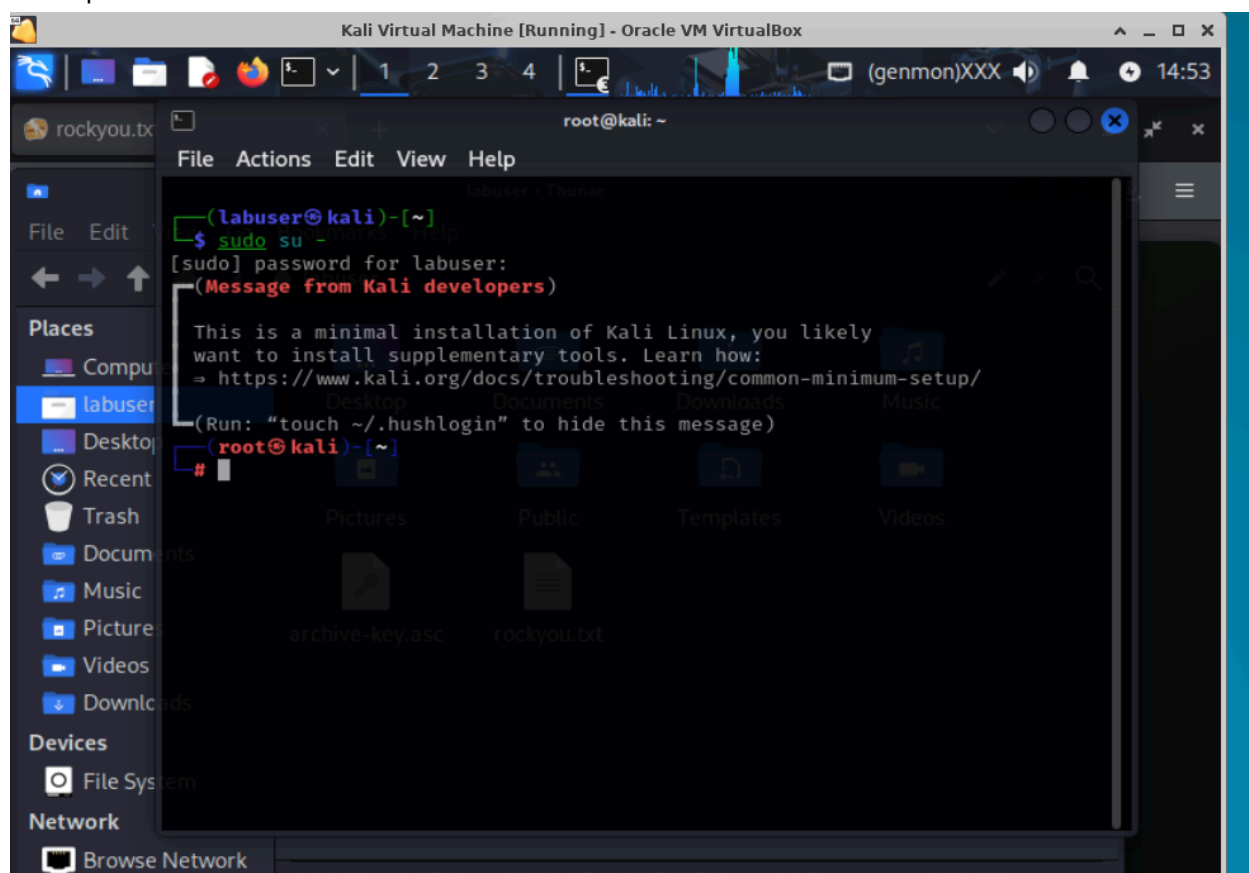
2.10 Right-click on the rockyou.txt.gz file and click on Extract here



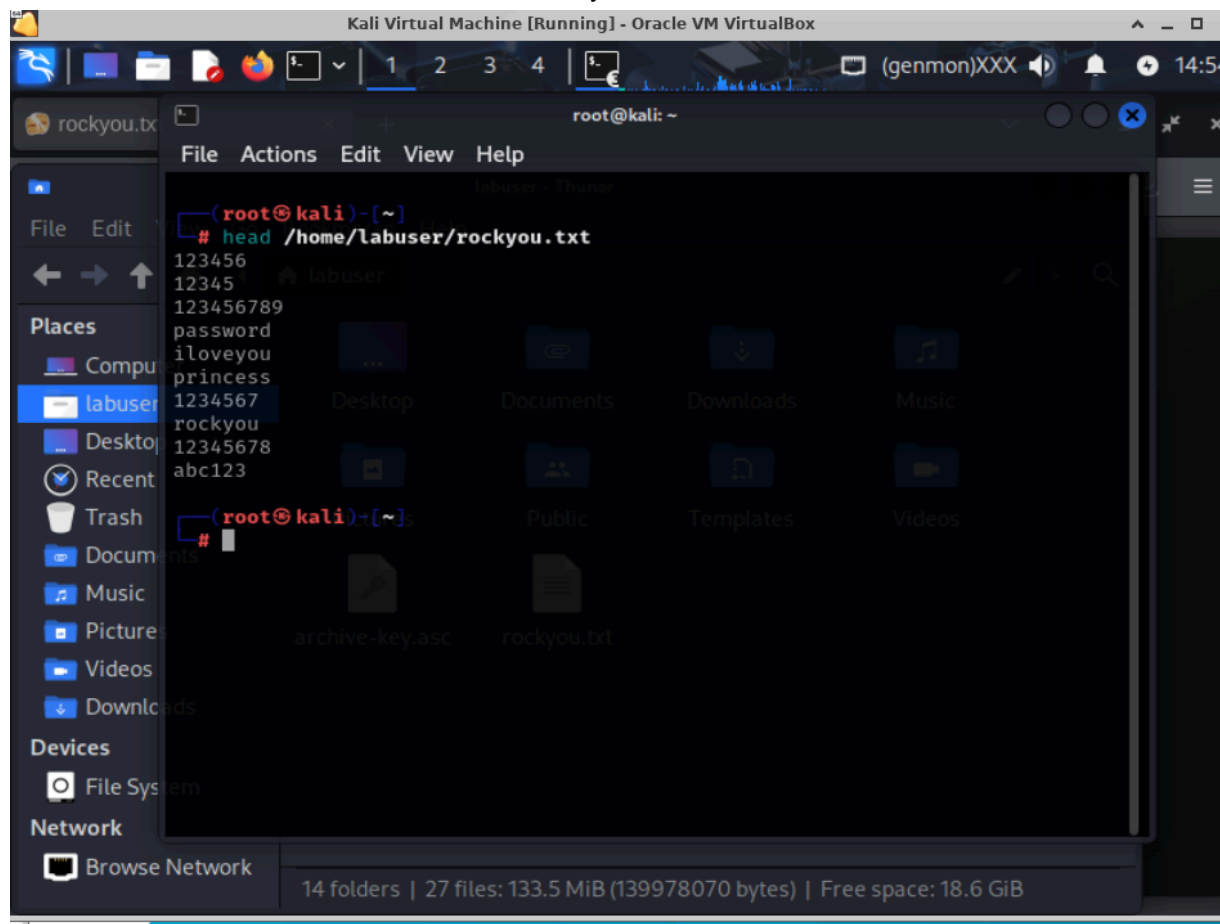
2.11 Navigate to labuser directory and paste the file here



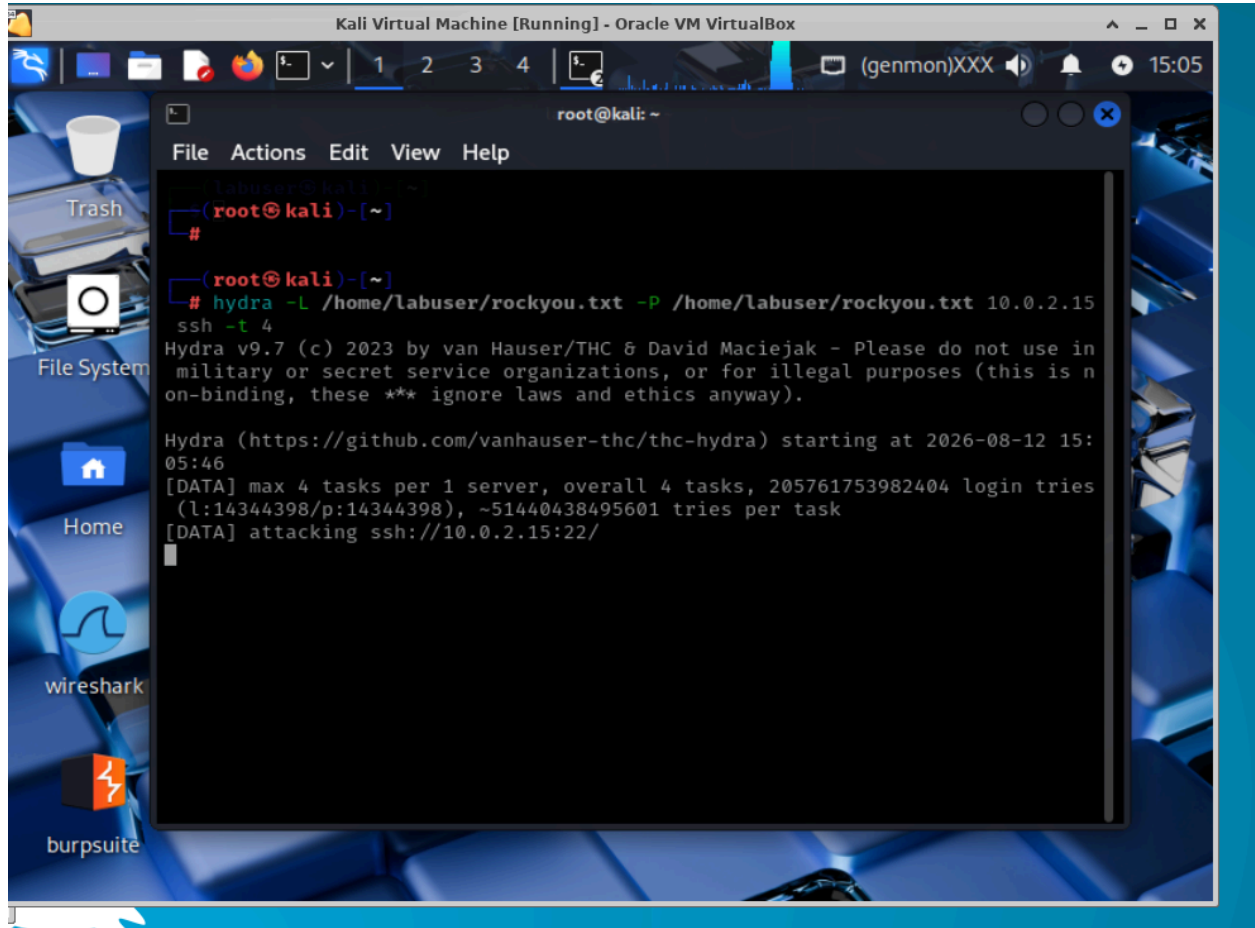
2.12 Open the terminal and use the command: `sudo su -`



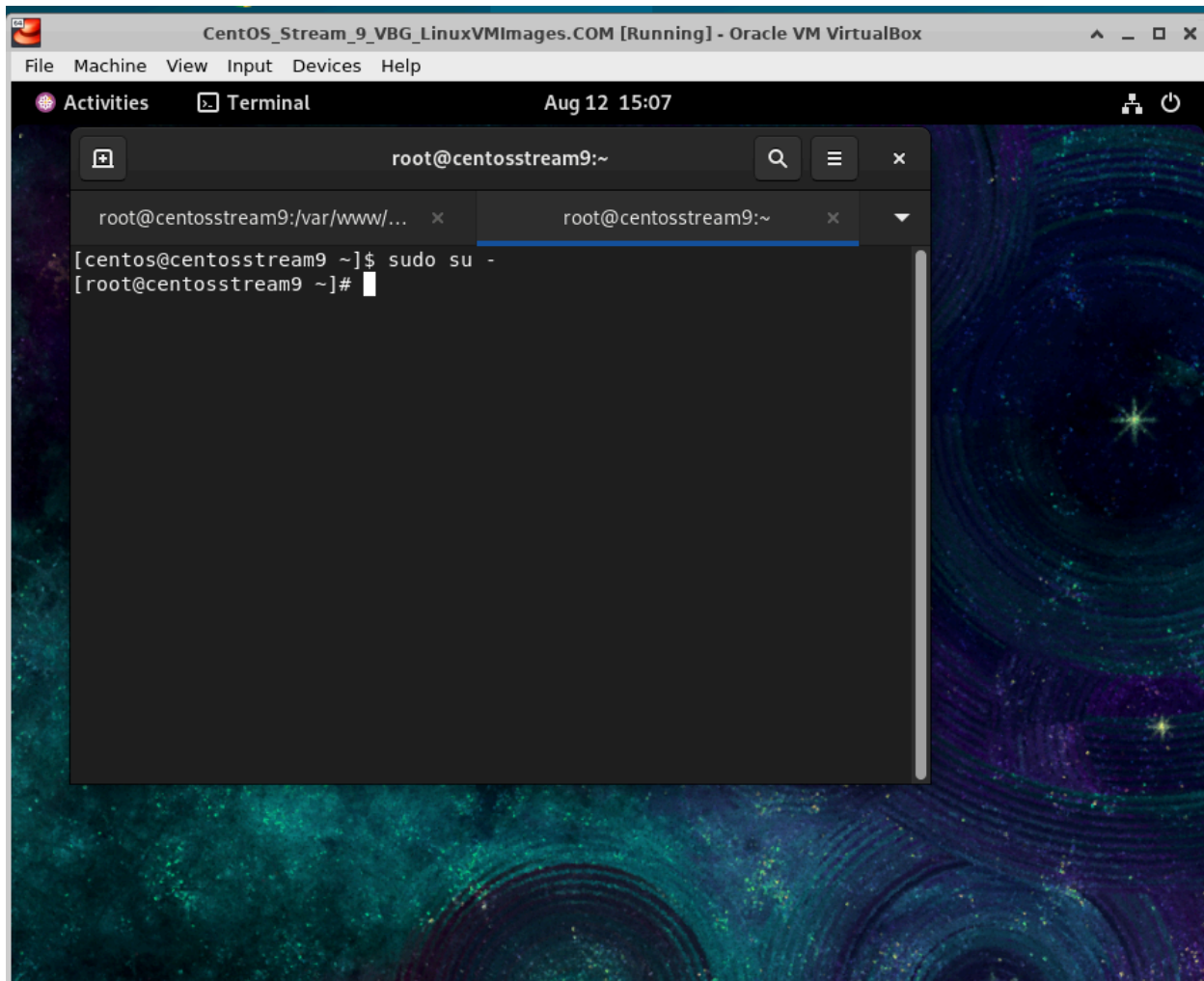
2.13 Use the command: head /home/labuser/rockyou.txt



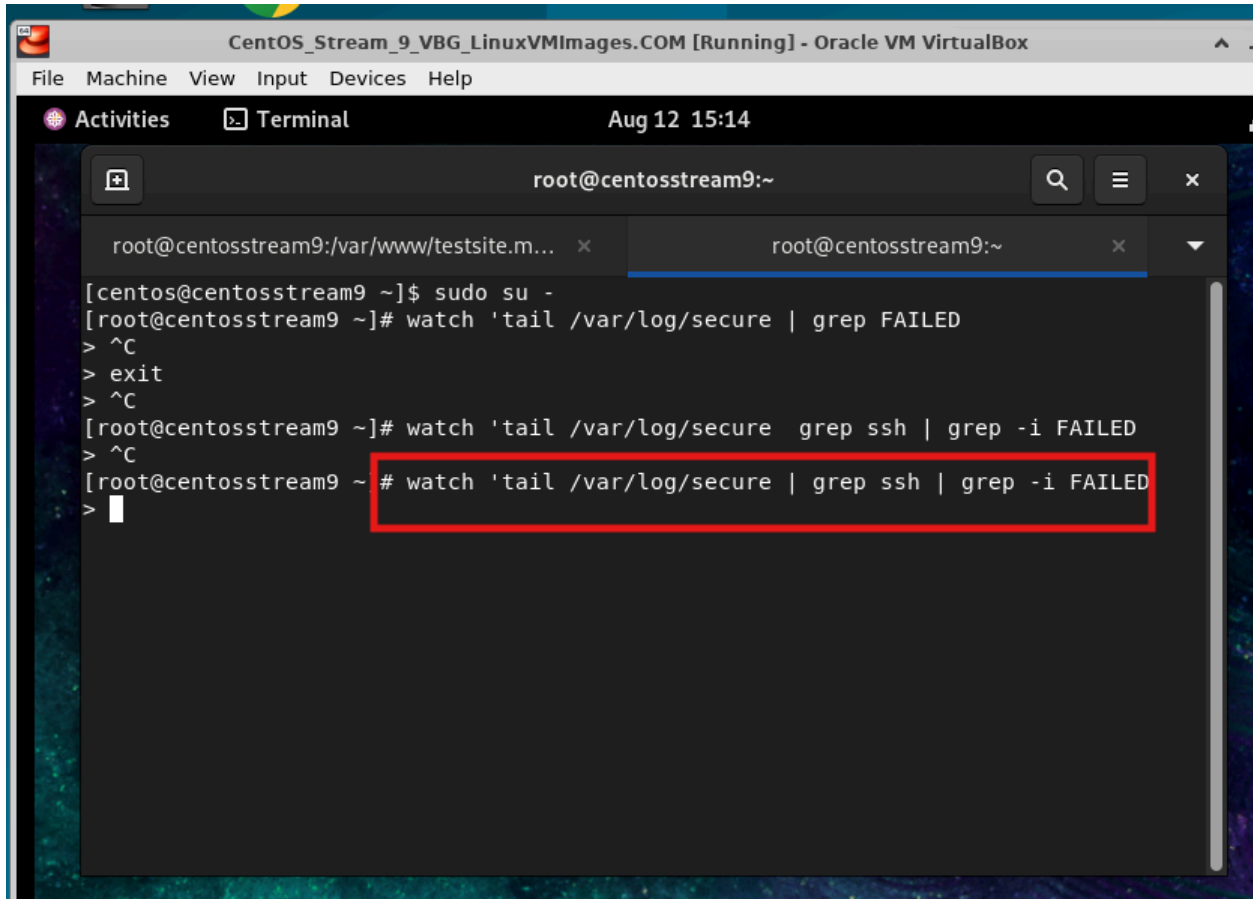
2.14 Use the command: `hydra -L /home/labuser/rockyou.txt -P /home/labuser/rockyou.txt 10.0.2.15 ssh -t 4`



2.15 Navigate to CentOS and open the terminal then use the command: `sudo su -`



2.16 Use the command: `watch 'tail /var/log/secure | grep FAILED'`

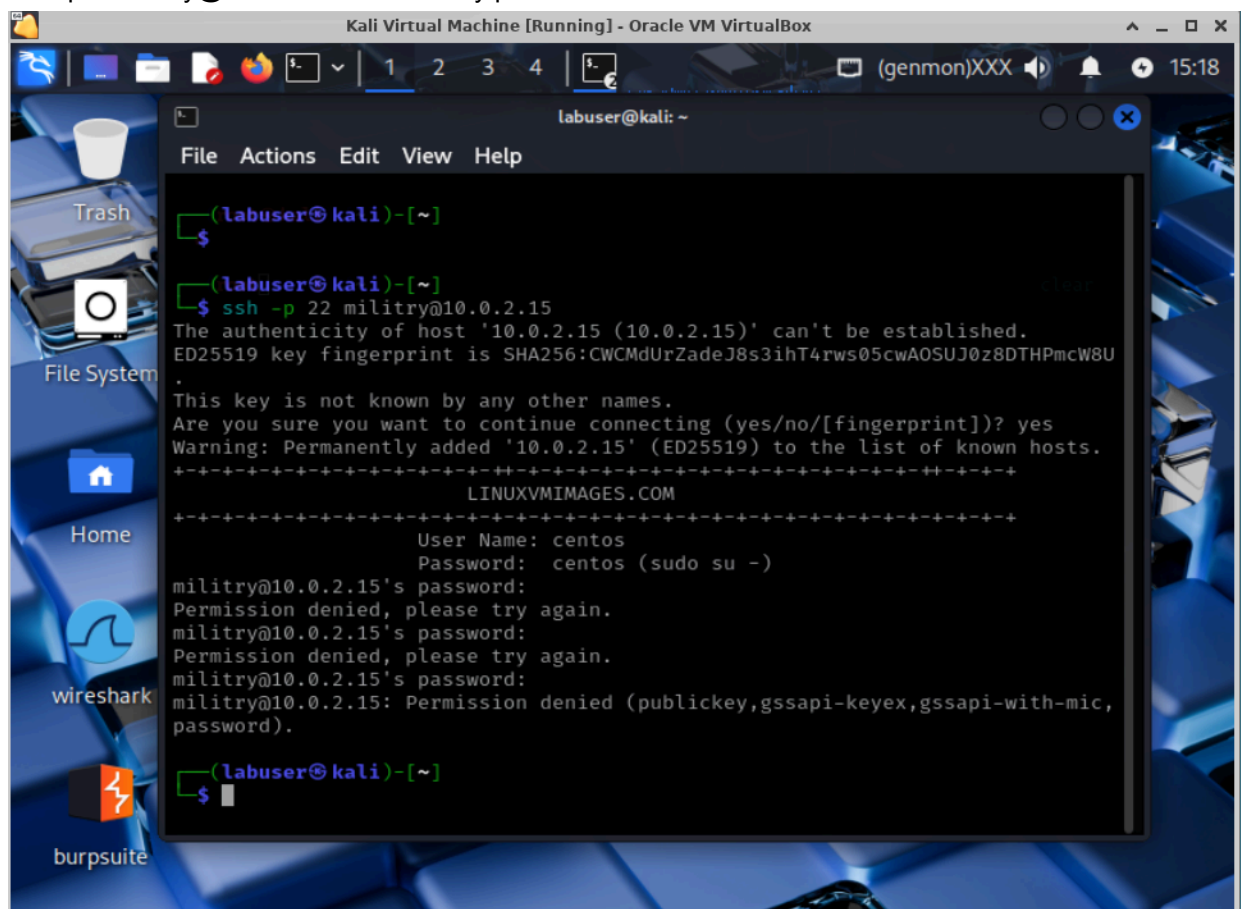


The screenshot shows a terminal window titled "CentOS_Stream_9_VBG_LinuxVMImages.COM [Running] - Oracle VM VirtualBox". The terminal is running as root on a CentOS Stream 9 system. The user has executed the following commands:

```
[centos@centosstream9 ~]$ sudo su -  
[root@centosstream9 ~]# watch 'tail /var/log/secure | grep FAILED'  
> ^C  
> exit  
> ^C  
[root@centosstream9 ~]# watch 'tail /var/log/secure  grep ssh | grep -i FAILED'  
> ^C  
[root@centosstream9 ~]# watch 'tail /var/log/secure | grep ssh | grep -i FAILED'  
> 
```

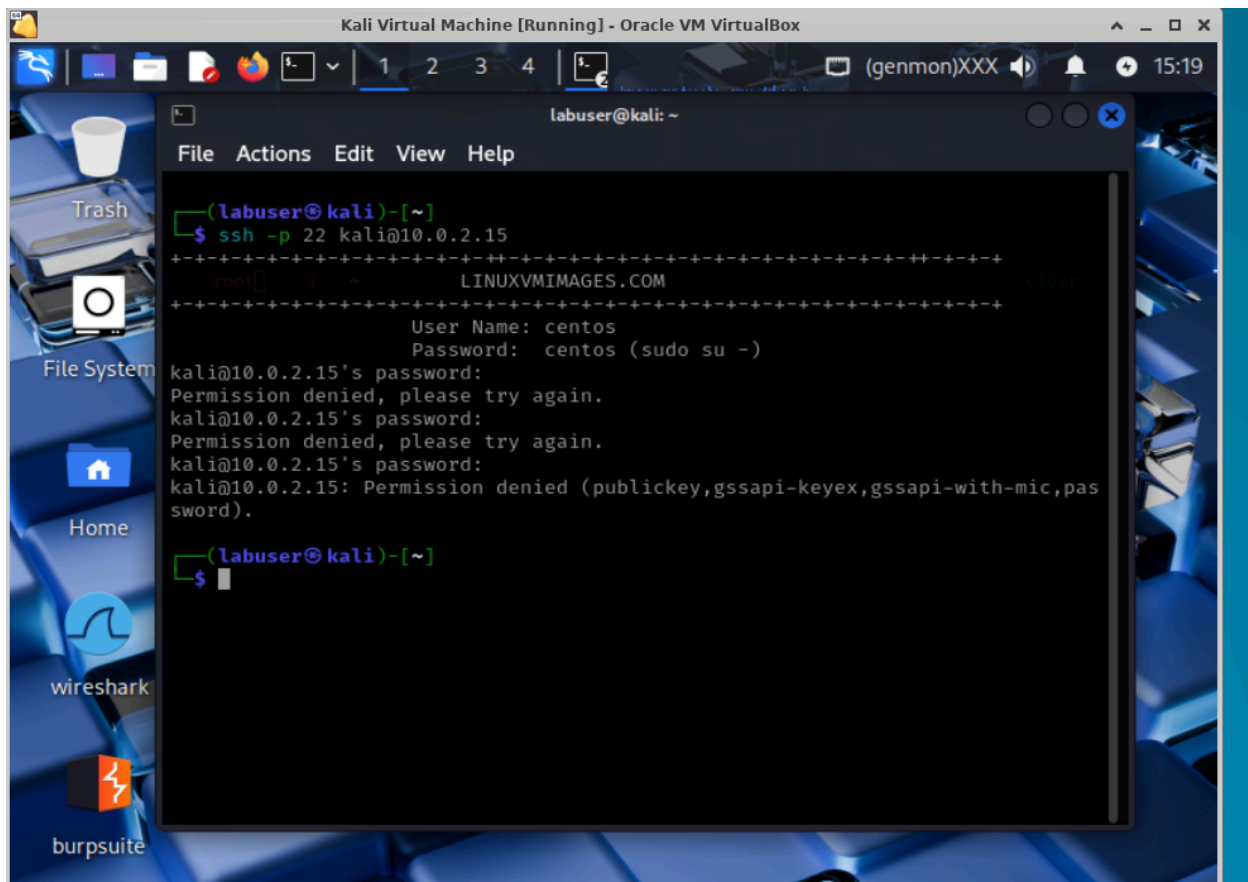
The last command, `watch 'tail /var/log/secure | grep ssh | grep -i FAILED'`, is highlighted with a red rectangle. The terminal also shows a tab for `root@centosstream9:/var/www/testsite.m...`.

2.17 Now use the wrong username and password to capture the failed login attempt Use the command: `ssh -p 22 military@10.0.2.15` and use any password.

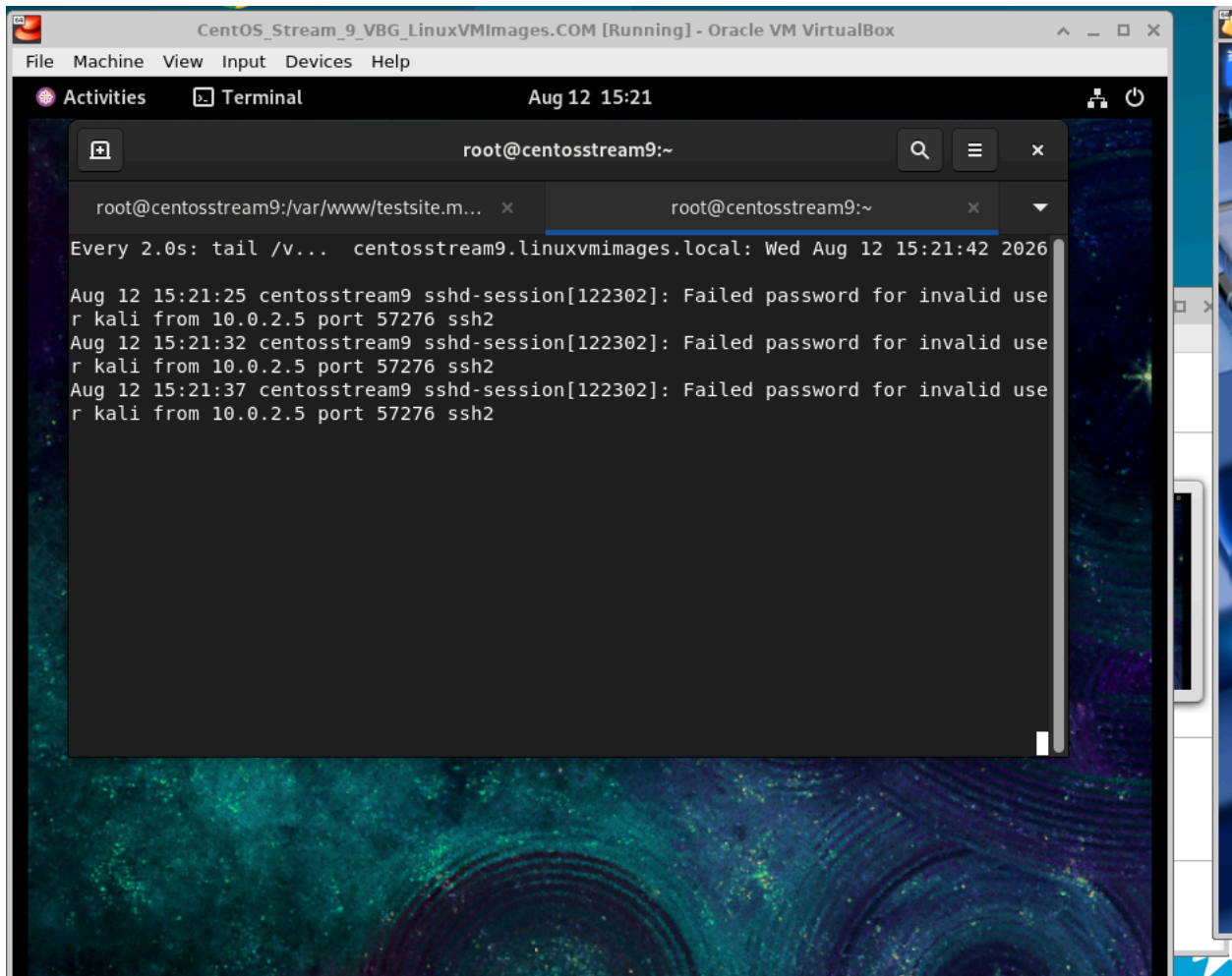


```
labuser@kali: ~  
File Actions Edit View Help  
(labuser@kali)-[~]  
$  
(labuser@kali)-[~]  
$ ssh -p 22 military@10.0.2.15  
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.  
ED25519 key fingerprint is SHA256:CWCMdUrZadeJ8s3ihT4rws05cwAOSUJ0z8DTHPmcW8U  
. This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.2.15' (ED25519) to the list of known hosts.  
+++++  
LINUXVMIMAGES.COM  
+++++  
User Name: centos  
Password: centos (sudo su -)  
military@10.0.2.15's password:  
Permission denied, please try again.  
military@10.0.2.15's password:  
Permission denied, please try again.  
military@10.0.2.15's password:  
military@10.0.2.15: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,  
password).  
(labuser@kali)-[~]  
$
```

2.18 Use the command: `ssh -p 22 kali@10.0.2.15` and use any password



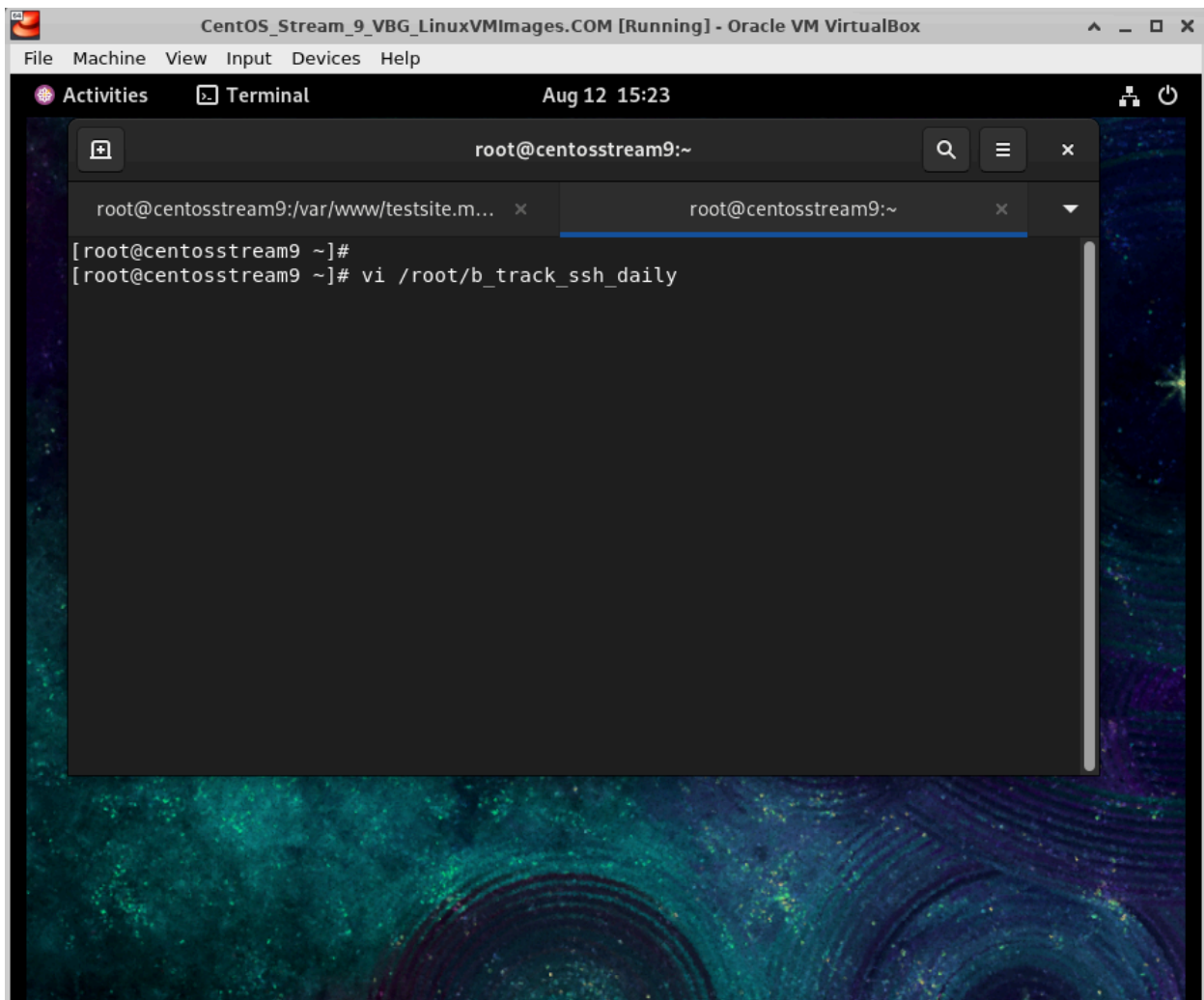
2.19 Now navigate to CentOS and see the Failed login attempt.



The screenshot shows a terminal window titled "CentOS_Stream_9_VBG_LinuxVMImages.COM [Running] - Oracle VM VirtualBox". The terminal is running as root on a CentOS Stream 9 machine. The prompt is "root@centosstream9:~". The terminal output shows three failed login attempts for the user "kali" from IP "10.0.2.5" on port "57276" using "ssh2". The messages are:

```
Every 2.0s: tail /v... centosstream9.linuxvmimages.local: Wed Aug 12 15:21:42 2026
Aug 12 15:21:25 centosstream9 sshd-session[122302]: Failed password for invalid use
r kali from 10.0.2.5 port 57276 ssh2
Aug 12 15:21:32 centosstream9 sshd-session[122302]: Failed password for invalid use
r kali from 10.0.2.5 port 57276 ssh2
Aug 12 15:21:37 centosstream9 sshd-session[122302]: Failed password for invalid use
r kali from 10.0.2.5 port 57276 ssh2
```

2.20 Edit the `b_track_ssh_daily` file using the command: `vi /root/b_track_ssh_daily`

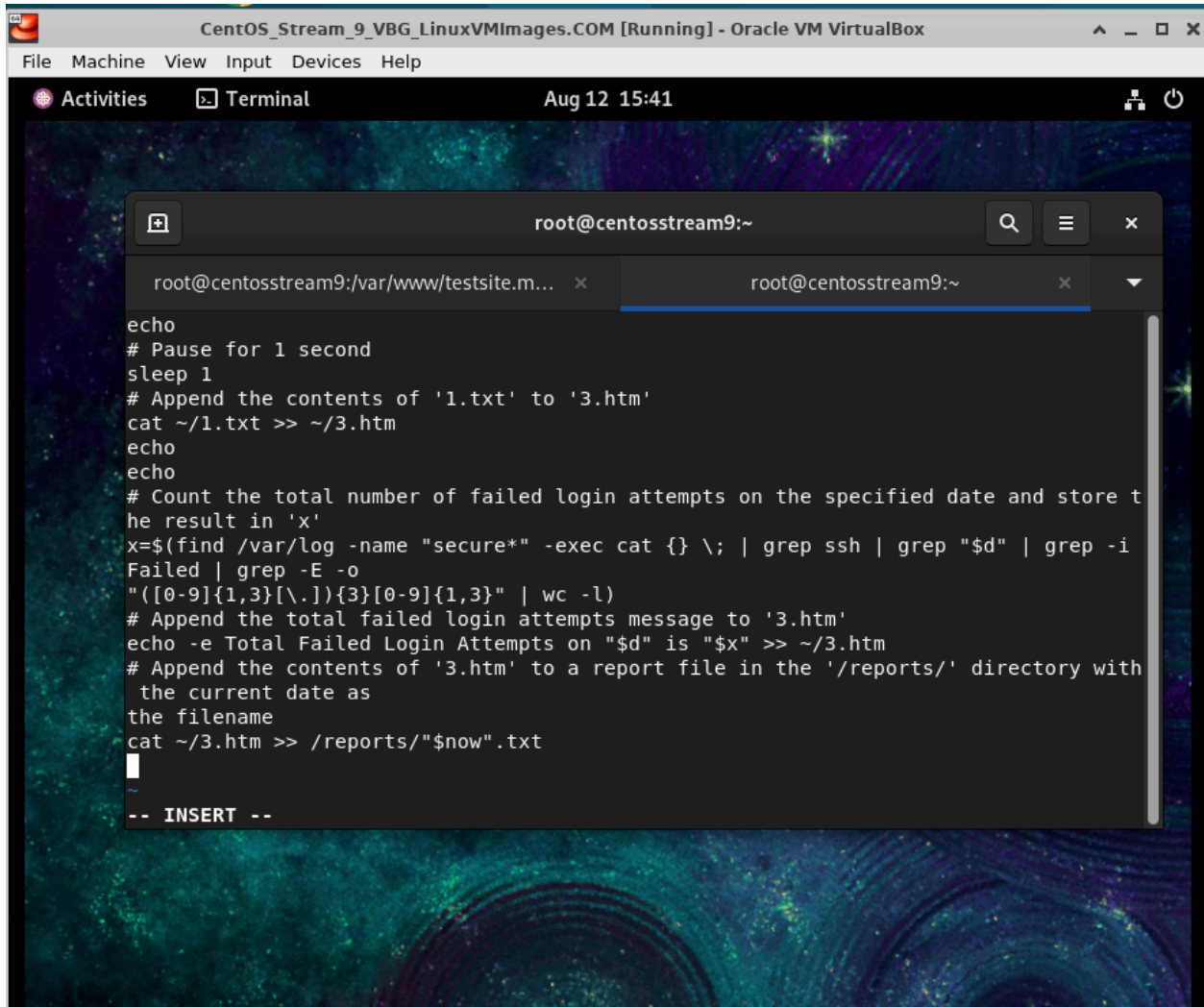


2.21 Use this script:

```
#!/bin/bash
# The current date in the format YYYY-MM-DD is stored in the variable 'now'
now=$(date +"%F")
# Extract the month and day from the current date using the 'date' command and 'awk'
m=$(date | awk '{print $2}')
dta=$(date | awk '{print $3}')
# Initialize variables 'x' and 'y' to 1 and 10 respectively
x=1
y=10
# Check if the current day is less than or equal to 10
if [ $dta -lt $y ] || [ $dta -eq $y ]
then
    # If true, set the 'd' variable to the previous day's date formatted as 'Month Day'
    d=$(date -d "1 day ago" '+%b %_d')
else
    # If false, set the 'd' variable to the previous day's date formatted as 'Month Day'
    d=$(date -d "1 day ago" '+%b %_d')
fi
# Print the date variable 'd'
echo "$d"
echo .
# Pause for 30 seconds
sleep 30
# Remove any previous instance of 'ssh_failed_sortlisted_logins.txt' file
rm ~/ssh_failed_sortlisted_logins.txt
# Search through all 'secure' logs for SSH failed login attempts on the previous day
# and save unique IP addresses to the 'ssh_failed_sortlisted_logins.txt' file
find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep "$d" | grep -i Failed | grep -E -o "([0-9]{1,3}[.]){3}[0-9]{1,3}" | sort -u > ~/ssh_failed_sortlisted_logins.txt
# Remove temporary files '1.txt', '2.txt', and '3.htm'
rm ~/1.txt
rm ~/2.txt
rm ~/3.htm
# Loop through each IP address in the 'ssh_failed_sortlisted_logins.txt' file
cat ~/ssh_failed_sortlisted_logins.txt | \
while read ip
do
```

```
# Perform a whois lookup to get the country information for each IP address
whois $ip | grep -i country
# If the country information is found, add the IP and country info to '1.txt'
if [ $? -eq 0 ]
then
    clear
    echo $ip >> ~/1.txt
    whois $ip | grep -i country | tr "a-z" "A-Z" | sort -u >> ~/1.txt
else
    # If no country info is found, log the IP with "COUNTRY: UNKNOWN"
    echo "$ip" >> ~/1.txt
    echo "COUNTRY:   UNKNOWN" >> ~/1.txt
fi
done
```

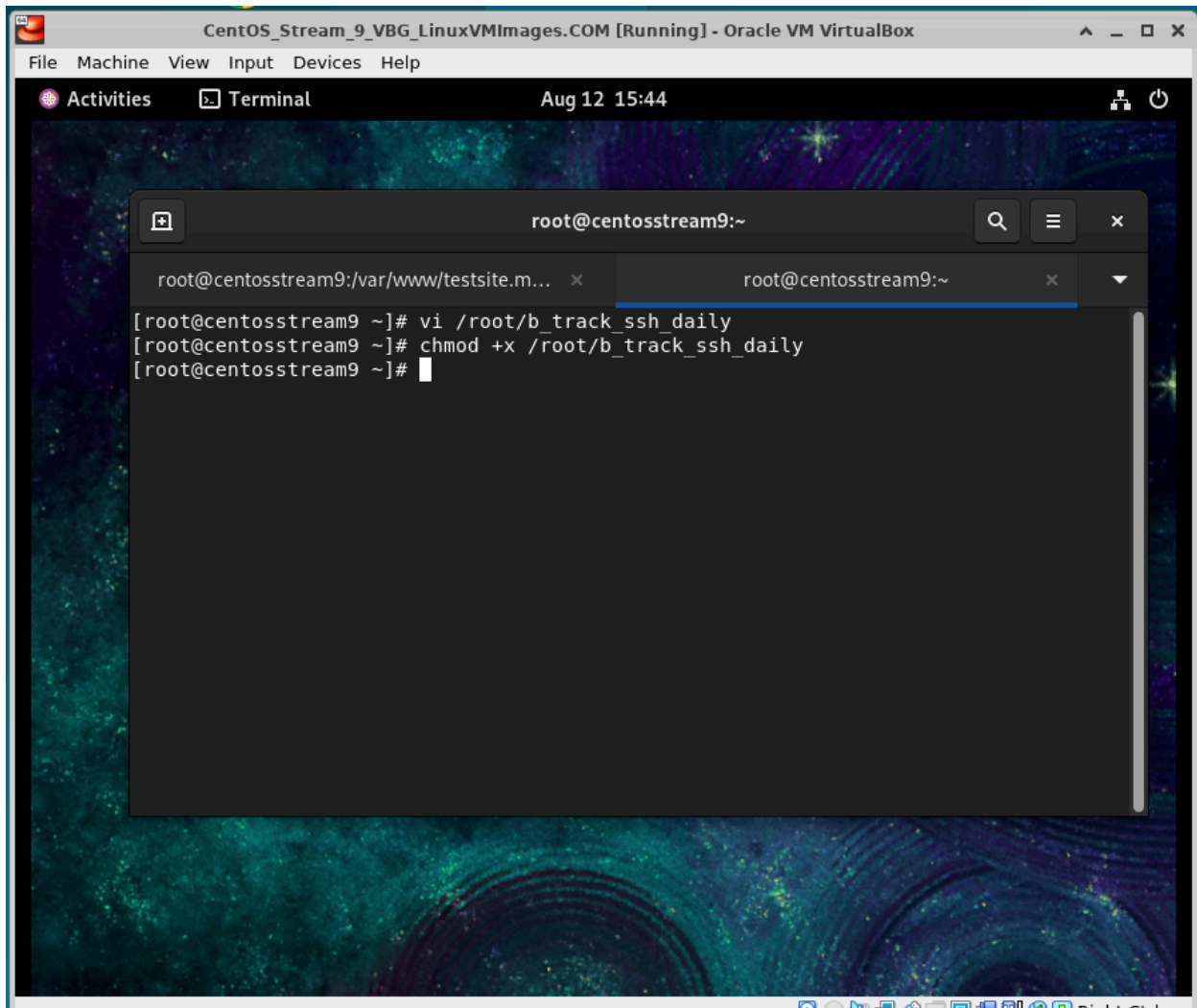
```
# Print a message to '3.htm' indicating the IPs were attacked and the countries on the specified
date
echo -e You were attacked by all IPs below from following countries on "$d" > ~/3.htm
echo
# Pause for 1 second
sleep 1
# Append the contents of '1.txt' to '3.htm'
cat ~/1.txt >> ~/3.htm
echo
echo
# Count the total number of failed login attempts on the specified date and store the result in 'x'
x=$(find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep "$d" | grep -i Failed | grep -E -o
"([0-9]{1,3}[\.]){3}[0-9]{1,3}" | wc -l)
# Append the total failed login attempts message to '3.htm'
echo -e Total Failed Login Attempts on "$d" is "$x" >> ~/3.htm
# Append the contents of '3.htm' to a report file in the '/reports/' directory with the current date as
the filename
cat ~/3.htm >> /reports/"$now".txt
```

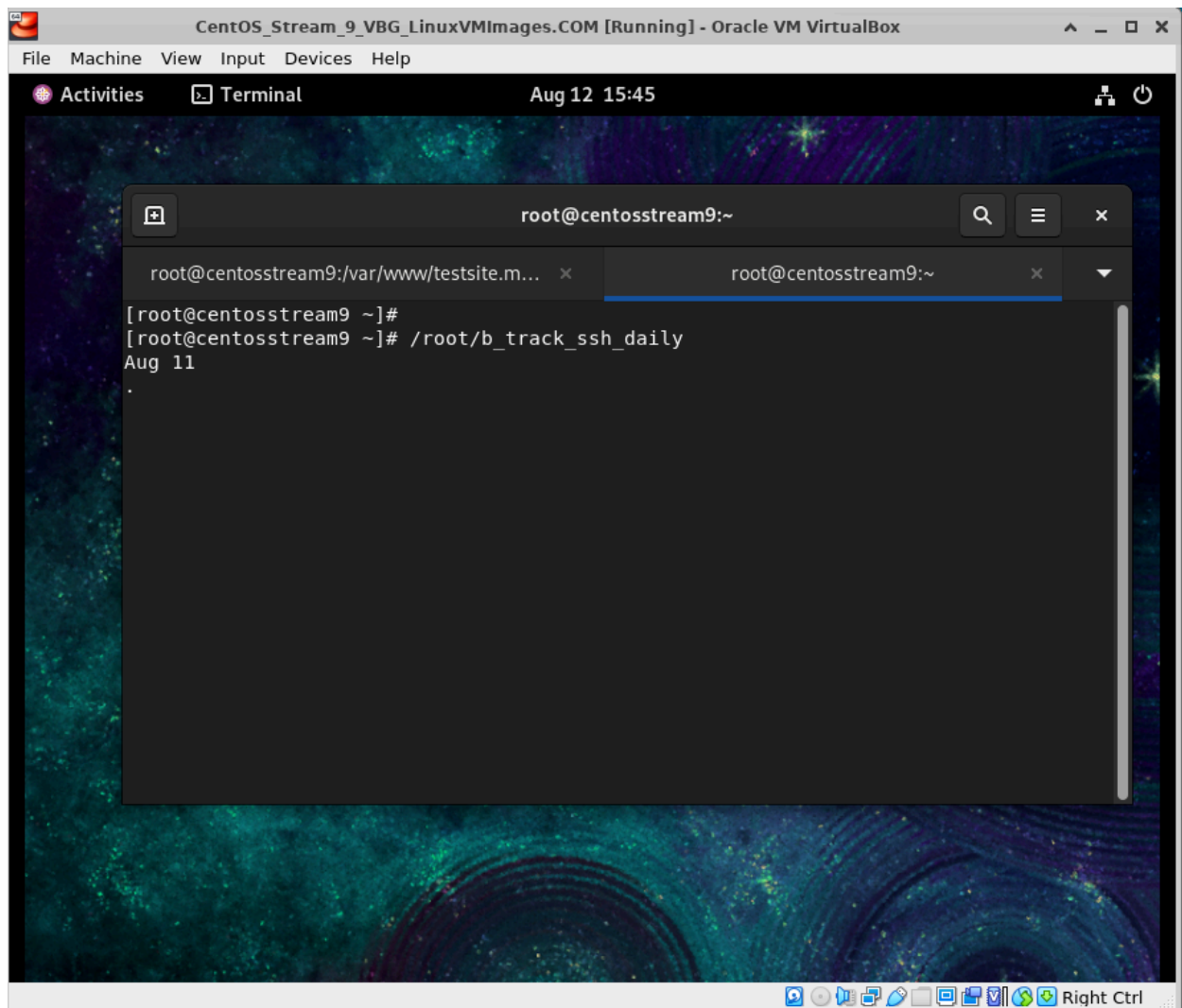
```
CentOS_Stream_9_VBG_LinuxVMImages.COM [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Aug 12 15:41

root@centosstream9:~
root@centosstream9:var/www/testsite.m... x root@centosstream9:~ x
echo
# Pause for 1 second
sleep 1
# Append the contents of '1.txt' to '3.htm'
cat ~/1.txt >> ~/3.htm
echo
echo
# Count the total number of failed login attempts on the specified date and store the result in 'x'
x=$(find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep "$d" | grep -i Failed | grep -E -o "[0-9]{1,3}[\.]){3}[0-9]{1,3}" | wc -l)
# Append the total failed login attempts message to '3.htm'
echo -e Total Failed Login Attempts on "$d" is "$x" >> ~/3.htm
# Append the contents of '3.htm' to a report file in the '/reports/' directory with the current date as the filename
cat ~/3.htm >> /reports/"$now".txt
-- INSERT --
```

2.22 Use the command: `chmod +x /root/b_track_ssh_daily`

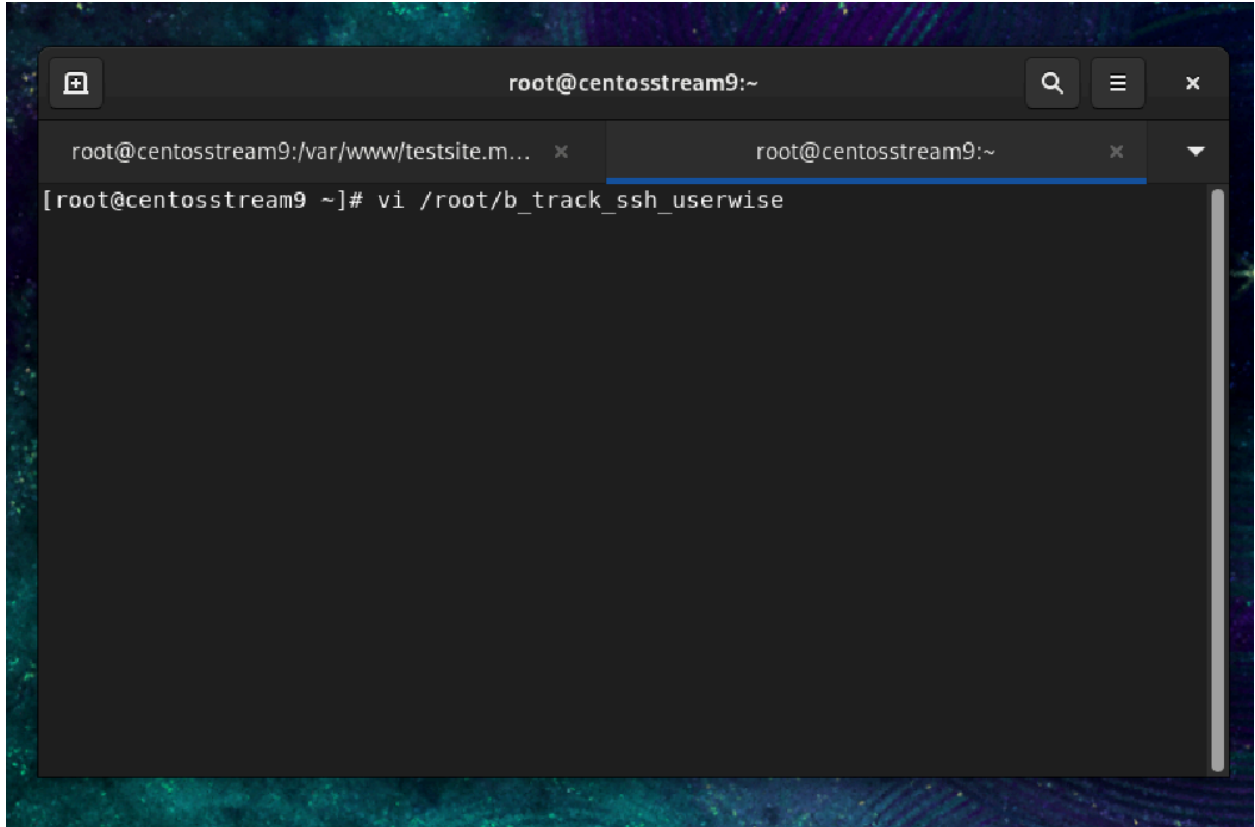


2.23 Use the command: `/root/b_track_ssh_daily`



Task 3: Implement IP Blocking to Mitigate Repeated Attacks

3.1 Edit the file using the command: `vi /root/b_track_ssh_userwise`

A terminal window with a dark background and a colorful, abstract pattern on the left side. The window has a title bar with the text "root@centosstream9:~". Below the title bar, there are two tabs: "root@centosstream9:/var/www/testsite.m..." and "root@centosstream9:~". The active tab is "root@centosstream9:~". The terminal prompt is "[root@centosstream9 ~]#". The command "vi /root/b_track_ssh_userwise" has been entered and is highlighted in blue. The terminal is currently empty, showing no output from the command.

```
root@centosstream9:~  
[root@centosstream9 ~]# vi /root/b_track_ssh_userwise
```

3.2 Use the following script:

```
#!/bin/bash
# Read date from the first argument and tokenize it
# Check if the second argument (day of the month) is less than 9 or equal to 9
if [ "$2" -lt 9 ] || [ "$2" -eq 9 ]
then
    # If the day is less than 9 or equal to 9, concatenate month and day with a period between
    d=$(echo "$1. $2")
else
    # Otherwise, concatenate the month and day with a space between
    d=$(echo "$1 $2")
fi
# Print the formatted date
echo "$d"
echo .
# Remove the previous failed login record file
rm ~/ssh_failed_sortlisted_logins.txt
# Search through system logs for failed SSH login attempts on the specified date
# and save the unique IP addresses to the 'ssh_failed_sortlisted_logins.txt' file
find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep "$d" | grep Failed | grep -E -o "([0-9]{1,3}[.]){3}[0-9]{1,3}" | sort -u > ~/ssh_failed_sortlisted_logins.txt
# Remove temporary files '1.txt' and '2.txt' if they exist
rm -f ~/1.txt
rm -f ~/2.txt
# Loop through each IP address in the 'ssh_failed_sortlisted_logins.txt' file
cat ~/ssh_failed_sortlisted_logins.txt | \
while read ip
do
    # Check again if the day of the month is less than 9 or equal to 9
    if [ "$2" -lt 9 ] || [ "$2" -eq 9 ]
    then
        # Perform a whois lookup to get the country information for each IP address
        whois $ip | grep -i country
        # If country information is found, store it in '1.txt'
        if [ $? -eq 0 ]
        then
            clear
            echo $ip >> ~/1.txt
            whois $ip | grep -i country | tr "a-z" "A-Z" | sort -u >> ~/1.txt
        else
            # If no country info is found, log the IP with "COUNTRY: UNKNOWN"
            echo $ip >> ~/1.txt
            echo "COUNTRY: UNKNOWN" >> ~/1.txt
        fi
    fi
done
```

```

fi
# Search for failed SSH login attempts for this IP in the logs and append to '1.txt'
find /var/log -name "secure*" -exec cat {} \; | grep "$d" | grep ssh | grep $ip | grep user | grep
Invalid | cut -d " " -f1-4,9 >> ~/1.txt
else
# If the day is greater than 9, perform the same actions but cut the logs differently
whois $ip | grep -i country
if [ $? -eq 0 ]
then
clear
echo $ip >> ~/1.txt
whois $ip | grep -i country | tr "a-z" "A-Z" | sort -u >> ~/1.txt
else
echo $ip >> ~/1.txt
echo "COUNTRY: UNKNOWN" >> ~/1.txt
fi

# Search for failed SSH login attempts for this IP in the logs and append to '1.txt'
find /var/log -name "secure*" -exec cat {} \; | grep "$d" | grep ssh | grep $ip | grep user | grep
Invalid | cut -d " " -f1-3,8 >> ~/1.txt
fi
done

# Display the report header with the formatted date
echo "REPORT:"$d""
echo -e "\E[47;31m""You were attacked by all IPs below from following
countries""\033[1m\033[0m"
echo

# Pause for 1 second
sleep 1

```

```
# If '1.txt' exists, display its contents
if [ -f ~/1.txt ]
then
    cat ~/1.txt
else
    # If '1.txt' does not exist, print an error message
    echo "I Guess You Entered Invalid Date"
    echo
    echo
fi
```

```
# Count the total number of failed SSH login attempts on the specified date
```

```
x=$(find /var/log -name "secure*" -exec cat {} \; | grep ssh | grep "$d" | grep -i Failed | grep -E -o
"([0-9]{1,3}[.]){3}[0-9]{1,3}" | wc -l)
```

```
# Display the total number of failed login attempts
```

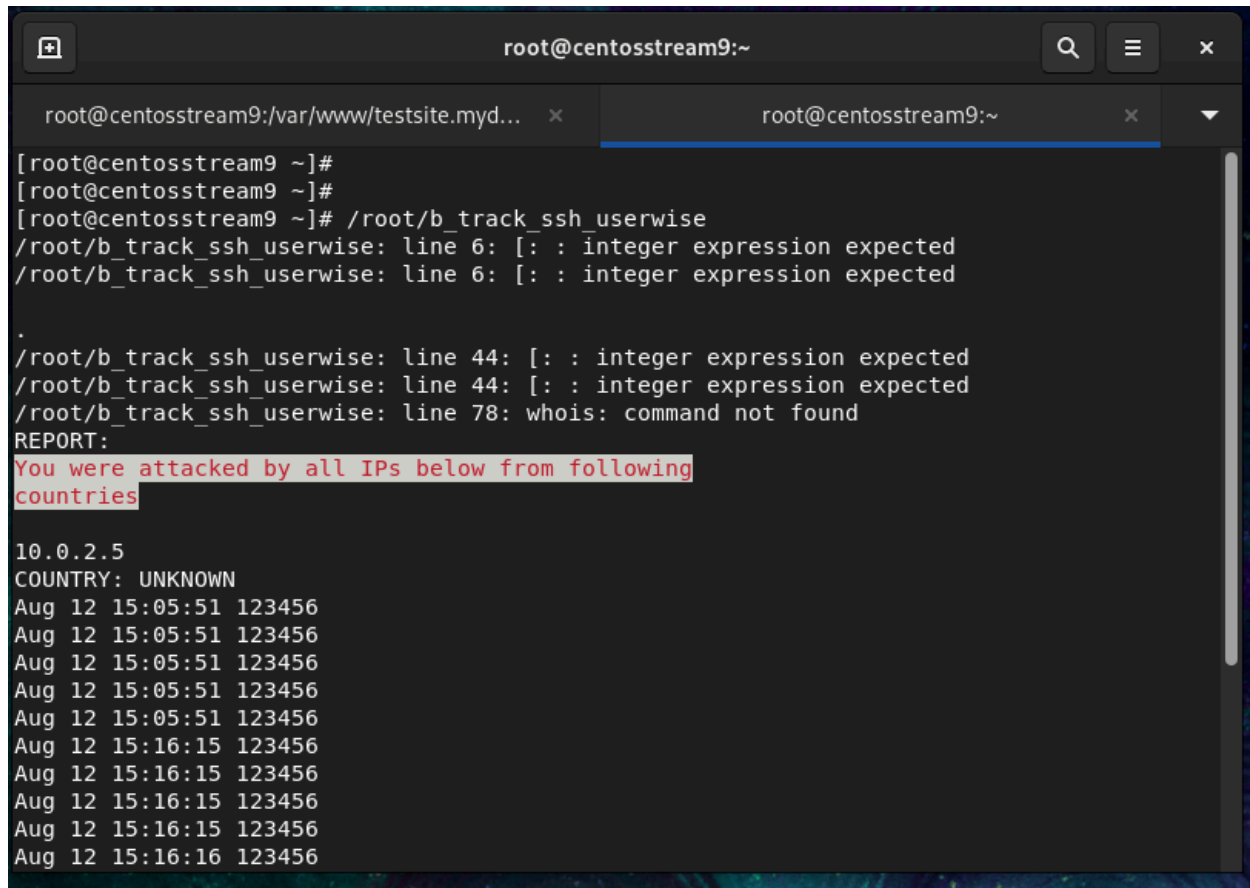
```
echo -e "\E[37;44m""\033[1m "Total Failed Login Attempts on "$d" is "$x""\033[0m" .
```

```
root@centosstream9:~  
#!/bin/bash  
  
# Read date from the first argument and tokenize it  
# Check if the second argument (day of the month) is less than 9 or equal to 9  
if [ "$2" -lt 9 ] || [ "$2" -eq 9 ]  
then  
    # If the day is less than 9 or equal to 9, concatenate month and day with a  
    period between  
    d=$(echo "$1. $2")  
else  
    # Otherwise, concatenate the month and day with a space between  
    d=$(echo "$1 $2")  
fi  
  
# Print the formatted date  
echo "$d"  
echo .  
  
# Remove the previous failed login record file  
rm ~/ssh_failed_sortlisted_logins.txt  
  
# Search through system logs for failed SSH login attempts on the specified date  
@@@  
-- INSERT -- 1,12 Top
```

3.3 Use the command: `chmod +x /root/b_track_ssh_userwise`

```
root@centosstream9:~  
root@centosstream9:/var/www/testsite.myd... x root@centosstream9:~ x  
[root@centosstream9 ~]#  
[root@centosstream9 ~]# chmod +x /root/b_track_ssh_userwise  
[root@centosstream9 ~]#
```


3.4 Now use the command: `/root/b_track_ssh_userwise`

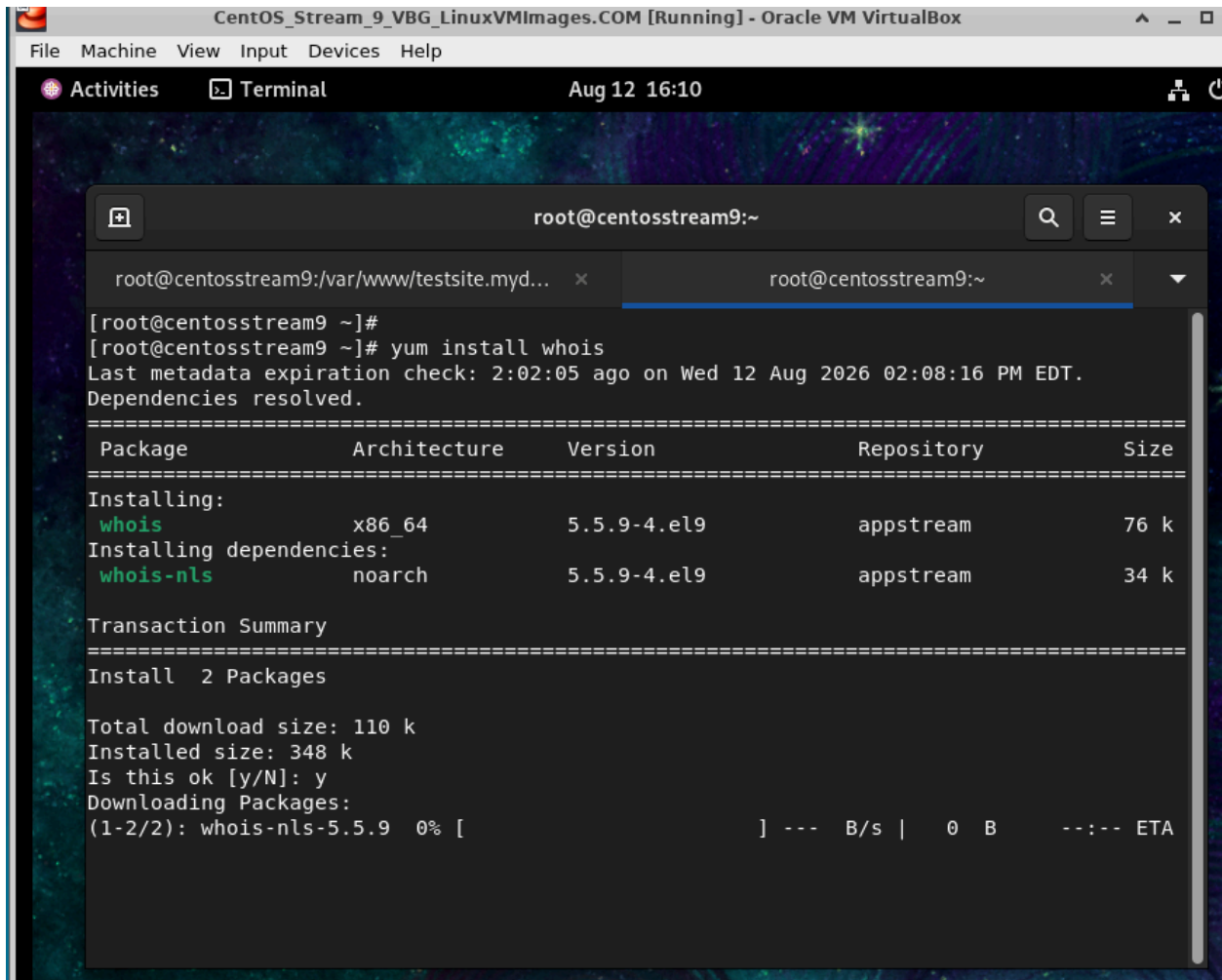


A terminal window titled 'root@centosstream9:~' with two tabs. The active tab shows the execution of the command `/root/b_track_ssh_userwise`. The output displays several syntax errors related to integer expressions on lines 6, 44, and 78, followed by a 'REPORT:' section. The report states: 'You were attacked by all IPs below from following countries' and lists the IP '10.0.2.5' with a 'COUNTRY: UNKNOWN' status. Below this, there is a list of timestamps and the number '123456'.

```
root@centosstream9:~/var/www/testsite.myd... x root@centosstream9:~ x
[root@centosstream9 ~]#
[root@centosstream9 ~]#
[root@centosstream9 ~]# /root/b_track_ssh_userwise
/root/b_track_ssh_userwise: line 6: [: : integer expression expected
/root/b_track_ssh_userwise: line 6: [: : integer expression expected
.
/root/b_track_ssh_userwise: line 44: [: : integer expression expected
/root/b_track_ssh_userwise: line 44: [: : integer expression expected
/root/b_track_ssh_userwise: line 78: whois: command not found
REPORT:
You were attacked by all IPs below from following
countries
10.0.2.5
COUNTRY: UNKNOWN
Aug 12 15:05:51 123456
Aug 12 15:05:51 123456
Aug 12 15:05:51 123456
Aug 12 15:05:51 123456
Aug 12 15:05:51 123456
Aug 12 15:16:15 123456
Aug 12 15:16:15 123456
Aug 12 15:16:15 123456
Aug 12 15:16:15 123456
Aug 12 15:16:15 123456
Aug 12 15:16:16 123456
```

Task 4: Analyze Attack Data and Generate Security Test Cases

4.1 Open the terminal in CentOS and install the whois using the command `yum install whois` press y to continue

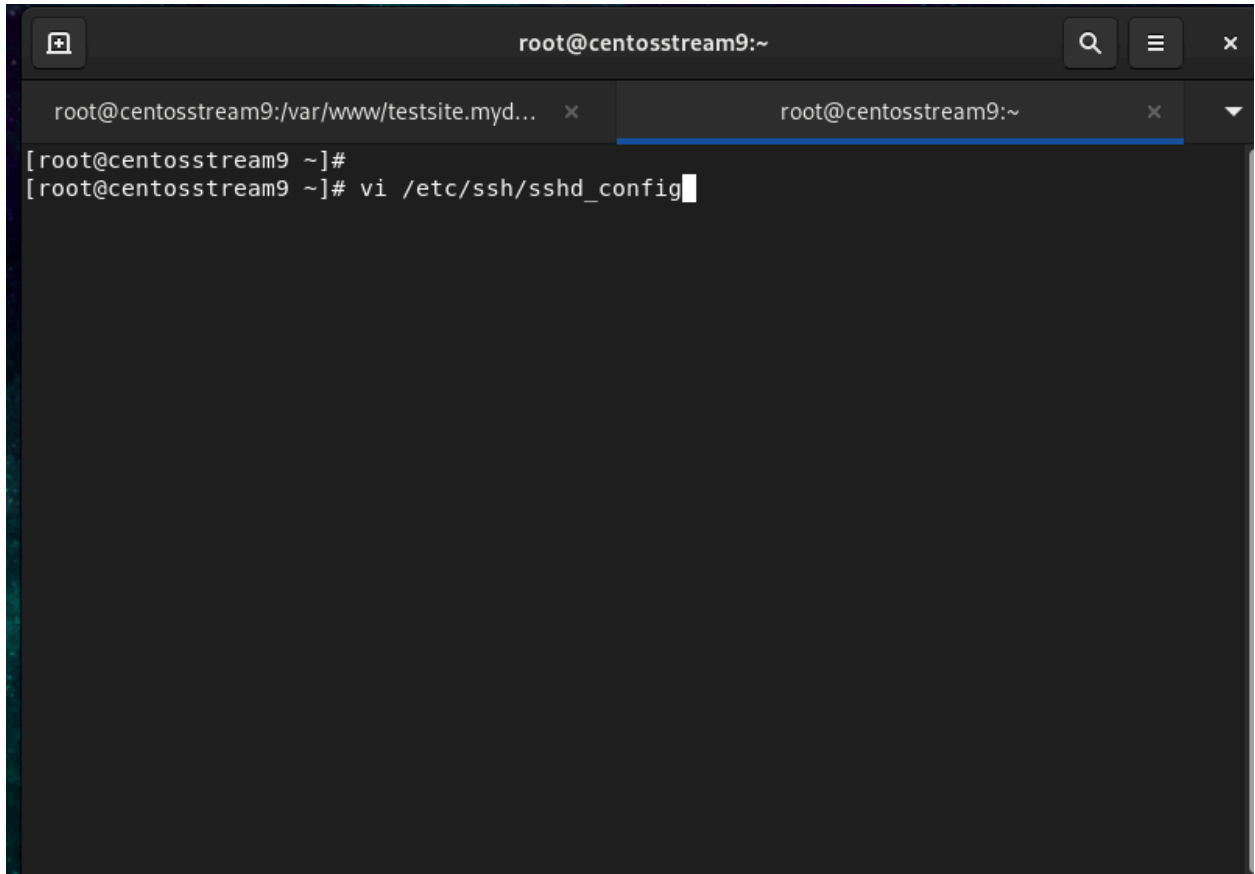


The screenshot shows a terminal window titled "CentOS_Stream_9_VBG_LinuxVMImages.COM [Running] - Oracle VM VirtualBox". The terminal is running as root on a CentOS Stream 9 system. The user has entered the command `yum install whois`. The output shows that the metadata expiration check passed, dependencies were resolved, and the packages `whois` and `whois-nls` are being installed. The transaction summary indicates a total download size of 110 k and an installed size of 348 k. The user has confirmed the installation with 'y'.

```
[root@centosstream9 ~]#
[root@centosstream9 ~]# yum install whois
Last metadata expiration check: 2:02:05 ago on Wed 12 Aug 2026 02:08:16 PM EDT.
Dependencies resolved.
=====
Package                Architecture  Version      Repository    Size
=====
Installing:
whois                   x86_64       5.5.9-4.el9  appstream     76 k
Installing dependencies:
whois-nls               noarch       5.5.9-4.el9  appstream     34 k
Transaction Summary
=====
Install 2 Packages

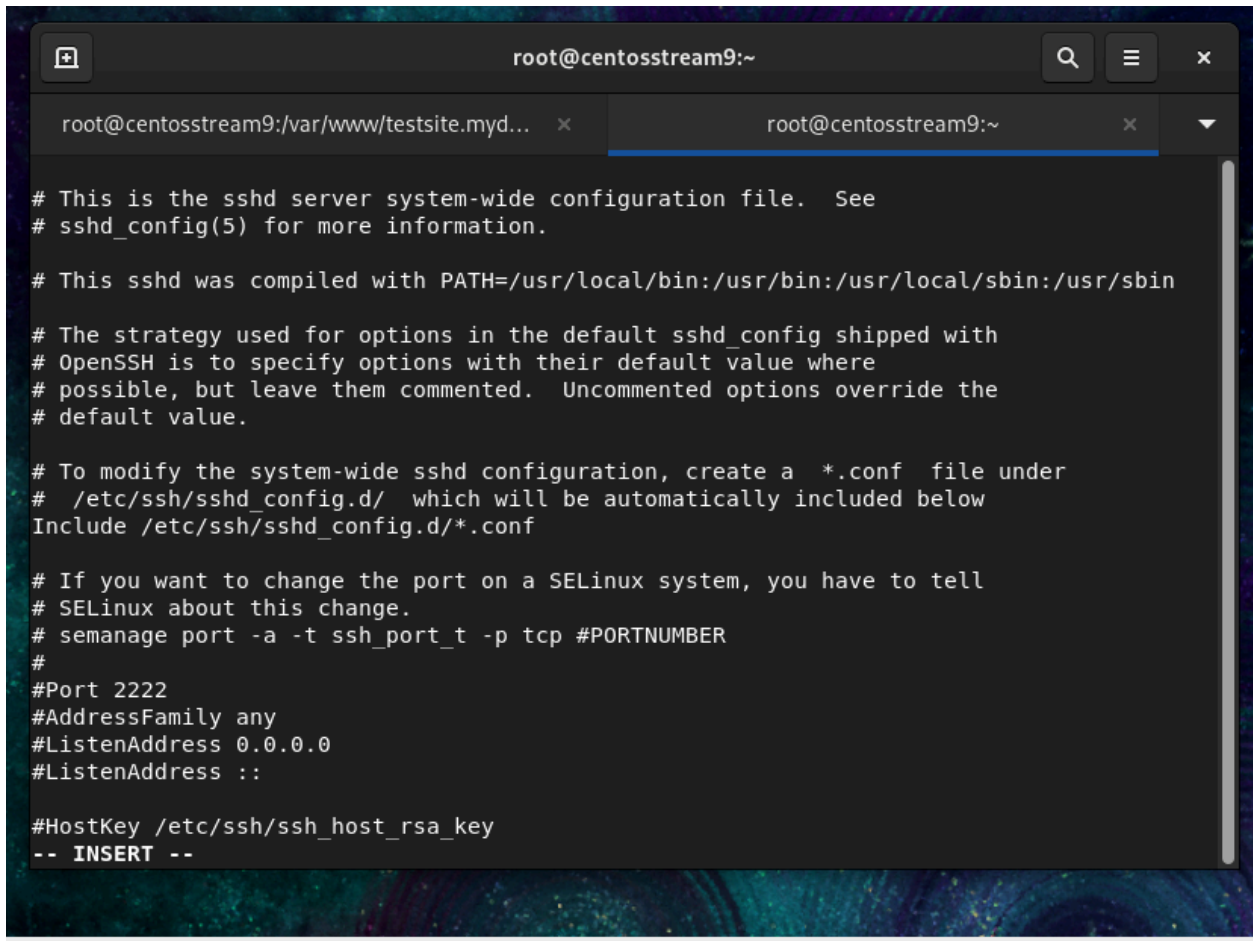
Total download size: 110 k
Installed size: 348 k
Is this ok [y/N]: y
Downloading Packages:
(1-2/2): whois-nls-5.5.9  0% [
```

4.2 Edit the ssh_config file using the command: vi /etc/ssh/sshd_config

A terminal window with a dark background. The title bar shows 'root@centosstream9:~' and search, menu, and close icons. There are two tabs: 'root@centosstream9:/var/www/testsite.myd...' and 'root@centosstream9:~'. The terminal shows the prompt '[root@centosstream9 ~]#' followed by the command 'vi /etc/ssh/sshd_config' with a cursor at the end.

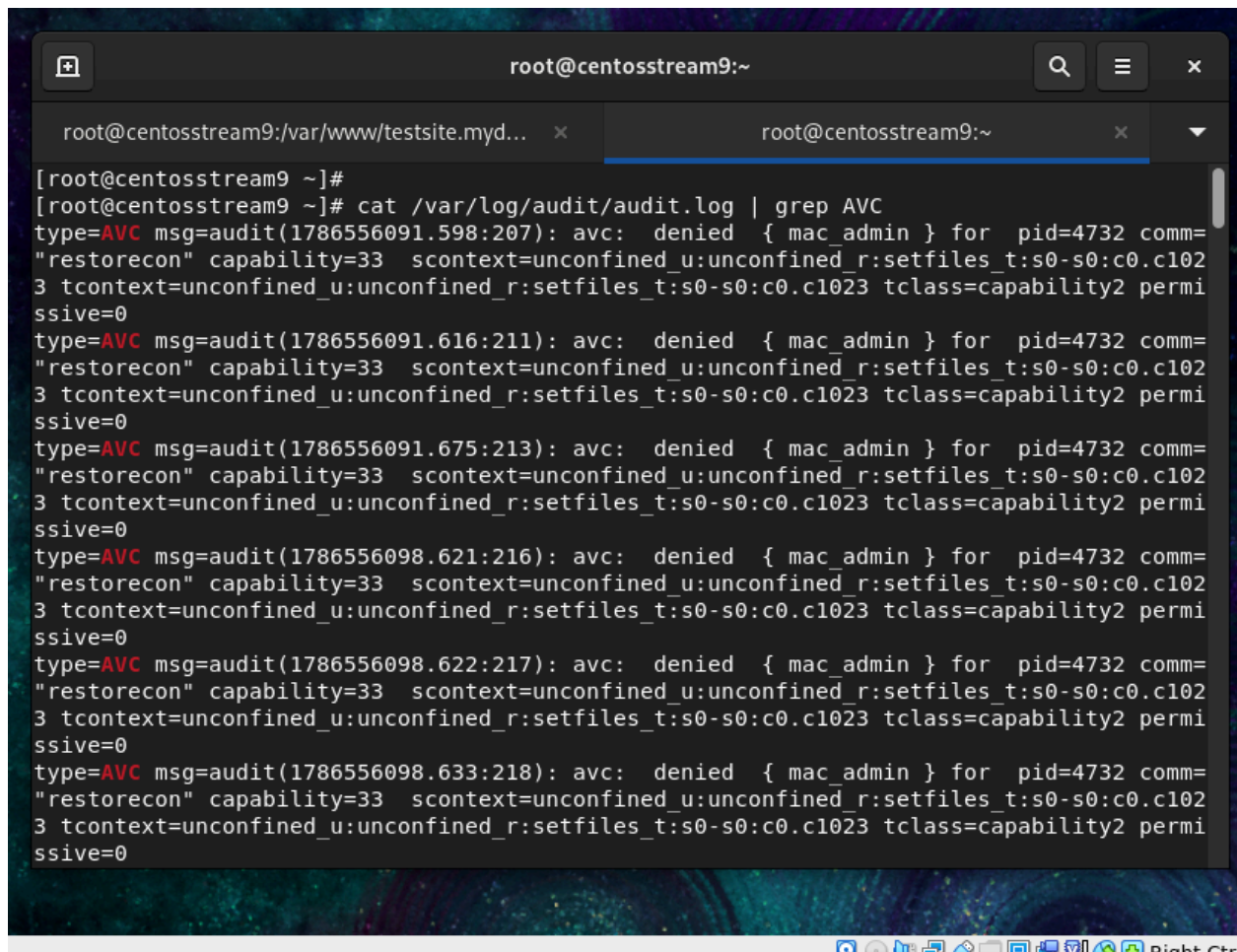
```
[root@centosstream9 ~]#  
[root@centosstream9 ~]# vi /etc/ssh/sshd_config
```

4.3 Type i to give insert permission and change #Port22 to Port2222



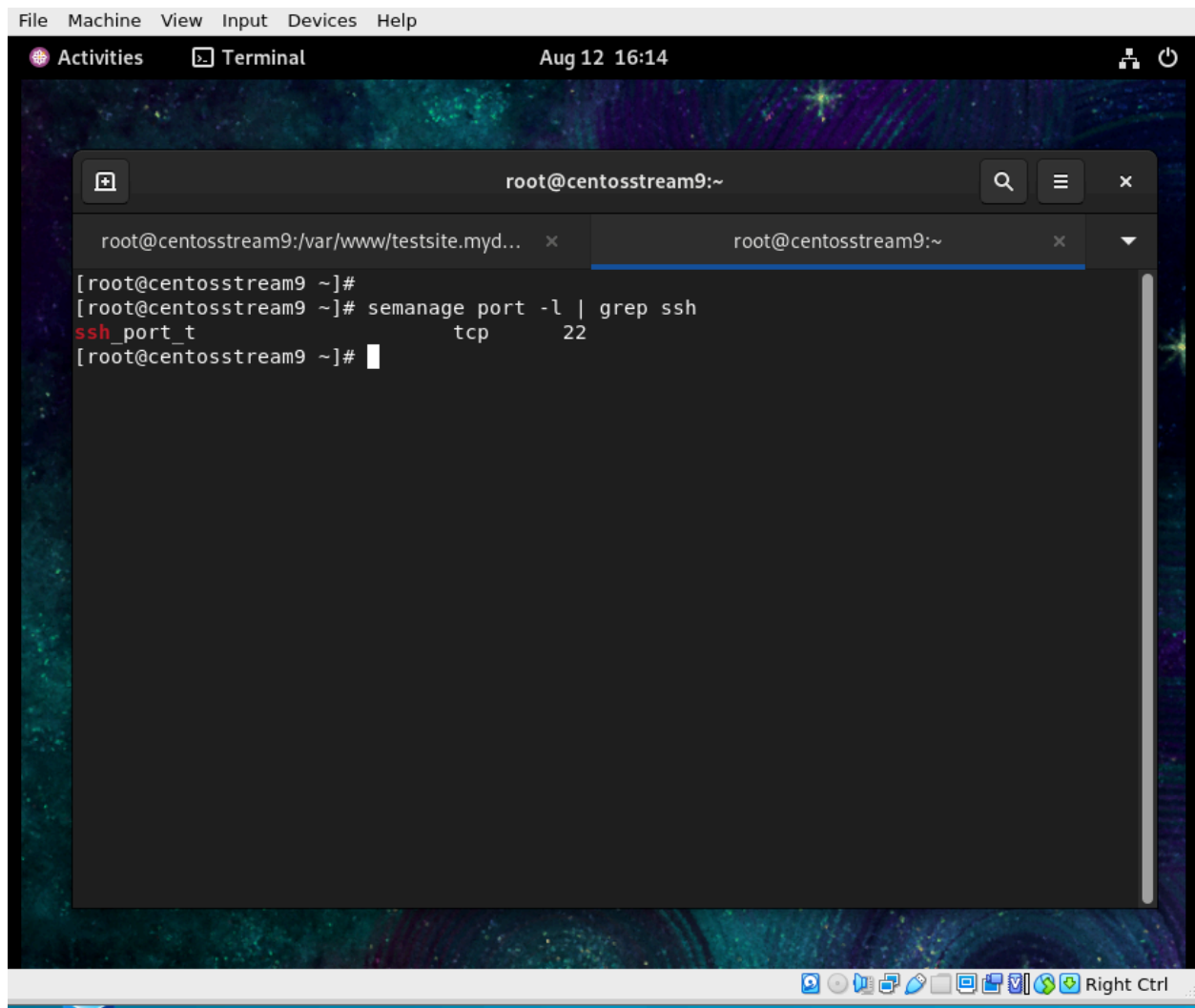
```
root@centosstream9:~  
root@centosstream9:/var/www/testsite.myd... x root@centosstream9:~ x  
# This is the sshd server system-wide configuration file. See  
# sshd_config(5) for more information.  
  
# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin  
  
# The strategy used for options in the default sshd_config shipped with  
# OpenSSH is to specify options with their default value where  
# possible, but leave them commented. Uncommented options override the  
# default value.  
  
# To modify the system-wide sshd configuration, create a *.conf file under  
# /etc/ssh/sshd_config.d/ which will be automatically included below  
Include /etc/ssh/sshd_config.d/*.conf  
  
# If you want to change the port on a SELinux system, you have to tell  
# SELinux about this change.  
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER  
#  
#Port 2222  
#AddressFamily any  
#ListenAddress 0.0.0.0  
#ListenAddress ::  
  
#HostKey /etc/ssh/ssh_host_rsa_key  
-- INSERT --
```

4.4 Use the command: cat /var/log/audit/audit.log | grep AVC



```
root@centosstream9:~  
root@centosstream9:~/var/www/testsite.myd... x root@centosstream9:~ x  
[root@centosstream9 ~]#  
[root@centosstream9 ~]# cat /var/log/audit/audit.log | grep AVC  
type=AVC msg=audit(1786556091.598:207): avc: denied { mac_admin } for pid=4732 comm=  
"restorecon" capability=33 scontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c102  
3 tcontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c1023 tclass=capability2 permi  
ssive=0  
type=AVC msg=audit(1786556091.616:211): avc: denied { mac_admin } for pid=4732 comm=  
"restorecon" capability=33 scontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c102  
3 tcontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c1023 tclass=capability2 permi  
ssive=0  
type=AVC msg=audit(1786556091.675:213): avc: denied { mac_admin } for pid=4732 comm=  
"restorecon" capability=33 scontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c102  
3 tcontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c1023 tclass=capability2 permi  
ssive=0  
type=AVC msg=audit(1786556098.621:216): avc: denied { mac_admin } for pid=4732 comm=  
"restorecon" capability=33 scontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c102  
3 tcontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c1023 tclass=capability2 permi  
ssive=0  
type=AVC msg=audit(1786556098.622:217): avc: denied { mac_admin } for pid=4732 comm=  
"restorecon" capability=33 scontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c102  
3 tcontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c1023 tclass=capability2 permi  
ssive=0  
type=AVC msg=audit(1786556098.633:218): avc: denied { mac_admin } for pid=4732 comm=  
"restorecon" capability=33 scontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c102  
3 tcontext=unconfined_u:unconfined_r:setfiles_t:s0-s0:c0.c1023 tclass=capability2 permi  
ssive=0
```

4.5 Use the command: semanage port -l | grep ssh

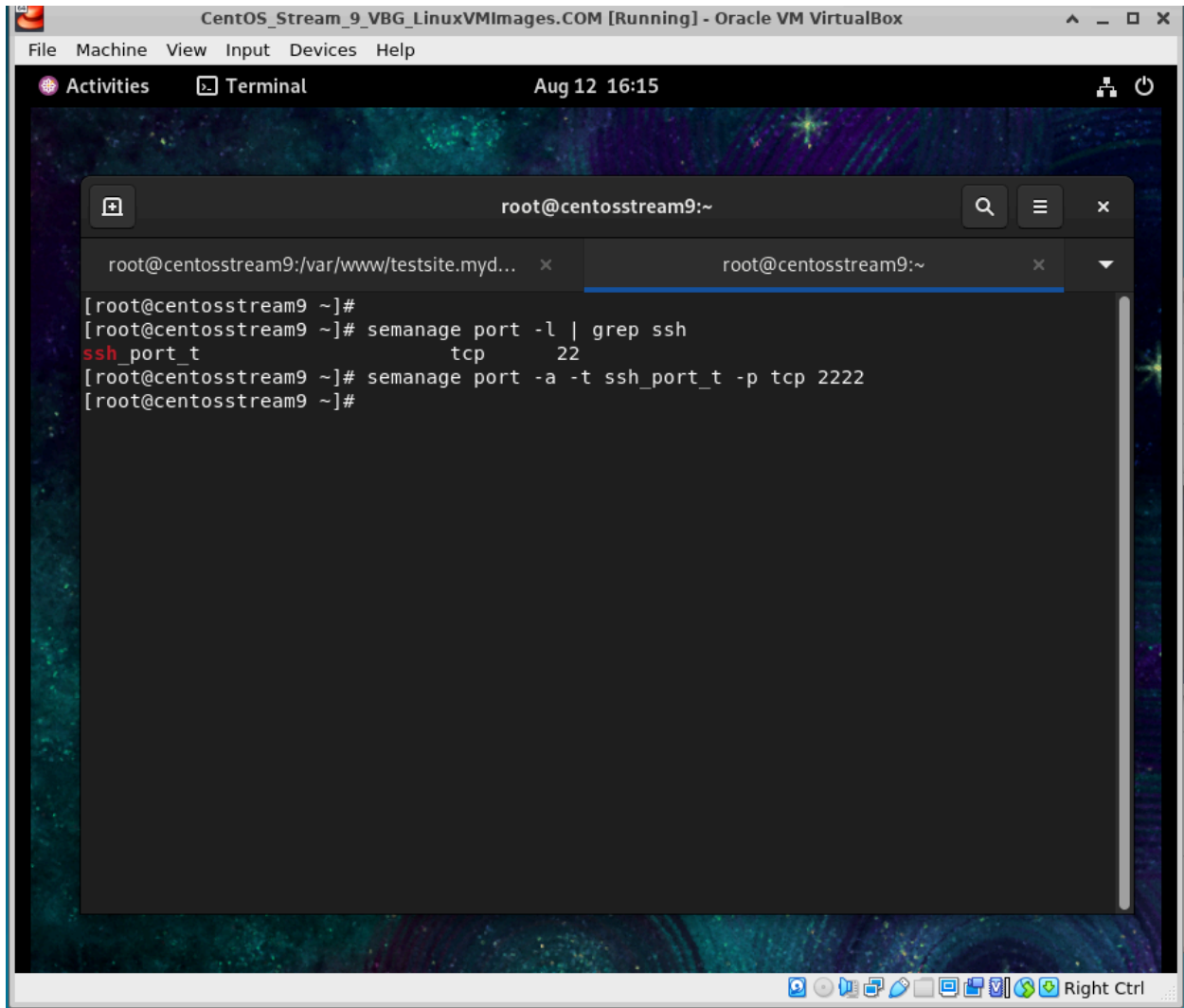


The screenshot shows a Linux desktop environment with a terminal window open. The terminal window has a title bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help' menus. The terminal title is 'root@centosstream9:~'. The terminal content shows the following commands and output:

```
[root@centosstream9 ~]#  
[root@centosstream9 ~]# semanage port -l | grep ssh  
ssh_port_t          tcp      22  
[root@centosstream9 ~]#
```

The terminal window also shows a tab for 'root@centosstream9:/var/www/testsite.myd...' and a search bar. The desktop background is a colorful abstract image, and the system tray at the bottom shows various icons and the text 'Right Ctrl'.

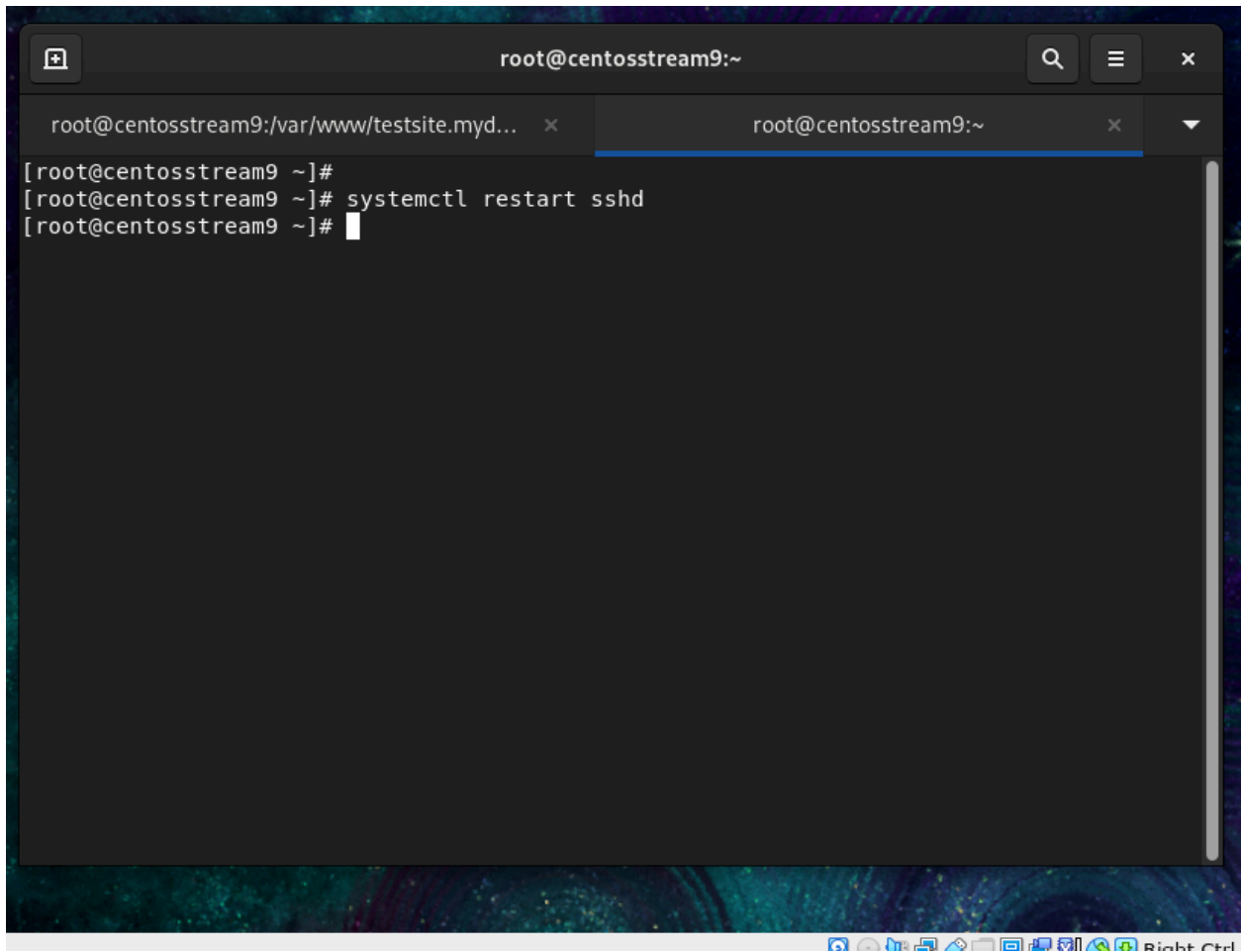
4.6 Use the command: `semanage port -a -t ssh_port_t -p tcp 2222`



```
CentOS_Stream_9_VBG_LinuxVMImages.COM [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Aug 12 16:15

root@centosstream9:~
root@centosstream9:/var/www/testsite.myd... x root@centosstream9:~ x
[root@centosstream9 ~]#
[root@centosstream9 ~]# semanage port -l | grep ssh
ssh_port_t tcp 22
[root@centosstream9 ~]# semanage port -a -t ssh_port_t -p tcp 2222
[root@centosstream9 ~]#
```

4.7 Use the command: `systemctl restart sshd`

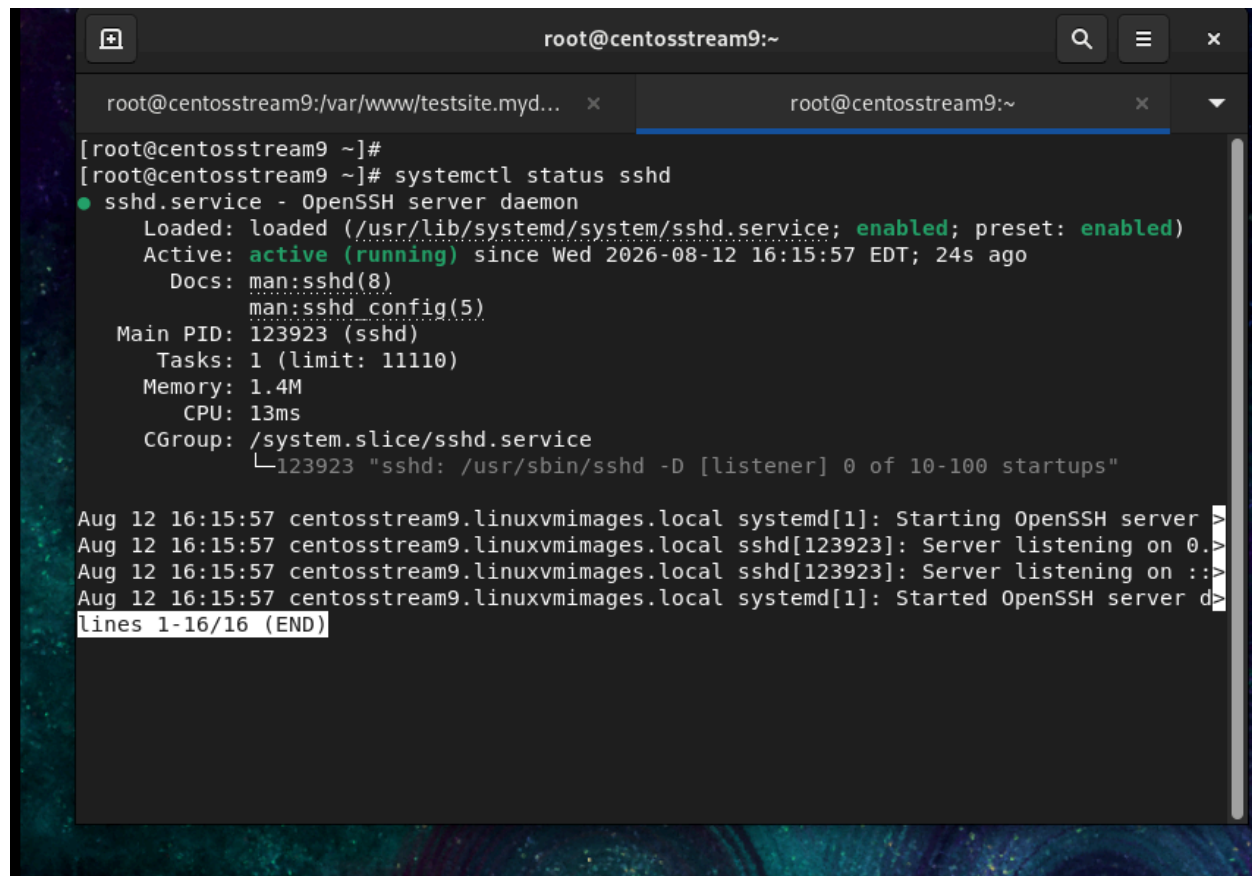


A terminal window titled `root@centosstream9:~` is shown. The window has a dark background and a light-colored border. The terminal output shows the following commands and prompts:

```
[root@centosstream9 ~]#  
[root@centosstream9 ~]# systemctl restart sshd  
[root@centosstream9 ~]#
```

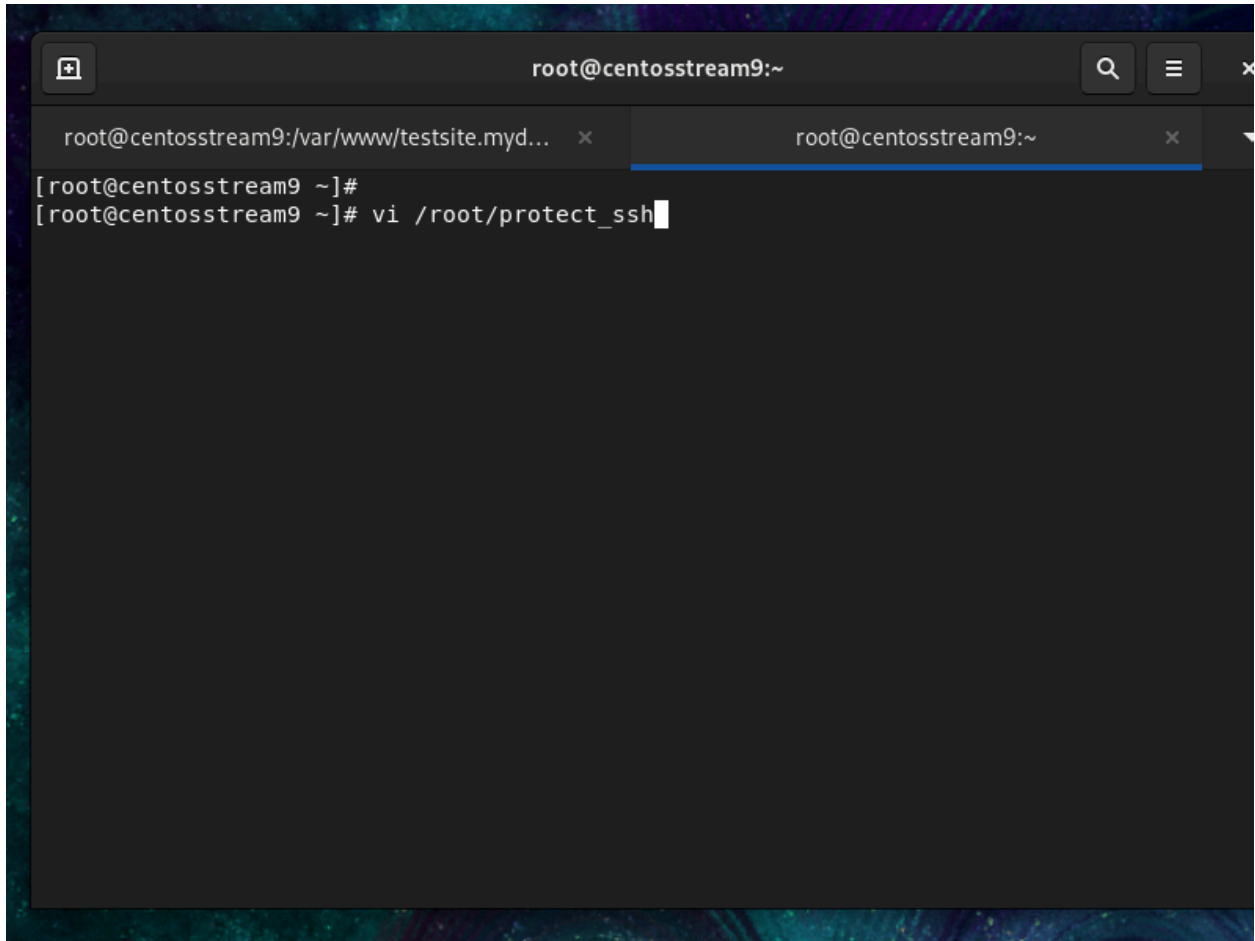
The terminal window is part of a desktop environment with a dark, abstract background. The window's title bar includes a search icon, a menu icon, and a close button. The terminal's status bar at the bottom shows various system icons and the text "Right Ctrl".

4.8 Use the command: systemctl status sshd



```
root@centosstream9:~  
root@centosstream9:~# systemctl status sshd  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)  
   Active: active (running) since Wed 2026-08-12 16:15:57 EDT; 24s ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
  Main PID: 123923 (sshd)  
    Tasks: 1 (limit: 11110)  
  Memory: 1.4M  
    CPU: 13ms  
   CGroup: /system.slice/sshd.service  
           └─123923 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Aug 12 16:15:57 centosstream9.linuxvmimages.local systemd[1]: Starting OpenSSH server  
Aug 12 16:15:57 centosstream9.linuxvmimages.local sshd[123923]: Server listening on 0.  
Aug 12 16:15:57 centosstream9.linuxvmimages.local sshd[123923]: Server listening on :.  
Aug 12 16:15:57 centosstream9.linuxvmimages.local systemd[1]: Started OpenSSH server d  
lines 1-16/16 (END)
```

4.9 Edit the file using the command: `vi /root/protect_ssh`

A terminal window titled 'root@centosstream9:~' with a search icon and a menu icon in the top right corner. The window has two tabs: 'root@centosstream9:/var/www/testsite.myd...' and 'root@centosstream9:~'. The active tab shows the command prompt '[root@centosstream9 ~]#' followed by the command 'vi /root/protect_ssh' being entered. The cursor is at the end of the command.

```
root@centosstream9:~  
[root@centosstream9 ~]#  
[root@centosstream9 ~]# vi /root/protect_ssh
```

4.10 Type i to give insert permission and use this script:

```
#!/bin/bash
# Backup the current /etc/hosts.deny file before proceeding. This command should be executed
first manually.
# "cp /etc/hosts.deny /etc/hosts.deny.bak"

# Restore the /etc/hosts.deny file from the backup
cat /etc/hosts.deny.bak > /etc/hosts.deny

# Extract unique IP addresses from all the reports in the /reports directory and save them in
/tmp/blacklist
cat /reports/* | grep -E -o "([0-9]{1,3}[\.]){3}[0-9]{1,3}" | sort -u > /tmp/blacklist

# Extract unique failed SSH login attempts from the /var/log/secure file and append them to the
/tmp/blacklist file
cat /var/log/secure | grep ssh | grep -i Failed | grep -E -o "([0-9]{1,3}[\.]){3}[0-9]{1,3}" | sort -u >>
/tmp/blacklist

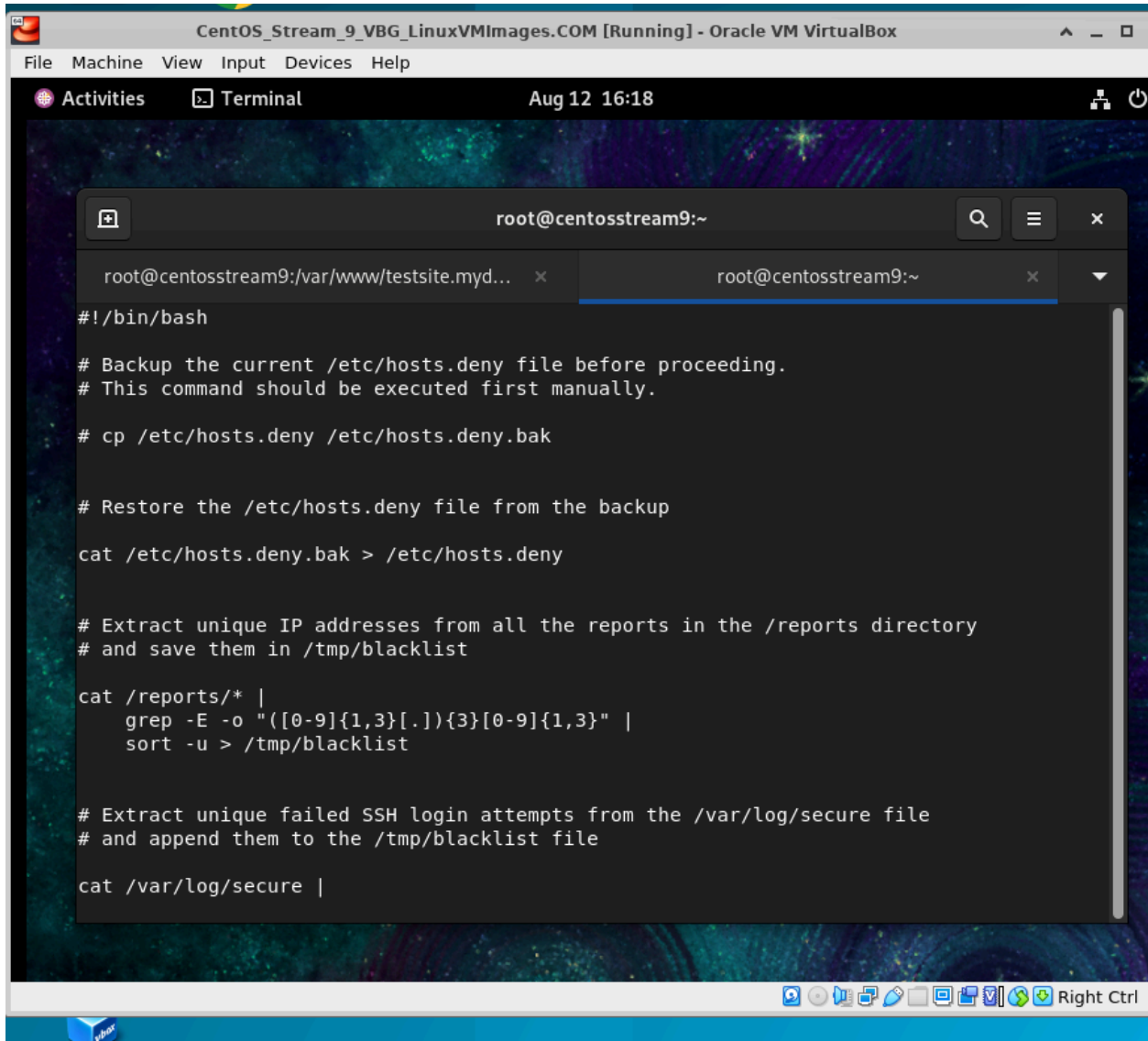
# Sort and remove duplicates from the /tmp/blacklist file, saving the result in /tmp/blacklist-
sorted
cat /tmp/blacklist | sort -u >> /tmp/blacklist-sorted

# Loop through each IP address in the /tmp/blacklist-sorted file and add it to /etc/hosts.deny
cat /tmp/blacklist-sorted | \
while read z
do
# For each IP, add an entry to the /etc/hosts.deny file to block SSH access
echo "sshd : $z" >> /etc/hosts.deny
done

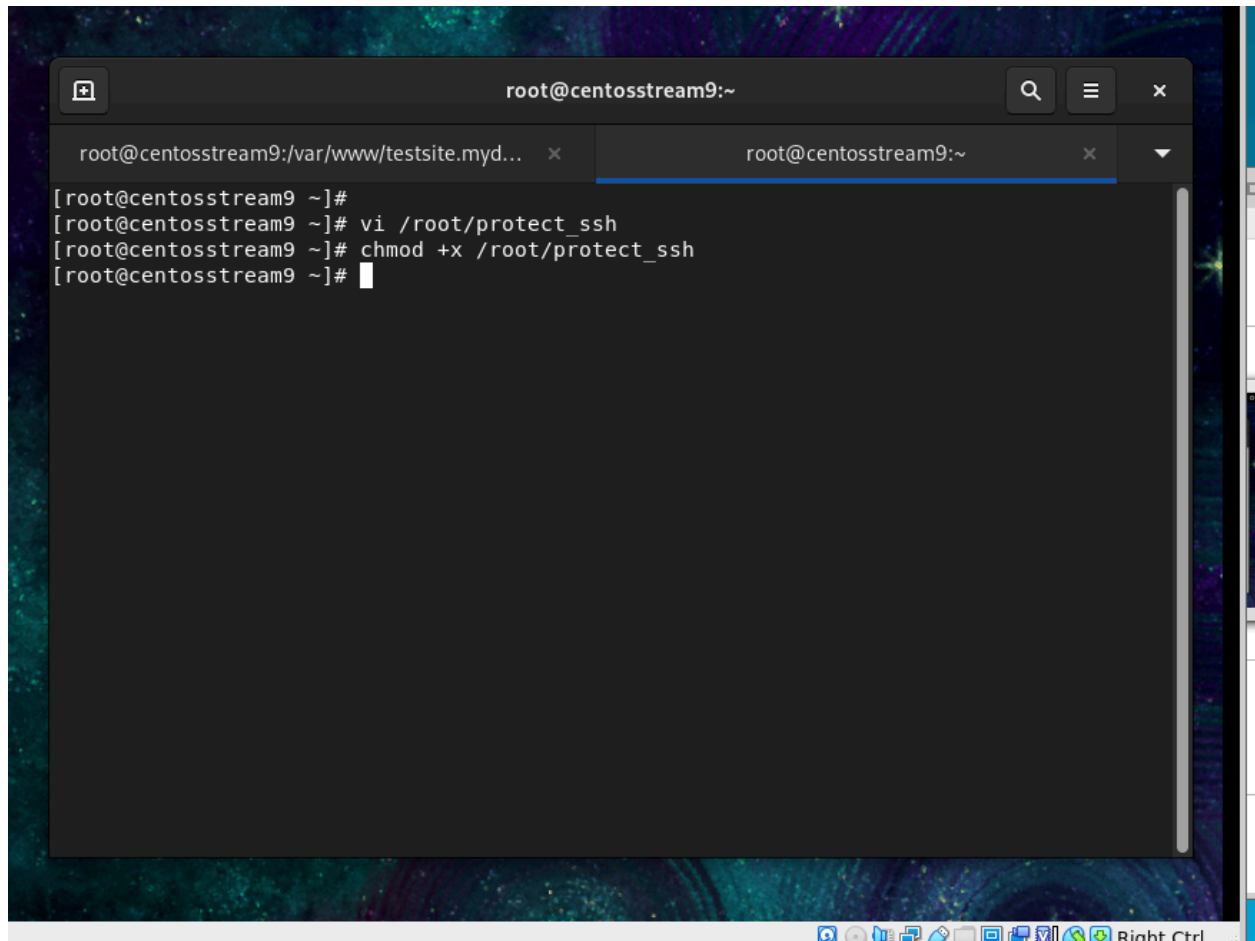
# Capture the current date and time
dt=$(date)

# Count the total number of attack entries (failed SSH login attempts) in /etc/hosts.deny
totalfailure=$(grep sshd /etc/hosts.deny | wc -l)

# Append a comment at the end of the /etc/hosts.deny file with the total number of attacks and
the date of the last update
echo "#Total attack is $totalfailure. Last updated on $dt" >> /etc/hosts.deny
```



4.11 Use the command: `chmod +x /root/protect_ssh`

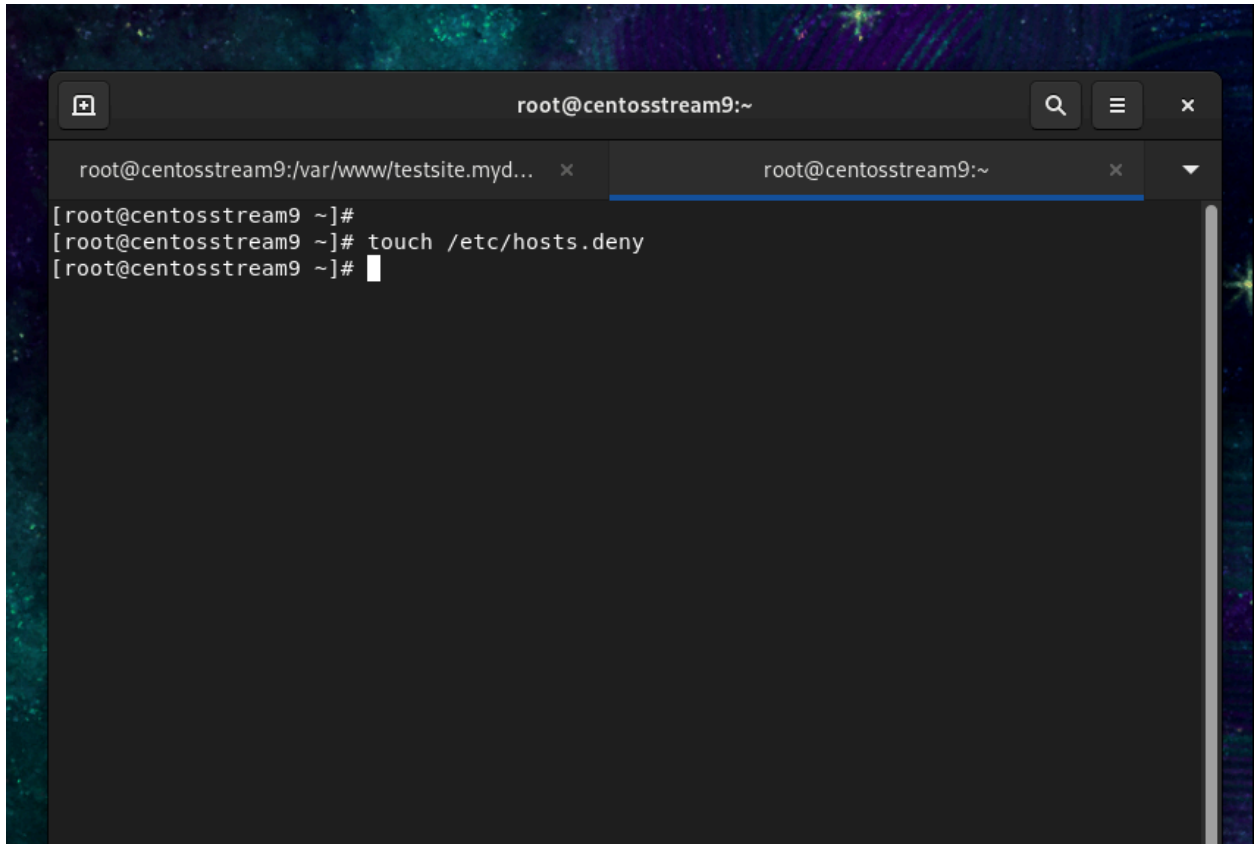


A terminal window titled "root@centosstream9:~" is shown. The window has a dark background and a light-colored text. The terminal output shows the following commands and their results:

```
[root@centosstream9 ~]#  
[root@centosstream9 ~]# vi /root/protect_ssh  
[root@centosstream9 ~]# chmod +x /root/protect_ssh  
[root@centosstream9 ~]#
```

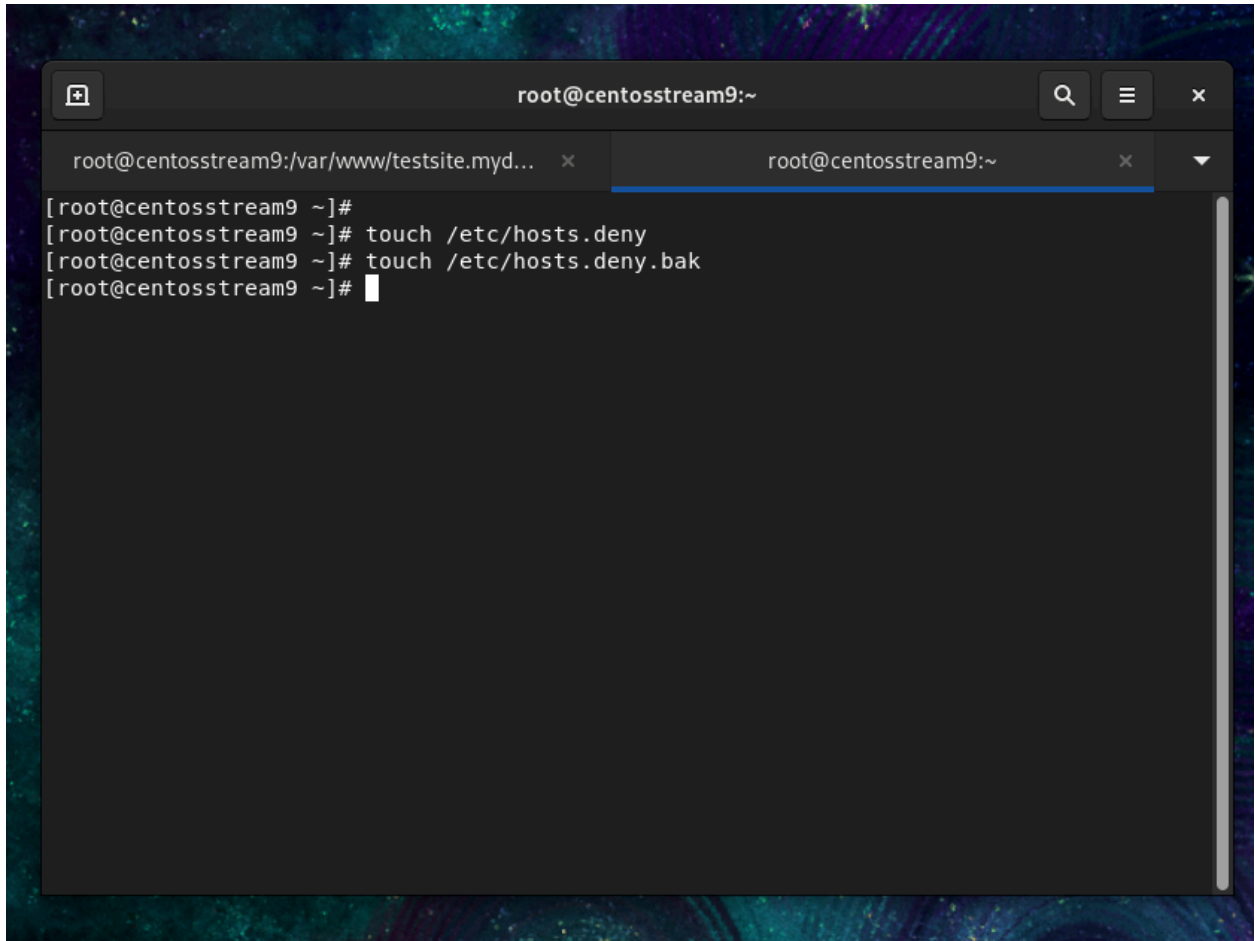
The terminal window is part of a desktop environment with a colorful, abstract background. The window's title bar includes a search icon, a menu icon, and a close button. The terminal output is displayed in a monospaced font.

4.12 Create a file using the command: touch /etc/hosts.deny

A terminal window titled 'root@centosstream9:~' is shown. The window has a dark theme and a search bar in the top right. The terminal content shows three lines: the first is the prompt '[root@centosstream9 ~]#', the second is the command 'touch /etc/hosts.deny', and the third is the prompt '[root@centosstream9 ~]#' followed by a cursor. The background of the terminal window is a dark space with colorful nebulae.

```
[root@centosstream9 ~]#  
[root@centosstream9 ~]# touch /etc/hosts.deny  
[root@centosstream9 ~]#
```

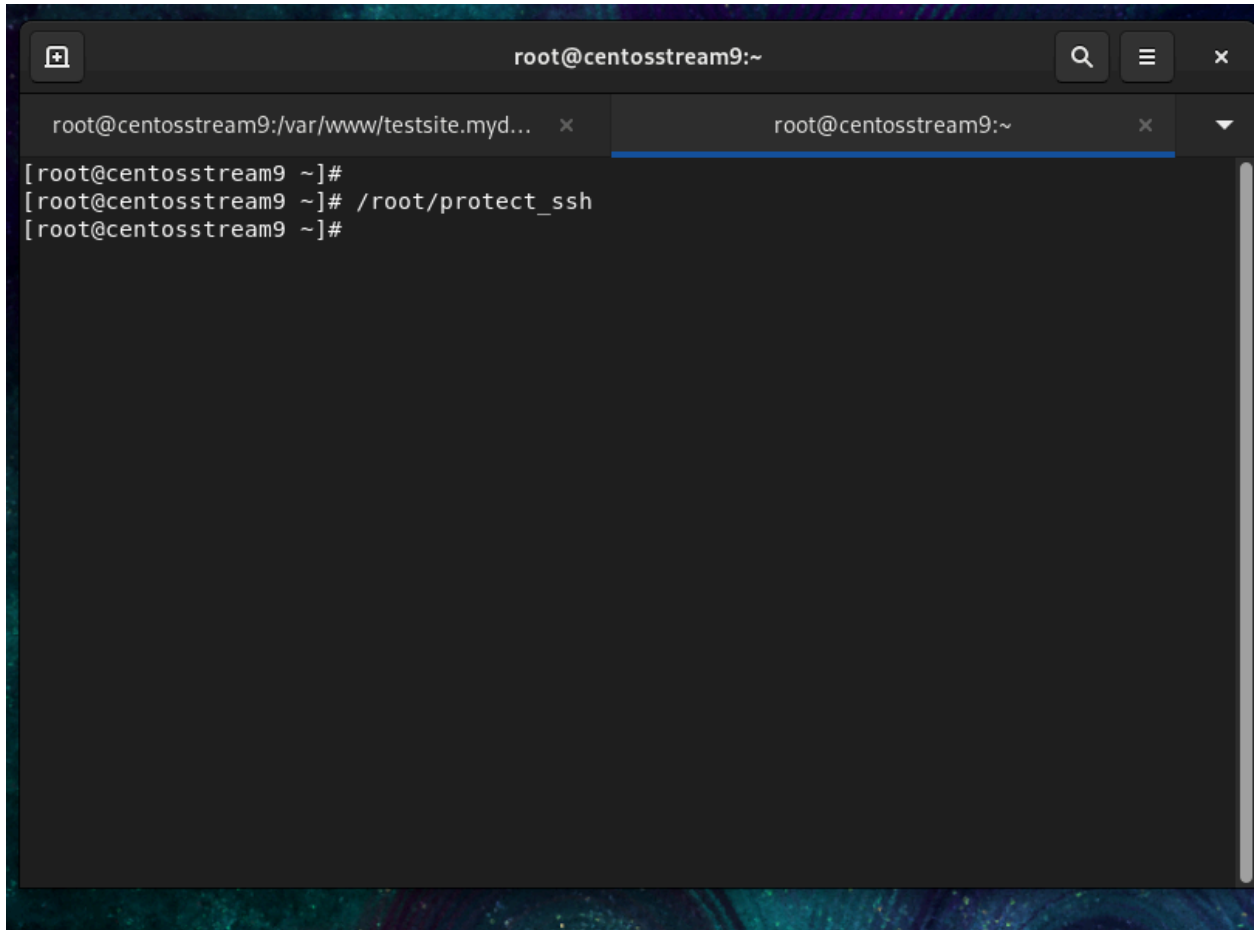
4.13 Create a new file using the command: touch /etc/hosts.deny.bak

A terminal window titled 'root@centosstream9:~' is shown. The window has a dark background and a light-colored text. The terminal output shows the following commands and their results:

```
[root@centosstream9 ~]#  
[root@centosstream9 ~]# touch /etc/hosts.deny  
[root@centosstream9 ~]# touch /etc/hosts.deny.bak  
[root@centosstream9 ~]#
```

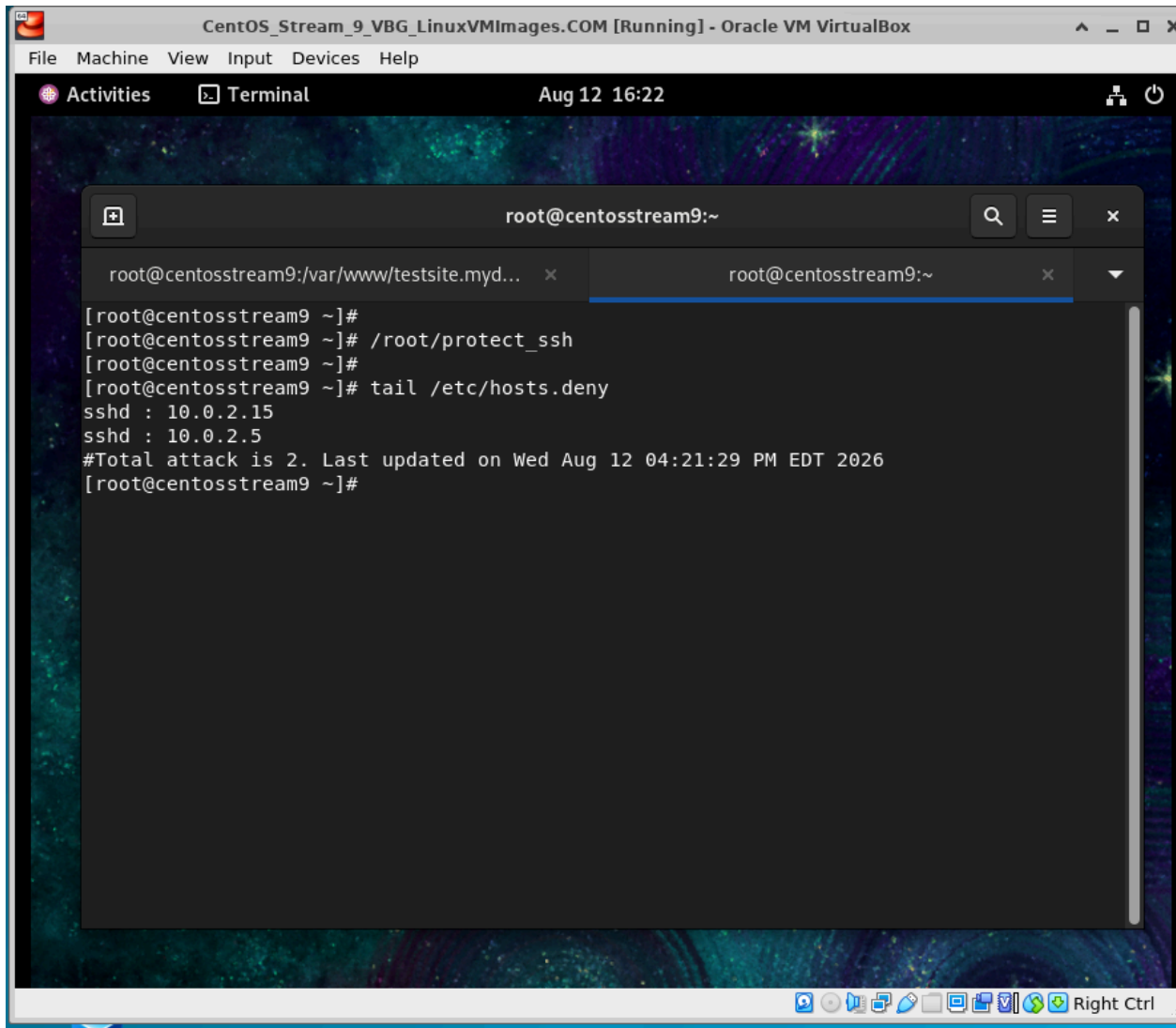
The terminal window is part of a desktop environment with a dark, abstract background. The terminal window has a title bar with a search icon, a menu icon, and a close icon. The terminal window has two tabs: 'root@centosstream9:/var/www/testsite.myd...' and 'root@centosstream9:~'. The 'root@centosstream9:~' tab is selected.

4.14 Use the command: /root/protect_ssh

A terminal window titled 'root@centosstream9:~' with a search icon, menu icon, and close button. It has two tabs: 'root@centosstream9:/var/www/testsite.myd...' and 'root@centosstream9:~'. The active tab shows a shell prompt '[root@centosstream9 ~]#' followed by the command '/root/protect_ssh' and a new prompt '[root@centosstream9 ~]#'.

```
root@centosstream9:~  
[root@centosstream9 ~]#  
[root@centosstream9 ~]# /root/protect_ssh  
[root@centosstream9 ~]#
```


4.15 Now use the command: `tail /etc/hosts.deny`



```
CentOS_Stream_9_VBG_LinuxVMImages.COM [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Aug 12 16:22

root@centosstream9:~
root@centosstream9:~# /root/protect_ssh
root@centosstream9:~# tail /etc/hosts.deny
sshd : 10.0.2.15
sshd : 10.0.2.5
#Total attack is 2. Last updated on Wed Aug 12 04:21:29 PM EDT 2026
root@centosstream9:~#
```

By following the above steps, I have successfully configured honeypots, secured SSH connections, implemented IP blocking, and analyzed intrusion patterns to mitigate brute-force attacks on Linux servers, ensuring enhanced security and operational stability.